

Laplace Transform

9.1 INTRODUCTION

In Chapters 3 to 6, we saw how frequency-domain methods are extremely useful in the study of signals and LTI systems. In these chapters, we demonstrated that Fourier analysis reduces the convolution operation required to compute the output of LTI systems to just the product of the Fourier transform of the input signal and the frequency response of the system. One of the problems we can run into is that many of the input signals we would like to use do not have Fourier transforms. Examples are $e^{at}u(t)$, $a > 0$; e^{-at} , $-\infty < t < \infty$; $tu(t)$; and other signals that are not absolutely integrable. The Fourier transform does not exist for signals that are not absolutely integrable, so the Fourier transform based methods cannot be employed in this class of problems. The difficulty could be resolved by generalizing the Fourier transform so that the signal $x(t)$ is expressed as a sum of complex exponentials e^{st} , where $s = \sigma + j\omega$ and thus is not restricted to the imaginary axis only. This is equivalent to multiplying the signal by an exponential convergent factor. For example, $e^{-\sigma t} e^{at}u(t)$ satisfies Dirichlet's conditions for $\sigma > \alpha$. This observation leads to generalization of the continuous-time Fourier transform, known as Laplace transform, named after the French mathematician Pierre Simon de Laplace.

The Laplace transform possesses a distinct set of properties for analysing signals and LTI systems. Many of these properties parallel those of the Fourier transform. The Laplace transform can be applied to the analysis of many unstable systems and consequently plays an important role in the investigation of the stability or instability of systems.

The Laplace transform comes in two varieties: *Bilateral*, or two-sided Laplace transform, and *unilateral*, or one-sided Laplace transform. The bilateral Laplace transform offers insight into the nature of system characteristics such as stability, causality, and frequency response. The unilateral Laplace transform is a convenient tool for solving differential equations with initial conditions.

9.2 THE BILATERAL (TWO-SIDED) LAPLACE TRANSFORM

Consider applying a complex exponential input $x(t) = e^{st}$ to an LTI system with impulse response $h(t)$. The system output is given by

$$\begin{aligned} y(t) &= h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \\ &= \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau \\ &= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau \\ y(t) &= H(s) e^{st} \end{aligned}$$

where

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

or equivalently,

$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-st} dt \quad (9.1)$$

$H(s)$ is known as the transfer function of the LTI system. We know that a signal for which the system output is a constant times the input is referred to as an *eigenfunction* of the system, and the amplitude factor is referred to as the system's *eigenvalue*. Hence, we identify e^{st} as an eigenfunction of the LTI system and $H(s)$ as the corresponding eigenvalue. For general values of the complex variable s , $H(s)$ is referred to as the *bilateral Laplace transform* or simply the *Laplace transform* of the impulse response $h(t)$. The bilateral Laplace transform in Eq. (9.1) involves an integration from $-\infty$ to $+\infty$, while the unilateral Laplace transform has a form similar to that in Eq. (9.1), but with limits of integration from 0^- to $+\infty$.

9.2.1 Inverse Laplace Transform

Substituting $s = \sigma + j\omega$ into Eq. (9.1) and using t as the variable of integration, we obtain

$$H(\sigma + j\omega) = \int_{-\infty}^{\infty} h(t) e^{-(\sigma + j\omega)t} dt = \int_{-\infty}^{\infty} [h(t) e^{-\sigma t}] e^{-j\omega t} dt$$

The above equation indicates that $H(\sigma + j\omega)$ is the Fourier transform of $[h(t)e^{-\sigma t}]$. Hence, the inverse Fourier transform of $H(\sigma + j\omega)$ must be $[h(t)e^{-\sigma t}]$, i.e.,

$$\begin{aligned} h(t)e^{-\sigma t} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\sigma + j\omega) e^{j\omega t} d\omega \\ h(t) &= e^{\sigma t} \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\sigma + j\omega) e^{j\omega t} d\omega \\ h(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} H(\sigma + j\omega) e^{(\sigma + j\omega)t} d\omega \end{aligned}$$

A change of variables is performed by letting $s = \sigma + j\omega$, which also yields $d\omega = ds/j$, $s \rightarrow (\sigma - j\infty)$ as $\omega \rightarrow -\infty$, and $s \rightarrow (\sigma + j\infty)$ as $\omega \rightarrow \infty$. Therefore,

$$h(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} H(s) e^{st} ds \quad (9.2)$$

Equation (9.2) expresses $h(t)$ as a function of $H(s)$. We say that $h(t)$ is the *inverse Laplace transform* of $H(s)$.

We have obtained the Laplace transform of the impulse response of a system. This relationship holds for an arbitrary signal. The Laplace transform of a general signal $x(t)$ is defined as

$$\mathcal{L}[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad (9.3)$$

and the inverse Laplace transform of $X(s)$ is

$$x(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} X(s) e^{st} ds \quad (9.4)$$

This equation states that $x(t)$ can be represented as a weighted integral of complex exponentials e^{st} . In practice, we usually do not evaluate this integral directly, since it requires techniques of contour integration. Instead, we determine inverse Laplace transform by exploiting the one-to-one relationship between $x(t)$ and $X(s)$.

We denote the transform relationship between $x(t)$ and $X(s)$ as

$$x(t) \longleftrightarrow X(s) \quad (9.5)$$

Finding the inverse Laplace transform by using Eq. (9.4) requires integration in the complex plane, a subject beyond the scope of this book. However, for the case of rational transforms, the inverse Laplace transform can be determined without directly evaluating Eq. (9.4) by using the technique of partial fraction expansion.

9.3 RELATIONSHIP BETWEEN LAPLACE TRANSFORM AND FOURIER TRANSFORM

Consider a continuous-time signal $x(t)$. Its Laplace transform is defined as

$$\mathcal{L}[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Substituting $s = \sigma + j\omega$ into the above equation, we obtain

$$\mathcal{L}[x(t)] = \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt = \int_{-\infty}^{\infty} [x(t) e^{-\sigma t}] e^{-j\omega t} dt = \mathcal{F}[x(t) e^{-\sigma t}]$$

$$\mathcal{L}[x(t)] = \mathcal{F}[x(t) e^{-\sigma t}] \quad (9.6)$$

Thus, the Laplace transform of $x(t)$ is the Fourier transform of $x(t) e^{-\sigma t}$. Now, if $\sigma = 0$, we obtain

$$\mathcal{L}[x(t)] = \mathcal{F}[x(t)], \quad \text{for } \sigma = 0 \quad (9.7)$$

9.4 REGION OF CONVERGENCE (ROC) FOR LAPLACE TRANSFORMS

Equation (9.6) indicates that the Laplace transform of $x(t)$ is the Fourier transform of $x(t) e^{-\sigma t}$. Hence, the Laplace transform is guaranteed to converge if $x(t) e^{-\sigma t}$ is absolutely integrable. That is, we must have

$$\int_{-\infty}^{\infty} |x(t) e^{-\sigma t}| dt < \infty \quad (9.8)$$

This guarantees that $X(s)$ will be finite, since

$$\mathcal{L}[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Substituting $s = \sigma + j\omega$ into the above equation, we obtain

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-(\sigma + j\omega)t} dt$$

$$X(s) = \int_{-\infty}^{\infty} [x(t) e^{-\sigma t}] e^{-j\omega t} dt$$

$$|X(s)| = \left| \int_{-\infty}^{\infty} [x(t) e^{-\sigma t}] e^{-j\omega t} dt \right|$$

$$|X(s)| \leq \int_{-\infty}^{\infty} |x(t) e^{-\sigma t} e^{-j\omega t}| dt$$

$$|X(s)| \leq \int_{-\infty}^{\infty} |x(t) e^{-\sigma t}| dt$$

So if

$$\int_{-\infty}^{\infty} |x(t) e^{-\sigma t}| dt < \infty$$

then

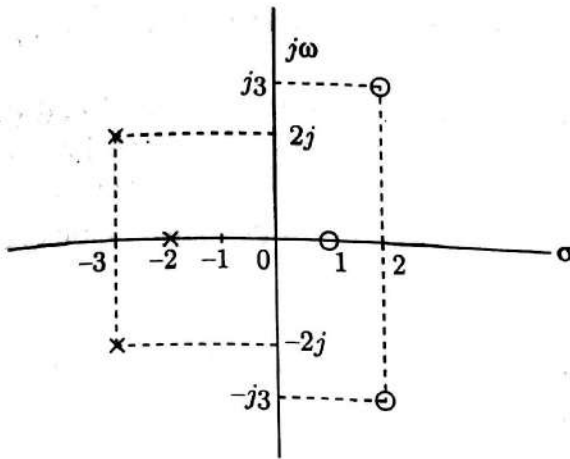
$$|X(s)| < \infty$$

The range of $\Re\{s\} = \sigma$ for which the Laplace transform converges is termed the *region of convergence* (ROC). That is, the ROC consists of those values of σ for which the Fourier transform of $x(t) e^{-\sigma t}$ converges. Note that the Laplace transform exists for some signals that do not have a Fourier transform. By limiting ourselves to a certain range of σ , we may ensure that $x(t) e^{-\sigma t}$ is absolutely integrable, even though $x(t)$ is not absolutely integrable by itself.

The ROC can also provide us with information about whether $x(t)$ is Fourier transformable or not. Since the Fourier transform is obtained from the bilateral Laplace transform by setting $\sigma = 0$, the ROC in this case is a single line (the $j\omega$ -axis). Therefore, if the ROC for $X(s)$ includes the $j\omega$ axis, $x(t)$ is Fourier transformable, and $X(\omega)$ can be obtained by replacing s in $X(s)$ by $j\omega$.

9.5 s -PLANE

It is convenient to represent the complex frequency s graphically in terms of a complex plane called the s -plane, as depicted in Fig. 9.1. The horizontal axis represents the real part of s (i.e., σ), and the vertical axis represents the imaginary part of s (i.e., $j\omega$).

Fig. 9.1 s -plane

The $j\omega$ -axis divides the s -plane into two equal halves. The region of the s -plane to the left of the $j\omega$ -axis is termed the *left half of the s -plane*, while the region of the s -plane to the right of the $j\omega$ -axis is termed the *right half of the s -plane*. The real part of s (i.e., σ) is negative in the left half of the s -plane and positive in the right half of the s -plane.

9.5.1 Poles and Zeros

The most commonly encountered form of the Laplace transform in engineering is a ratio of two polynomials in s , i.e.,

$$H(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

It is often convenient to factor the polynomials in the numerator and denominator, and to write the transfer function in terms of those factors:

$$H(s) = \frac{N(s)}{D(s)} = K \frac{(s - z_1)(s - z_2) \cdots (s - z_{m-1})(s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_{n-1})(s - p_n)} \quad (9.9)$$

where the numerator and denominator polynomials, $N(s)$ and $D(s)$, have real coefficients and $K = b_m/a_n$. As written in the above equation, the z_i 's are the roots of the equation $N(s) = 0$, and are defined as zeros, and the p_i 's are the roots of the equation $D(s) = 0$, and are defined as poles. In Eq. (9.9) the factors in the numerator and denominator are written so that when $s = z_i$ the numerator $N(s) = 0$ and the transfer function vanishes, i.e.,

$$\lim_{s \rightarrow z_i} H(s) = 0$$

and similarly when $s = p_i$ the denominator polynomial $D(s) = 0$ and the value of the transfer function becomes unbounded, that is,

$$\lim_{s \rightarrow p_i} H(s) = \infty$$

All the coefficients of polynomials $N(s)$ and $D(s)$ are real, and therefore the poles and zeros must either be purely real or appear in complex conjugate pairs. The existence of a single complex pole without a corresponding conjugate pole would generate complex

coefficients in the polynomial $D(s)$. Similarly, the zeros are either real or appear in complex conjugate pairs.

We denote the locations of zeros in the s -plane with the 'o' symbol and the location of the poles with the 'x' symbol, as illustrated in Fig. 9.1. The locations of the poles and zeros in the s -plane uniquely specify $H(s)$, except for the constant gain factor K . In Fig. 9.1, zeros are depicted at $s = 1$ and $s = 2 \pm j3$, and poles are depicted at $s = -2$ and $s = -3 \pm 2j$.

Example 9.1 For each of the following integrals, specify the values of the real parameter σ which ensures that the integral converges:

1. $\int_0^{\infty} e^{-5t} e^{-(\sigma+j\omega)t} dt$
2. $\int_{-\infty}^0 e^{-5t} e^{-(\sigma+j\omega)t} dt$
3. $\int_{-5}^5 e^{-5t} e^{-(\sigma+j\omega)t} dt$
4. $\int_{-\infty}^{\infty} e^{-5t} e^{-(\sigma+j\omega)t} dt$
5. $\int_{-\infty}^{\infty} e^{-5|t|} e^{-(\sigma+j\omega)t} dt$

Solution

1. Consider the given integral

$$\begin{aligned} \int_0^{\infty} e^{-5t} e^{-(\sigma+j\omega)t} dt &= \int_0^{\infty} e^{-(\sigma+5)t} e^{-j\omega t} dt \\ &= -\frac{1}{(5+\sigma+j\omega)} e^{-(\sigma+5)t} e^{-j\omega t} \Big|_0^{\infty} \end{aligned}$$

Note that $|e^{-j\omega t}| = 1$ regardless of the value of ωt . Therefore, as $t \rightarrow \infty$, $e^{-(\sigma+5)t} e^{-j\omega t} \rightarrow 0$ only if $(\sigma+5) > 0$ or, $\sigma > -5$, and $e^{-(\sigma+5)t} e^{-j\omega t} \rightarrow \infty$ only if $(\sigma+5) < 0$ or, $\sigma < -5$. Thus,

$$\lim_{t \rightarrow \infty} e^{-(\sigma+5)t} e^{-j\omega t} = \begin{cases} 0, & (\sigma+5) > 0 \rightarrow \sigma > -5 \\ \infty, & (\sigma+5) < 0 \rightarrow \sigma < -5 \end{cases}$$

Therefore,

$$\begin{aligned} \int_0^{\infty} e^{-(\sigma+5)t} e^{-j\omega t} dt &= -\frac{1}{(5+\sigma+j\omega)} [0 - 1] \\ &= \frac{1}{(5+\sigma+j\omega)} \quad \text{for } \sigma > -5 \end{aligned}$$

2. Consider the given integral

$$\begin{aligned}\int_{-\infty}^0 e^{-5t} e^{-(\sigma+j\omega)t} dt &= \int_{-\infty}^0 e^{-(\sigma+5)t} e^{-j\omega t} dt \\ &= -\frac{1}{(5+\sigma+j\omega)} e^{-(\sigma+5)t} e^{-j\omega t} \Big|_{-\infty}^0\end{aligned}$$

Note that $|e^{-j\omega t}| = 1$ regardless of the value of ωt . Therefore, as $t \rightarrow -\infty$, $e^{-(\sigma+5)t} e^{-j\omega t} \rightarrow 0$ only if $(\sigma+5) < 0$ or, $\sigma < -5$, and $e^{-(\sigma+5)t} e^{-j\omega t} \rightarrow \infty$ only if $(\sigma+5) > 0$ or, $\sigma > -5$. Thus,

$$\lim_{t \rightarrow -\infty} e^{-(\sigma+5)t} e^{-j\omega t} = \begin{cases} 0, & (\sigma+5) < 0 \rightarrow \sigma < -5 \\ \infty, & (\sigma+5) > 0 \rightarrow \sigma > -5 \end{cases}$$

Therefore,

$$\begin{aligned}\int_{-\infty}^0 e^{-(\sigma+5)t} e^{-j\omega t} dt &= -\frac{1}{(5+\sigma+j\omega)} [1 - 0] \\ &= -\frac{1}{(5+\sigma+j\omega)} \quad \text{for } \sigma < -5\end{aligned}$$

3. The given integral may be written as

$$\int_{-5}^5 e^{-(5+\sigma)t} e^{-j\omega t} dt$$

Since this integral has a finite range of limit, it has a finite value for all finite values of σ .

4. The given integral may be written as

$$\int_{-\infty}^{\infty} e^{-(5+\sigma)t} e^{-j\omega t} dt = \int_{-\infty}^0 e^{-(5+\sigma)t} e^{-j\omega t} dt + \int_0^{\infty} e^{-(5+\sigma)t} e^{-j\omega t} dt$$

The first integral diverges for $\sigma < -5$. The second integral diverges for $\sigma > -5$. If $\sigma = -5$, then both the integrals still diverge. Therefore, the integral does not converge for any value of σ .

5. The given integral may be written as

$$\int_{-\infty}^0 e^{-(-5+\sigma)t} e^{-j\omega t} dt + \int_0^{\infty} e^{-(5+\sigma)t} e^{-j\omega t} dt$$

The first integral converges for $\sigma < 5$. The second integral converges for $\sigma > -5$. Therefore, the integral converges for $|\sigma| < 5$.

Example 9.2 Determine the Laplace transform of the causal signal [shown in Fig. 9.2(a)]

$$x(t) = e^{-at}u(t)$$

and depict the ROC and the locations of poles and zeros in the s -plane.

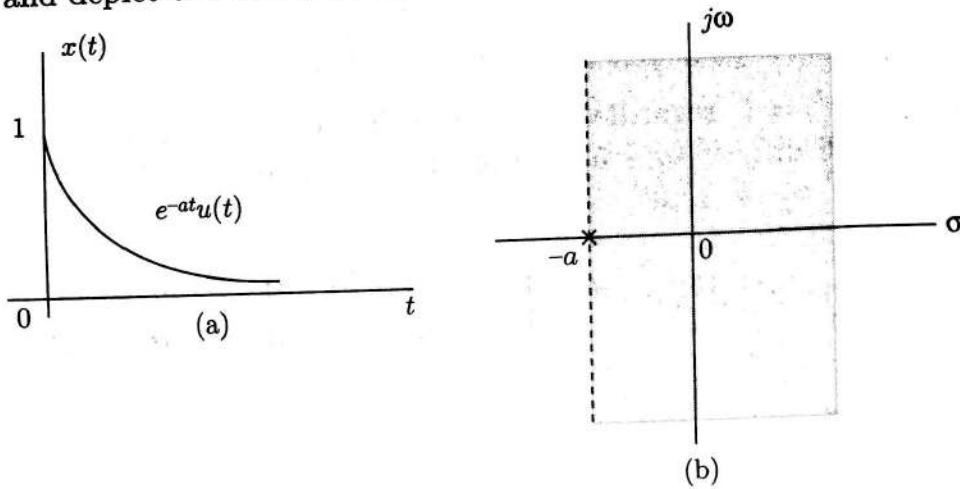


Fig. 9.2 (a) Signal $x(t) = e^{-at}u(t)$ and (b) its ROC

Solution

By definition

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} e^{-at}u(t) e^{-st} dt$$

Since, $u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$, we obtain

$$\begin{aligned} X(s) &= \int_0^{\infty} e^{-at} e^{-st} dt = \int_0^{\infty} e^{-(s+a)t} dt \\ X(s) &= -\frac{1}{s+a} e^{-(s+a)t} \Big|_0^{\infty} \end{aligned}$$

Substitute $s = \sigma + j\omega$ into the above equation to obtain

$$X(s) = -\frac{1}{\sigma + j\omega + a} e^{-(\sigma+a)t} e^{-j\omega t} \Big|_0^{\infty}$$

Now, if $(\sigma + a) > 0$ or $\sigma > -a$, then $e^{-(\sigma+a)t} \rightarrow 0$ as $t \rightarrow \infty$, and

$$X(s) = -\frac{1}{\sigma + j\omega + a} [0 - 1] \quad \sigma > -a$$

$$X(s) = \frac{1}{\sigma + j\omega + a} \quad \sigma > -a$$

$$X(s) = \frac{1}{s+a} \quad \Re\{s\} > -a$$

Therefore,

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a \quad (9.10)$$

The ROC for $X(s)$ is $\Re\{s\} > -a$, as shown in the shaded area in Fig. 9.2(b).

Example 9.3 Determine the Laplace transform of an anticausal signal [shown in Fig. 9.3(a)]

$$x(t) = -e^{-at}u(-t)$$

and depict the ROC and the locations of poles and zeros in the s -plane.

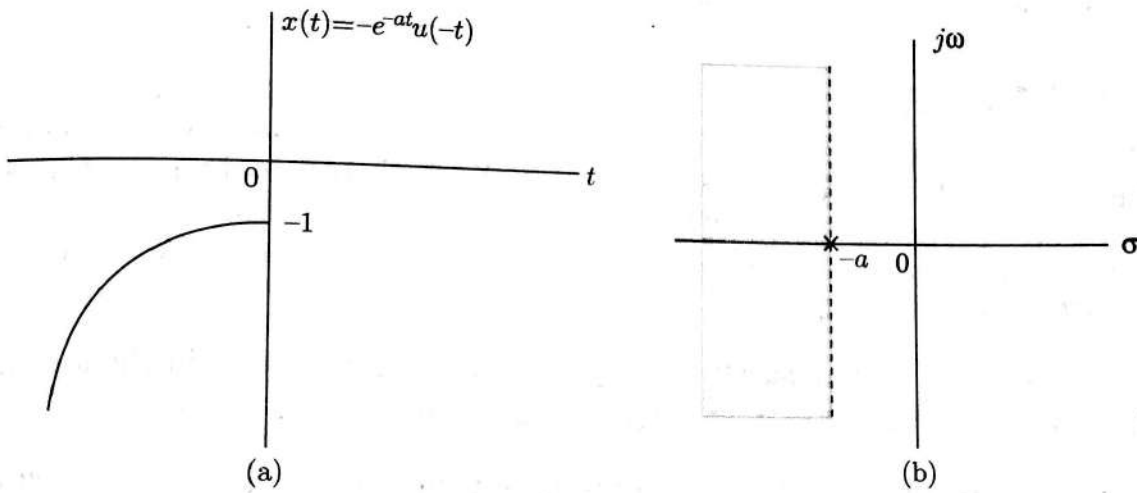


Fig. 9.3 (a) Signal $x(t) = -e^{-at}u(-t)$ and (b) its ROC

Solution

By definition

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} -e^{-at}u(-t) e^{-st} dt$$

Since, $u(-t) = \begin{cases} 0, & t > 0 \\ 1, & t < 0 \end{cases}$, we obtain

$$X(s) = - \int_{-\infty}^0 e^{-at} e^{-st} dt = - \int_{-\infty}^0 e^{-(s+a)t} dt$$

$$X(s) = \frac{1}{s+a} e^{-(s+a)t} \Big|_{-\infty}^0$$

Substitute $s = \sigma + j\omega$ into the above equation to obtain

$$X(s) = \frac{1}{\sigma + j\omega + a} e^{-(\sigma+a)t} e^{-j\omega t} \Big|_{-\infty}^0$$

Now, if $(\sigma + a) < 0$ or $\sigma < -a$, then $e^{-(\sigma+a)t} \rightarrow 0$ as $t \rightarrow -\infty$, and

$$X(s) = \frac{1}{\sigma + j\omega + a} [1 - 0] \quad \sigma < -a$$

$$X(s) = \frac{1}{\sigma + j\omega + a} \quad \sigma < -a$$

$$X(s) = \frac{1}{s + a} \quad \Re\{s\} < -a$$

Therefore,

$$-e^{-at}u(-t) \longleftrightarrow \frac{1}{s + a} \quad \Re\{s\} < -a \quad (9.11)$$

The ROC for $X(s)$ is $\Re\{s\} < -a$, as shown in the shaded area in Fig. 9.3(b).

9.6 PROPERTIES OF ROC

Note that the Laplace transforms for the signals $e^{-at}u(t)$ and $-e^{-at}u(-t)$ are identical except for their ROCs. Therefore, for a given $X(s)$, there may be more than one inverse transform, depending on the ROC. In other words, unless the ROC is specified, there is no one-to-one correspondence between $X(s)$ and $x(t)$. In this section, we explain properties of the ROC of various classes of signals.

1. The ROC of $X(s)$ consists of strips parallel to the $j\omega$ -axis in the s -plane.
2. If $X(s)$ is rational, then the ROC must not contain any poles.
3. If $x(t)$ is of finite duration and is absolutely integrable (i.e., $\int_{-\infty}^{\infty} |x(t)| dt < \infty$), then the ROC is the entire s -plane.
4. If $x(t)$ is right-sided and of infinite duration (i.e., $x(t) = 0$ for all $t < T_1$ for some finite T_1), then the ROC is the region in the s -plane to the right of the rightmost pole.
5. If $x(t)$ is left-sided and of infinite duration (i.e., $x(t) = 0$ for all $t > T_2$ for some finite T_2), then the ROC is the region in the s -plane to the left of the leftmost pole.
6. If $x(t)$ is two-sided and of infinite duration (i.e., the signal is of infinite extent for both $t < 0$ and $t > 0$), then the ROC will consist of a strip in the s -plane.

Example 9.4 Determine the Laplace transform of

$$x(t) = e^{-2t}u(t) - e^{-3t}u(t)$$

and depict the ROC and the locations of poles and zeros in the s -plane.

Solution

Consider the given signal

$$x(t) = e^{-2t}u(t) - e^{-3t}u(t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[e^{-2t}u(t) - e^{-3t}u(t)]$$

$$X(s) = \mathcal{L}[e^{-2t}u(t)] + \mathcal{L}[e^{-3t}u(t)]$$

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From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a$$

Therefore,

$$e^{-2t}u(t) \longleftrightarrow \frac{1}{s+2} \quad \Re\{s\} > -2$$

$$e^{-3t}u(t) \longleftrightarrow \frac{1}{s+3} \quad \Re\{s\} > -3$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > -2$, and thus we obtain

$$X(s) = \frac{1}{s+2} - \frac{1}{s+3} \quad \Re\{s\} > -2$$

$$X(s) = \frac{1}{s^2 + 5s + 6} \quad \Re\{s\} > -2$$

The ROC for $X(s)$ is $\Re\{s\} > -2$, as shown in the shaded area in Fig. 9.4. For a right-sided infinite duration signal, the ROC is the region in the s -plane to the right of the rightmost pole.

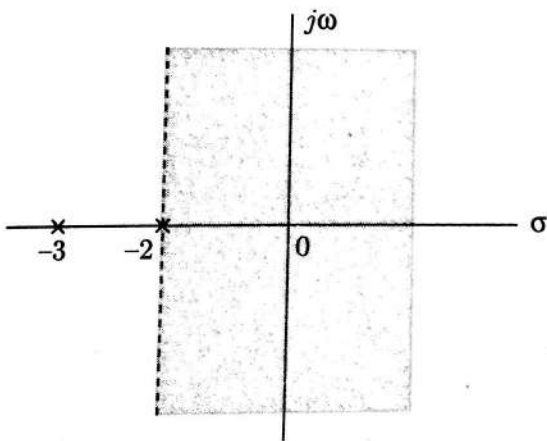


Fig. 9.4 Pole-zero plot and ROC for Example 9.4

Example 9.5 Determine the Laplace transform of

$$x(t) = -e^{-2t}u(-t) + e^{-3t}u(-t)$$

and depict the ROC and the locations of poles and zeros in the s -plane.

Solution

Consider the given signal

$$x(t) = -e^{-2t}u(-t) + e^{-3t}u(-t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[-e^{-2t}u(-t) + e^{-3t}u(-t)]$$

$$X(s) = \mathcal{L}[-e^{-2t}u(-t)] - \mathcal{L}[e^{-3t}u(-t)]$$

From Eq. (9.11), we have

$$-e^{-at}u(-t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} < -a$$

Therefore,

$$-e^{-2t}u(-t) \longleftrightarrow \frac{1}{s+2} \quad \Re\{s\} < -2$$

$$-e^{-3t}u(-t) \longleftrightarrow \frac{1}{s+3} \quad \Re\{s\} < -3$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} < -3$, and thus we obtain

$$X(s) = \frac{1}{s+2} - \frac{1}{s+3} \quad \Re\{s\} < -3$$

$$X(s) = \frac{1}{s^2 + 5s + 6} \quad \Re\{s\} < -3$$

The ROC for $X(s)$ is $\Re\{s\} < -3$, as shown in the shaded area in Fig. 9.5. For a left-sided infinite duration signal, the ROC is the region in the s -plane to the left of the leftmost pole.

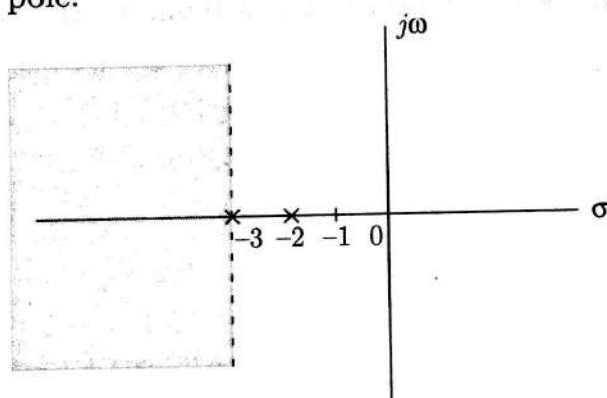


Fig. 9.5 Pole-zero plot and ROC for Example 9.5

Example 9.6 Determine the Laplace transform of the finite duration signal

$$x(t) = \begin{cases} e^{-at}, & 0 < t < T \\ 0, & \text{otherwise} \end{cases}$$

Solution

By definition

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_0^T e^{-at} e^{-st} dt$$

$$= \int_0^T e^{-(s+a)t} dt$$

$$X(s) = \frac{1}{s+a} [1 - e^{-(s+a)T}]$$

The ROC is the entire s -plane. For a finite duration signal, the ROC is the entire s -plane. At $s = -a$, both numerator and denominator are zero in the above expression. To determine $X(s)$ at $s = -a$, we can use the L'Hôpital's rule to obtain

$$\lim_{s \rightarrow -a} X(s) = \lim_{s \rightarrow -a} \left[\frac{d/dt(1 - e^{-(s+a)T})}{d/dt(s+a)} \right]$$

$$X(-a) = \lim_{s \rightarrow -a} T e^{-(s+a)T}$$

$$X(-a) = T$$

Example 9.7 Determine the Laplace transform of

- (a) a unit impulse function $x(t) = \delta(t)$
- (b) a unit step function $x(t) = u(t)$
- (c) a unit ramp signal $x(t) = r(t)$

Solution

(a) Given that $x(t) = \delta(t)$. By definition

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} \delta(t) e^{-st} dt = e^{-st} \Big|_{t=0} = 1$$

Therefore,

$$\delta(t) \longleftrightarrow 1 \quad \text{ROC is the entire } s\text{-plane} \quad (9.12)$$

(b) Given that $x(t) = u(t)$. By definition

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} u(t) e^{-st} dt = \int_0^{\infty} e^{-st} dt \\ &= -\frac{1}{s} e^{-st} \Big|_0^{\infty} \\ &= -\frac{1}{s} [0 - 1] \quad \Re\{s\} > 0 \\ X(s) &= \frac{1}{s} \quad \Re\{s\} > 0 \end{aligned}$$

Therefore,

$$u(t) \longleftrightarrow \frac{1}{s} \quad \Re\{s\} > 0 \quad (9.13)$$

(c) Given that $x(t) = r(t) = tu(t)$. By definition

$$\begin{aligned}
 X(s) &= \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} tu(t) e^{-st} dt = \int_0^{\infty} t e^{-st} dt \\
 &= -\frac{1}{s} t e^{-st} \Big|_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt \\
 &= 0 - \frac{1}{s^2} e^{-st} \Big|_0^{\infty} \\
 &= -\frac{1}{s^2} [0 - 1] \quad \Re\{s\} > 0 \\
 X(s) &= \frac{1}{s^2} \quad \Re\{s\} > 0
 \end{aligned}$$

Therefore,

$$r(t) \longleftrightarrow \frac{1}{s^2} \quad \Re\{s\} > 0 \quad (9.14)$$

Example 9.8 Determine the Laplace transform of the following signals:

(a) $x(t) = \sin(\omega_0 t)u(t)$

(b) $x(t) = \cos(\omega_0 t)u(t)$

Solution

(a) Given that

$$x(t) = \sin(\omega_0 t)u(t)$$

$$x(t) = \frac{1}{2j} [e^{j\omega_0 t} u(t) - e^{-j\omega_0 t} u(t)]$$

Taking the Laplace transform of the above equation, we get

$$\mathcal{L}[x(t)] = \frac{1}{2j} (\mathcal{L}[e^{j\omega_0 t} u(t)] - \mathcal{L}[e^{-j\omega_0 t} u(t)])$$

From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a$$

Therefore,

$$e^{j\omega_0 t}u(t) \longleftrightarrow \frac{1}{s-j\omega_0} \quad \Re\{s\} > 0$$

$$e^{-j\omega_0 t}u(t) \longleftrightarrow \frac{1}{s+j\omega_0} \quad \Re\{s\} > 0$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > 0$, and thus we obtain

$$X(s) = \frac{1}{2j} \left(\frac{1}{s - j\omega_0} - \frac{1}{s + j\omega_0} \right) = \frac{1}{2j} \left(\frac{s + j\omega_0 - s + j\omega_0}{s^2 + \omega_0^2} \right) = \frac{\omega_0}{s^2 + \omega_0^2}$$

Therefore,

$$\sin(\omega_0 t)u(t) \longleftrightarrow \frac{\omega_0}{s^2 + \omega_0^2} \quad \Re\{s\} > 0 \quad (9.15)$$

The ROC for $X(s)$ is $\Re\{s\} > 0$, as shown in the shaded area in Fig. 9.6(a).

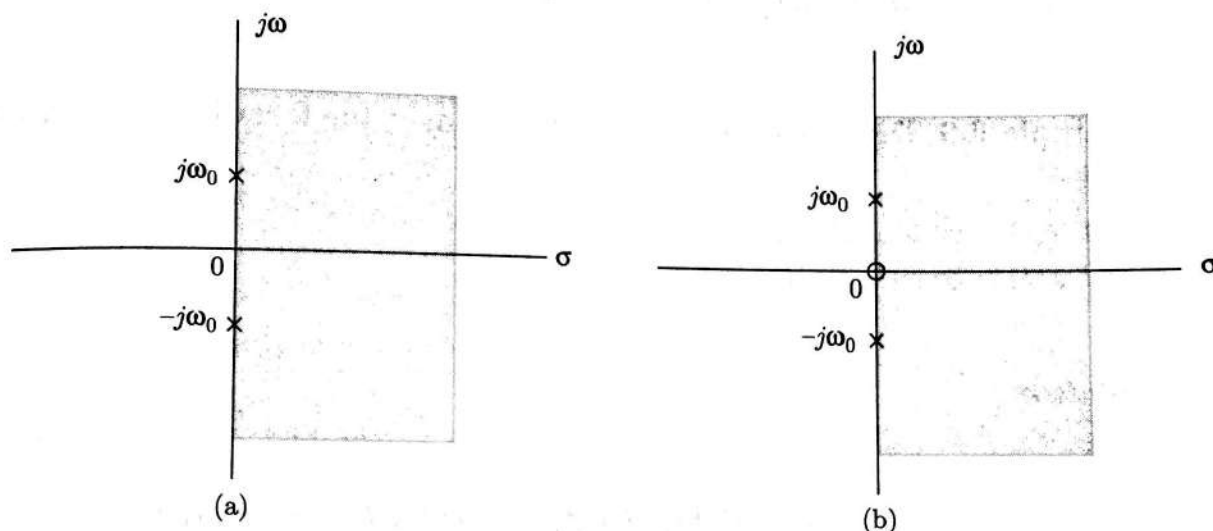


Fig. 9.6 Pole-zero plot and ROC for (a) $x(t) = \sin(\omega_0 t)u(t)$ and (b) $x(t) = \cos(\omega_0 t)u(t)$

(b) Given that

$$x(t) = \cos(\omega_0 t)u(t)$$

$$x(t) = \frac{1}{2} [e^{j\omega_0 t}u(t) + e^{-j\omega_0 t}u(t)]$$

Taking the Laplace transform of the above equation, we get

$$\mathcal{L}[x(t)] = \frac{1}{2} (\mathcal{L}[e^{j\omega_0 t}u(t)] + \mathcal{L}[e^{-j\omega_0 t}u(t)])$$

From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s + a} \quad \Re\{s\} > -a$$

Therefore,

$$e^{j\omega_0 t}u(t) \longleftrightarrow \frac{1}{s - j\omega_0} \quad \Re\{s\} > 0$$

$$e^{-j\omega_0 t}u(t) \longleftrightarrow \frac{1}{s + j\omega_0} \quad \Re\{s\} > 0$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > 0$, and thus we obtain

$$X(s) = \frac{1}{2} \left(\frac{1}{s - j\omega_0} + \frac{1}{s + j\omega_0} \right) = \frac{1}{2} \left(\frac{s + j\omega_0 + s - j\omega_0}{s^2 + \omega_0^2} \right) = \frac{s}{s^2 + \omega_0^2}$$

Therefore,

$$\cos(\omega_0 t)u(t) \longleftrightarrow \frac{s}{s^2 + \omega_0^2} \quad \Re\{s\} > 0 \quad (9.16)$$

The ROC for $X(s)$ is $\Re\{s\} > 0$, as shown in the shaded area in Fig. 9.6(b).

Example 9.9 Determine the Laplace transform of the following hyperbolic sinusoidal signals:

(a) $x(t) = \sinh(\omega_0 t)u(t)$

(b) $x(t) = \cosh(\omega_0 t)u(t)$

Solution

(a) Given that

$$x(t) = \sinh(\omega_0 t)u(t) = \frac{1}{2} [e^{\omega_0 t}u(t) - e^{-\omega_0 t}u(t)]$$

Taking the Laplace transform of the above equation, we get

$$\mathcal{L}[x(t)] = \frac{1}{2} (\mathcal{L}[e^{\omega_0 t}u(t)] - \mathcal{L}[e^{-\omega_0 t}u(t)])$$

From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s + a} \quad \Re\{s\} > -a$$

Therefore,

$$e^{\omega_0 t}u(t) \longleftrightarrow \frac{1}{s - \omega_0} \quad \Re\{s\} > \omega_0$$

$$e^{-\omega_0 t}u(t) \longleftrightarrow \frac{1}{s + \omega_0} \quad \Re\{s\} > -\omega_0$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > \omega_0$, and thus we obtain

$$X(s) = \frac{1}{2} \left(\frac{1}{s - \omega_0} - \frac{1}{s + \omega_0} \right) = \frac{1}{2} \left(\frac{s + \omega_0 - s + \omega_0}{s^2 - \omega_0^2} \right) = \frac{\omega_0}{s^2 - \omega_0^2}$$

Therefore,

$$\sinh(\omega_0 t)u(t) \longleftrightarrow \frac{\omega_0}{s^2 - \omega_0^2} \quad \Re\{s\} > \omega_0 \quad (9.17)$$

The ROC for $X(s)$ is $\Re\{s\} > \omega_0$, as shown in the shaded area in Fig. 9.7(a).

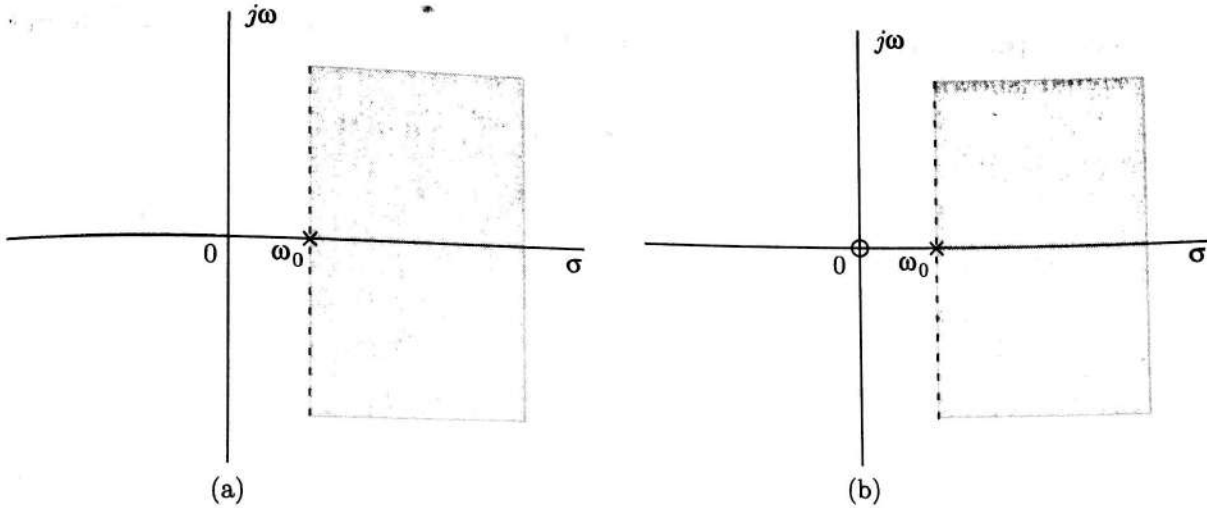


Fig. 9.7 Pole-zero plot and ROC for (a) $x(t) = \sinh(\omega_0 t)u(t)$ and (b) $x(t) = \cosh(\omega_0 t)u(t)$

(b) Given that

$$x(t) = \cosh(\omega_0 t)u(t) = \frac{1}{2}[e^{\omega_0 t}u(t) + e^{-\omega_0 t}u(t)]$$

Taking the Laplace transform of the above equation, we get

$$\mathcal{L}[x(t)] = \frac{1}{2}(\mathcal{L}[e^{\omega_0 t}u(t)] + \mathcal{L}[e^{-\omega_0 t}u(t)])$$

From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a$$

Therefore,

$$\begin{aligned} e^{\omega_0 t}u(t) &\longleftrightarrow \frac{1}{s - \omega_0} \quad \Re\{s\} > \omega_0 \\ e^{-\omega_0 t}u(t) &\longleftrightarrow \frac{1}{s + \omega_0} \quad \Re\{s\} > -\omega_0 \end{aligned}$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > \omega_0$, and thus we obtain

$$X(s) = \frac{1}{2} \left(\frac{1}{s - \omega_0} + \frac{1}{s + \omega_0} \right) = \frac{1}{2} \left(\frac{s + \omega_0 + s - \omega_0}{s^2 - \omega_0^2} \right) = \frac{s}{s^2 - \omega_0^2}$$

Therefore,

$$\cosh(\omega_0 t)u(t) \longleftrightarrow \frac{s}{s^2 - \omega_0^2} \quad \Re\{s\} > \omega_0 \quad (9.18)$$

The ROC for $X(s)$ is $\Re\{s\} > \omega_0$, as shown in the shaded area in Fig. 9.7(b).

Example 9.10 Determine the Laplace transform of the two-sided infinite duration signal [shown in Fig. 9.8(a)]:

$$x(t) = e^{-a|t|}, \quad a > 0$$

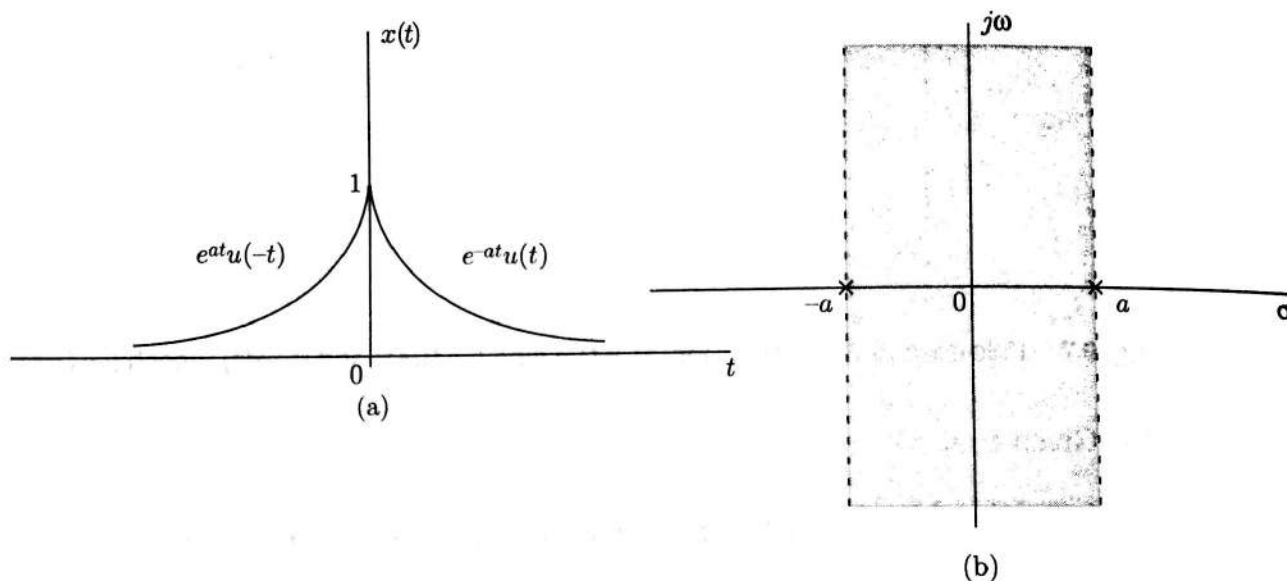


Fig. 9.8 (a) Signal $x(t) = e^{-a|t|}$, $a > 0$ and (b) its ROC

Solution

The signal $x(t)$ is defined as

$$\begin{aligned} x(t) &= e^{-a|t|} = \begin{cases} e^{at}, & t < 0 \\ e^{-at}, & t \geq 0 \end{cases} \\ &= e^{at}u(-t) + e^{-at}u(t) \end{aligned}$$

From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s + a} \quad \Re\{s\} > -a$$

From Eq. (9.11), we have

$$-e^{-at}u(-t) \longleftrightarrow \frac{1}{s + a} \quad \Re\{s\} < -a$$

Therefore,

$$e^{at}u(-t) \longleftrightarrow -\frac{1}{s - a} \quad \Re\{s\} < a$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $-a < \Re\{s\} < a$, and thus we obtain

$$X(s) = \frac{1}{s+a} - \frac{1}{s-a} = \frac{-2a}{s^2 - a^2}$$

Therefore,

$$e^{-a|t|} \longleftrightarrow \frac{-2a}{s^2 - a^2} \quad -a < \Re\{s\} < a \quad (9.19)$$

The ROC for $X(s)$ is $\Re\{s\} > 0$, as shown in the shaded area in Fig. 9.8(b). For a two-sided infinite duration signal, the ROC is a strip in the s -plane.

Example 9.11 Find the Laplace transform of the signal $x(t) = t^n u(t)$.

Solution

By definition

$$\begin{aligned} X(s) = \mathcal{L}[t^n u(t)] &= \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} t^n u(t) e^{-st} dt \\ &= \int_0^{\infty} t^n e^{-st} dt \\ &= -\frac{1}{s} t^n e^{-st} \Big|_0^{\infty} + \frac{n}{s} \int_0^{\infty} t^{n-1} e^{-st} dt \quad \text{for } n \geq 1 \\ &= 0 + \frac{n}{s} \mathcal{L}[t^{n-1} u(t)] \\ \mathcal{L}[t^n u(t)] &= \frac{n}{s} \mathcal{L}[t^{n-1} u(t)] \quad \text{for } n \geq 1 \end{aligned}$$

Similarly, we can obtain

$$\begin{aligned} \mathcal{L}[t^{n-1} u(t)] &= \frac{n-1}{s} \mathcal{L}[t^{n-2} u(t)] \quad \text{for } n \geq 2 \\ \mathcal{L}[t^{n-2} u(t)] &= \frac{n-2}{s} \mathcal{L}[t^{n-3} u(t)] \quad \text{for } n \geq 3 \\ &\vdots \end{aligned}$$

Therefore,

$$\begin{aligned} \mathcal{L}[t^n u(t)] &= \frac{n}{s} \frac{n-1}{s} \frac{n-2}{s} \frac{n-3}{s} \dots \frac{2}{s} \frac{1}{s} \mathcal{L}[t^{n-n} u(t)] = \frac{n!}{s^n} \frac{1}{s} \\ \mathcal{L}[t^n u(t)] &= \frac{n!}{s^{n+1}} \quad \Re\{s\} > 0 \end{aligned}$$

Hence,

$$t^n u(t) \longleftrightarrow \frac{n!}{s^{n+1}} \quad \Re\{s\} > 0 \quad (9.20)$$

or, equivalently,

$$\frac{t^n}{n!} u(t) \longleftrightarrow \frac{1}{s^{n+1}} \quad \Re\{s\} > 0 \quad (9.21)$$

and

$$\frac{t^{n-1}}{(n-1)!} u(t) \longleftrightarrow \frac{1}{s^n} \quad \Re\{s\} > 0 \quad (9.22)$$

Example 9.12 Consider the signal

$$x(t) = e^{-5t} u(t-1)$$

and denote its Laplace transform by $X(s)$.

- Evaluate $X(s)$ and find its ROC.
- Determine the values of the finite numbers A and t_0 such that the Laplace transform $G(s)$ of

$$g(t) = A e^{-5t} u(-t - t_0)$$

has the same algebraic form as $X(s)$. What is the ROC corresponding to $G(s)$?

Solution

(a) By definition

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt = \int_{-\infty}^{\infty} e^{-5t} u(t-1) e^{-st} dt$$

Since

$$u(t-1) = \begin{cases} 1, & t > 1 \\ 0, & t < 1 \end{cases}$$

we have

$$\begin{aligned} X(s) &= \int_1^{\infty} e^{-5t} e^{-st} dt = \int_1^{\infty} e^{-(s+5)t} dt \\ &= \left. \frac{e^{-(s+5)t}}{-(s+5)} \right|_1^{\infty} \\ &= -1(s+5)[0 - e^{-(s+5)}] \quad \Re\{s\} > -5 \\ X(s) &= \frac{e^{-(s+5)}}{s+5} \quad \Re\{s\} > -5 \end{aligned}$$

(b) By definition

$$G(s) = \int_{-\infty}^{\infty} g(t) e^{-st} dt = \int_{-\infty}^{\infty} A e^{-5t} u(-t - t_0) e^{-st} dt$$

Since

$$u(-t - t_0) = \begin{cases} 1, & (-t - t_0) > 0 \longrightarrow t < -t_0 \\ 0, & (-t - t_0) < 0 \longrightarrow t > -t_0 \end{cases}$$

we have

$$\begin{aligned} G(s) &= \int_{-\infty}^{-t_0} A e^{-5t} e^{-st} dt = \int_{-\infty}^{-t_0} A e^{-(s+5)t} dt \\ &= A \frac{e^{-(s+5)t}}{-(s+5)} \Big|_{-\infty}^{-t_0} \\ &= \frac{-A}{s+5} [0 - e^{(s+5)t_0}] \quad \Re\{s\} < -5 \\ G(s) &= \frac{A e^{(s+5)t_0}}{s+5} \quad \Re\{s\} < -5 \end{aligned}$$

Clearly, $G(s)$ has the same algebraic form as $X(s)$ for $A = 1$ and $t_0 = -1$.

Example 9.13 Find the inverse Laplace transform of

$$X(s) = \frac{-3}{(s+2)(s-1)}$$

if the ROC is

- (a) $\Re\{s\} > 1$
- (b) $\Re\{s\} < -2$
- (c) $-2 < \Re\{s\} < 1$

Solution

The partial fraction expansion of $X(s)$ yields

$$X(s) = \frac{1}{s+2} - \frac{1}{s-1}$$

$X(s)$ has poles at -2 and 1 .

(a) The ROC and the locations of the poles are depicted in Fig. 9.9(a). The ROC, $\Re\{s\} > 1$, is to the right of the rightmost pole, so both poles correspond to causal

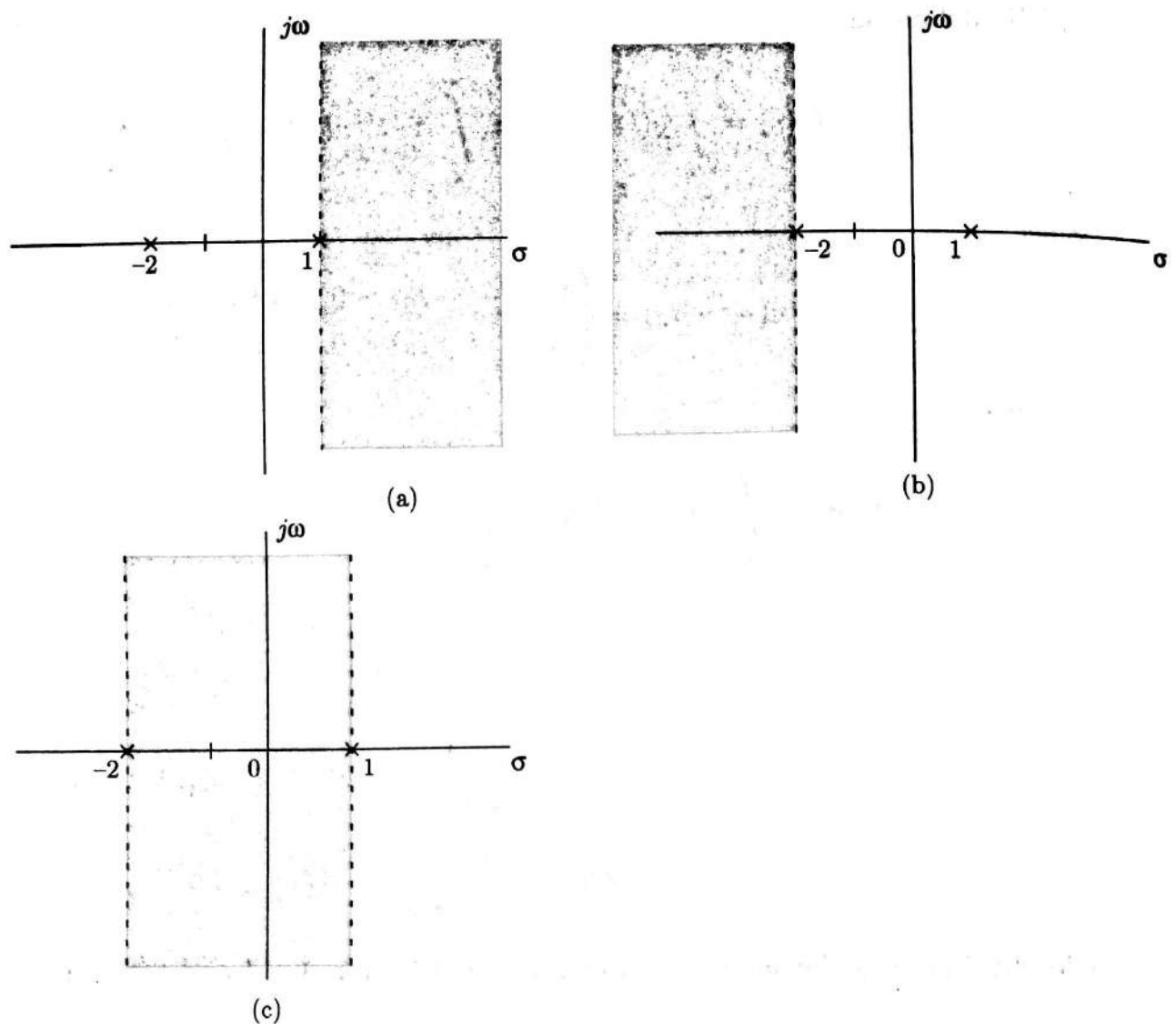


Fig. 9.9 ROC for Example 9.13

(right-sided) signals. Therefore,

$$e^{-2t}u(t) \longleftrightarrow \frac{1}{s+2}$$

$$e^t u(t) \longleftrightarrow \frac{1}{s-1}$$

and

$$x(t) = e^{-2t}u(t) - e^t u(t)$$

(b) The ROC and the locations of the poles are depicted in Fig. 9.9(b). The ROC, $\Re\{s\} < -2$, is to the left of the leftmost pole, so both poles correspond to anticausal (left-sided) signals. Therefore,

$$-e^{-2t}u(-t) \longleftrightarrow \frac{1}{s+2}$$

$$-e^t u(-t) \longleftrightarrow \frac{1}{s-1}$$

and

$$x(t) = -e^{-2t}u(-t) + e^t u(-t)$$

(c) The ROC and the locations of the poles are depicted in Fig. 9.9(c). The ROC, $-2 < \Re\{s\} < 1$, is a strip. The pole of the first term is at -2 . The ROC lies to the right of this pole, so this pole corresponds to a causal (right-sided) signal. Therefore,

$$e^{-2t}u(t) \longleftrightarrow \frac{1}{s+2}$$

The second term has a pole at $s = 1$. Here the ROC is to the left of this pole, so this pole corresponds to the anticausal (left-sided) signal. Therefore,

$$-e^t u(-t) \longleftrightarrow \frac{1}{s-1}$$

and hence we obtain

$$x(t) = e^{-2t}u(t) + e^t u(-t)$$

Example 9.14 Find the inverse Laplace transform of

$$X(s) = \frac{-5s - 7}{(s+1)(s-1)(s+2)}$$

if the ROC is

- (a) $\Re\{s\} > 1$
- (b) $\Re\{s\} < -2$
- (c) $-1 < \Re\{s\} < 1$
- (d) $-2 < \Re\{s\} < -1$

Solution

Using the partial fraction expansion, we obtain

$$X(s) = \frac{1}{s+1} - \frac{2}{s-1} + \frac{1}{s+2}$$

$X(s)$ has poles at -1 , 1 , and -2 .

(a) The ROC and the locations of the poles are depicted in Fig. 9.10(a). The ROC, $\Re\{s\} > 1$, is to the right of the rightmost pole, so all the three poles correspond to causal (right-sided) signals. Therefore,

$$e^{-t}u(t) \longleftrightarrow \frac{1}{s+1}$$

$$2e^t u(t) \longleftrightarrow \frac{2}{s-1}$$

$$e^{-2t}u(t) \longleftrightarrow \frac{1}{s+2}$$

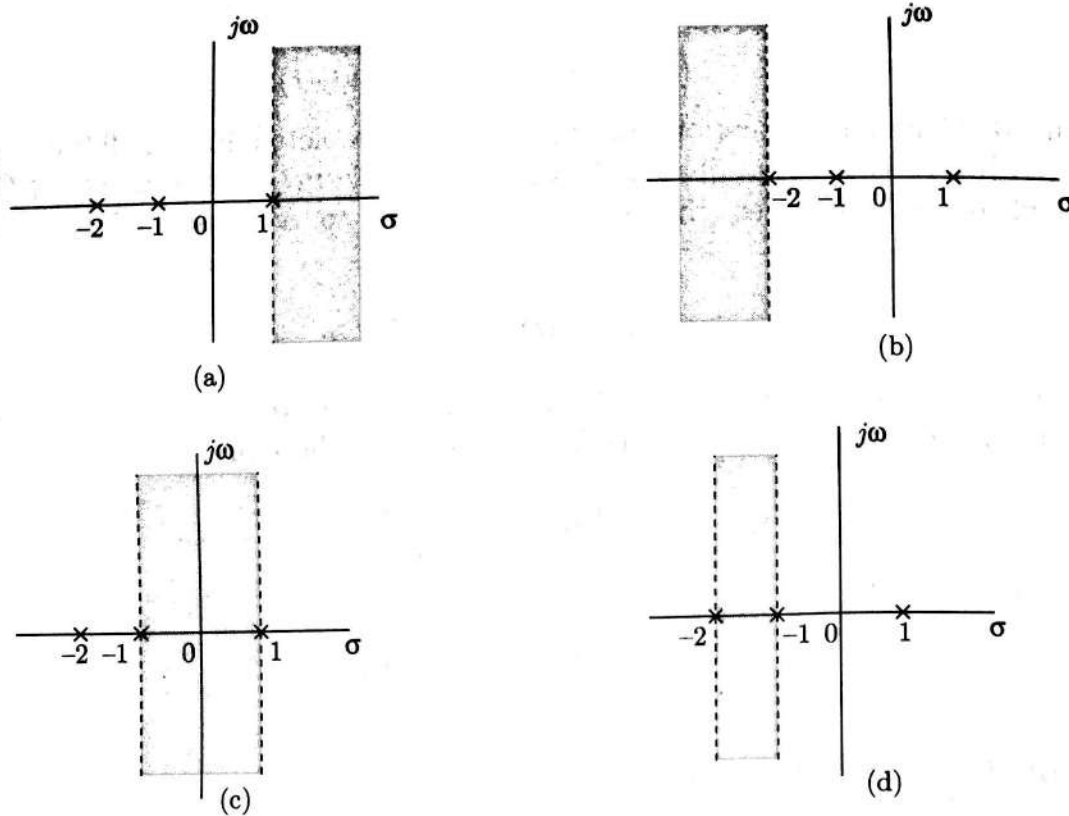


Fig. 9.10 ROC for Example 9.14

and hence,

$$x(t) = e^{-t}u(t) - 2e^t u(t) + e^{-2t}u(t)$$

(b) The ROC and the locations of the poles are depicted in Fig. 9.10(b). The ROC, $\Re\{s\} < -2$, is to the left of the leftmost pole, so all the three poles correspond to anti-causal (left-sided) signals. Therefore,

$$-e^{-t}u(-t) \longleftrightarrow \frac{1}{s+1}$$

$$-2e^t u(-t) \longleftrightarrow \frac{2}{s-1}$$

$$-e^{-2t}u(-t) \longleftrightarrow \frac{1}{s+2}$$

and hence,

$$x(t) = -e^{-t}u(-t) + 2e^t u(-t) - e^{-2t}u(-t)$$

(c) The ROC and the locations of the poles are depicted in Fig. 9.10(c). The ROC, $-1 < \Re\{s\} < 1$, is a strip. The pole of the first term is at -1 . The ROC lies to the right of this pole, so this pole corresponds to a causal (right-sided) signal. Therefore,

$$e^{-t}u(t) \longleftrightarrow \frac{1}{s+1}$$

The second term has a pole at $s = 1$. Here the ROC is to the left of this pole, so this pole corresponds to an anticausal (left-sided) signal. Therefore,

$$-2e^t u(-t) \longleftrightarrow \frac{2}{s-1}$$

The third term has a pole at $s = -2$. Here the ROC is to the right of this pole, so this pole corresponds to a causal (right-sided) signal. Therefore,

$$e^{-2t} u(t) \longleftrightarrow \frac{1}{s+2}$$

and hence we obtain

$$x(t) = e^{-t} u(t) + 2e^t u(-t) + e^{-2t} u(t)$$

(d) The ROC and the locations of the poles are depicted in Fig. 9.10(d). The ROC, $-2 < \Re\{s\} < -1$, is a strip. The pole of the first term is at -1 . The ROC lies to the left of this pole, so this pole corresponds to an anticausal (left-sided) signal. Therefore,

$$-e^{-t} u(-t) \longleftrightarrow \frac{1}{s+1}$$

The second term has a pole at $s = 1$. Here the ROC is to the left of this pole, so this pole corresponds to an anti-causal (left-sided) signal. Therefore,

$$-2e^t u(-t) \longleftrightarrow \frac{2}{s-1}$$

The third term has a pole at $s = -2$. Here the ROC is to the right of this pole, so this pole corresponds to causal (right-sided) signal. Therefore,

$$e^{-2t} u(t) \longleftrightarrow \frac{1}{s+2}$$

and hence we obtain

$$x(t) = -e^{-t} u(-t) + 2e^t u(-t) + e^{-2t} u(t)$$

9.7 PROPERTIES OF THE LAPLACE TRANSFORM

The properties of the bilateral Laplace transform (or Laplace transform) are similar to those of the Fourier transform. The derivations of many of these properties are analogous to those of the corresponding properties for the Fourier transform. The properties described in this section are specifically for the bilateral Laplace transform. The unilateral and bilateral transforms have many properties in common, although there are important differences.

9.7.1 Linearity

If

$$x_1(t) \longleftrightarrow X_1(s) \quad \text{with ROC} = R_1$$

and

$$x_2(t) \longleftrightarrow X_2(s) \quad \text{with ROC} = R_2$$

then

$$ax_1(t) + bx_2(t) \longleftrightarrow aX_1(s) + bX_2(s) \quad \text{with ROC containing } R_1 \cap R_2 \quad (9.23)$$

As indicated, the ROC of $X(s)$ is at least the intersection of R_1 and R_2 , which could be empty, in which case $X(s)$ has no ROC, i.e., $x(t)$ has no Laplace transform.

Proof The Laplace transform of $ax_1(t) + bx_2(t)$ is given by

$$\begin{aligned} \mathcal{L}[ax_1(t) + bx_2(t)] &= \int_{-\infty}^{\infty} [ax_1(t) + bx_2(t)] e^{-st} dt \\ &= a \int_{-\infty}^{\infty} x_1(t) e^{-st} dt + b \int_{-\infty}^{\infty} x_2(t) e^{-st} dt \\ &= aX_1(s) + bX_2(s) \end{aligned}$$

9.7.2 Time Shifting

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$x(t - t_0) \longleftrightarrow X(s) e^{-st_0} \quad \text{with ROC} = R \quad (9.24)$$

Proof The Laplace transform of $x(t - t_0)$ is given by

$$\mathcal{L}[x(t - t_0)] = \int_{-\infty}^{\infty} x(t - t_0) e^{-st} dt$$

A change of variables is performed by letting $\tau = t - t_0$, which also yields $d\tau = dt$, $\tau \rightarrow -\infty$ as $t \rightarrow -\infty$, and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\mathcal{L}[x(t - t_0)] = \int_{-\infty}^{\infty} x(\tau) e^{-s(\tau+t_0)} d\tau$$

$$= e^{-st_0} \int_{-\infty}^{\infty} x(\tau) e^{-s\tau} d\tau$$

$$\mathcal{L}[x(t - t_0)] = X(s) e^{-st_0}$$

Example 9.15 Determine the Laplace transform of the following signals:

- (a) $x_1(t) = \delta(t + T) + \delta(t - T)$
- (b) $x_2(t) = u(t + T) - u(t - T)$
- (c) $x_3(t) = r(t + T) - r(t - T)$

The signals $x_1(t)$, $x_2(t)$, and $x_3(t)$ are shown in Figs 9.11(a), (b), and (c), respectively.

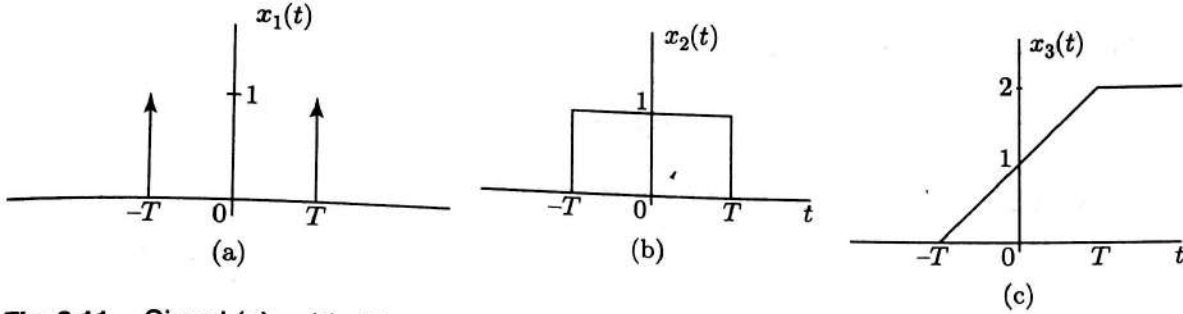


Fig. 9.11 Signal (a) $x_1(t)$, (b) $x_2(t)$, and (c) $x_3(t)$ for Example 9.15

Solution:

(a) Given that

$$x_1(t) = \delta(t + T) + \delta(t - T)$$

$$\mathcal{L}[x_1(t)] = X_1(s) = \mathcal{L}[\delta(t + T)] + \mathcal{L}[\delta(t - T)]$$

We know that [Eq. (9.12)]

$$\delta(t) \longleftrightarrow 1 \quad \text{ROC is the entire } s\text{-plane}$$

Now, using the time-shifting property, we obtain

$$\delta(t + T) \longleftrightarrow e^{sT} \quad \text{and} \quad \delta(t - T) \longleftrightarrow e^{-sT}$$

Therefore,

$$X_1(s) = e^{sT} + e^{-sT} \quad \text{ROC is the entire } s\text{-plane}$$

(b) Given that

$$x_2(t) = u(t + T) - u(t - T)$$

$$\mathcal{L}[x_2(t)] = X_2(s) = \mathcal{L}[u(t + T)] - \mathcal{L}[u(t - T)]$$

We know that [Eq. (9.13)]

$$u(t) \longleftrightarrow \frac{1}{s} \quad \Re\{s\} > 0$$

Now, using the time-shifting property, we obtain

$$u(t + T) \longleftrightarrow \frac{1}{s} e^{sT} \quad \text{and} \quad u(t - T) \longleftrightarrow \frac{1}{s} e^{-sT}$$

Therefore,

$$X_2(s) = \frac{1}{s}(e^{sT} - e^{-sT}) \quad \Re\{s\} > 0$$

(c) Given that

$$\begin{aligned} x_3(t) &= r(t+T) - r(t-T) \\ \mathcal{L}[x_3(t)] &= X_3(s) = \mathcal{L}[r(t+T)] - \mathcal{L}[r(t-T)] \end{aligned}$$

We know that [Eq. (9.14)]

$$r(t) \longleftrightarrow \frac{1}{s^2} \quad \Re\{s\} > 0$$

Now, using the time-shifting property, we obtain

$$r(t+T) \longleftrightarrow \frac{1}{s^2} e^{sT} \quad \text{and} \quad r(t-T) \longleftrightarrow \frac{1}{s^2} e^{-sT}$$

Therefore,

$$X_3(s) = \frac{1}{s^2}(e^{sT} - e^{-sT}) \quad \Re\{s\} > 0$$

Example 9.16 A zero-order hold system has the impulse response $h(t)$ as shown in Fig. 9.12. Find the Laplace transform of this impulse response.

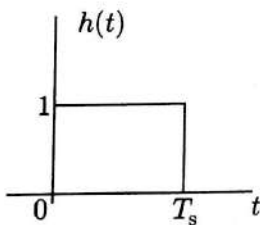


Fig. 9.12 Impulse response of a zero-order hold system

Solution:

From Fig. 9.12, the impulse response $h(t)$ can be written as

$$h(t) = u(t) - u(t - T_s)$$

where $u(t)$ is the unit step function. Taking the Laplace transform of the above equation, we obtain

$$H(s) = \mathcal{L}[h(t)] = \mathcal{L}[u(t)] - \mathcal{L}[u(t - T_s)]$$

$$H(s) = \frac{1}{s} - \frac{1}{s} e^{-sT}$$

$$H(s) = \frac{1}{s}(1 - e^{-sT})$$

9.7.3 Shifting in the s -Domain

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$x(t) e^{s_0 t} \longleftrightarrow X(s - s_0) \quad \text{with ROC} = R + \Re\{s_0\} \quad (9.25)$$

Proof The Laplace transform of $x(t) e^{s_0 t}$ is given by

$$\begin{aligned} \mathcal{L}[x(t) e^{s_0 t}] &= \int_{-\infty}^{\infty} [x(t) e^{s_0 t}] e^{-st} dt \\ &= \int_{-\infty}^{\infty} x(t) e^{-(s-s_0)t} dt \end{aligned}$$

$$\mathcal{L}[x(t) e^{s_0 t}] = X(s - s_0)$$

That is, the ROC associated with $X(s - s_0)$ is that of $X(s)$, shifted by $\Re\{s_0\}$. Note that if $X(s)$ has a pole or zero at $s = a$, then $X(s - s_0)$ has a pole or zero at $(s - s_0) = a$, i.e., $s = (a + s_0)$. ■

Example 9.17 Find the Laplace transform and ROC of the following damped sinusoidal signals:

(a) $g_1(t) = e^{-at} \cos(\omega_0 t) u(t)$

(b) $g_2(t) = e^{-at} \sin(\omega_0 t) u(t)$

Solution:

(a) Consider the given signal

$$g_1(t) = e^{-at} \cos(\omega_0 t) u(t) = e^{-at} x(t)$$

where $x(t) = \cos(\omega_0 t) u(t)$ and its Laplace transform is given by [Eq. (9.16)]

$$X(s) = \frac{s}{s^2 + \omega_0^2} \quad \Re\{s\} > 0$$

Now, taking the Laplace transform of the given signal $g_1(t)$ and using shifting in the s -domain property, we obtain

$$\mathcal{L}[g_1(t)] = \mathcal{L}[e^{-at} x(t)]$$

$$G_1(s) = X(s + a) = \frac{s + a}{(s + a)^2 + \omega_0^2} \quad \Re\{s\} > -a$$

Therefore,

$$e^{-at} \cos(\omega_0 t) u(t) \longleftrightarrow \frac{s+a}{(s+a)^2 + \omega_0^2} \quad \Re\{s\} > -a \quad (9.26)$$

The ROC of $g_1(t) = e^{-at} \cos(\omega_0 t) u(t)$ is shown in Fig. 9.13(a).

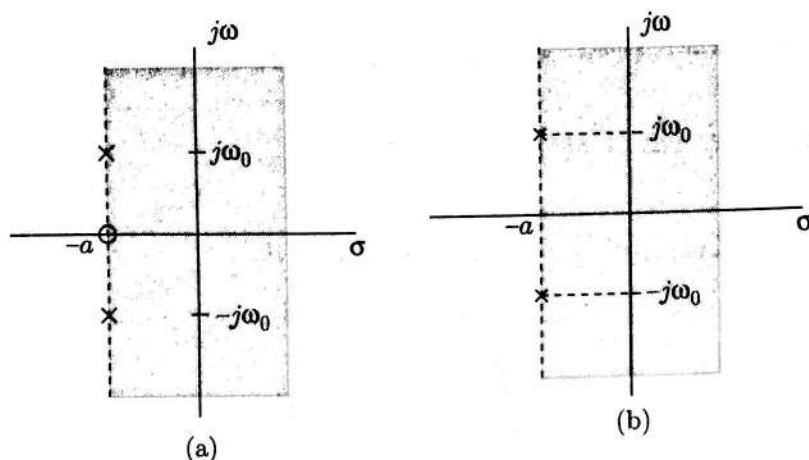


Fig. 9.13 Pole-zero plot and ROC for (a) $g_1(t) = e^{-at} \cos(\omega_0 t) u(t)$ and (b) $g_2(t) = e^{-at} \sin(\omega_0 t) u(t)$

(b) Consider the given signal

$$g_2(t) = e^{-at} \sin(\omega_0 t) u(t) = e^{-at} x(t)$$

where $x(t) = \sin(\omega_0 t) u(t)$ and its Laplace transform is given by [Eq. (9.15)]

$$X(s) = \frac{\omega_0}{s^2 + \omega_0^2} \quad \Re\{s\} > 0$$

Now, taking the Laplace transform of the given signal $g_2(t)$ and using shifting in the s -domain property, we obtain

$$\mathcal{L}[g_2(t)] = \mathcal{L}[e^{-at} x(t)]$$

$$G_2(s) = X(s+a) = \frac{\omega_0}{(s+a)^2 + \omega_0^2} \quad \Re\{s\} > -a$$

Therefore,

$$e^{-at} \sin(\omega_0 t) u(t) \longleftrightarrow \frac{\omega_0}{(s+a)^2 + \omega_0^2} \quad \Re\{s\} > -a \quad (9.27)$$

The ROC of $g_1(t) = e^{-at} \sin(\omega_0 t) u(t)$ is shown in Fig. 9.13(b).

Example 9.18 Find the Laplace transform and ROC of the signal $g(t) = t^n e^{-at} u(t)$.

Solution:

Consider the given signal

$$g(t) = t^n e^{-at} u(t) = e^{-at} x(t)$$

where $x(t) = t^n u(t)$ and its Laplace transform is given by [Eq. (9.20)]

$$X(s) = \frac{n!}{s^{n+1}} \quad \Re\{s\} > 0$$

Now, taking the Laplace transform of the given signal $g(t)$ and using shifting in the s -domain property, we obtain

$$\mathcal{L}[g(t)] = \mathcal{L}[e^{-at}x(t)]$$

$$G(s) = X(s+a) = \frac{n!}{(s+a)^{n+1}} \quad \Re\{s\} > -a$$

Therefore,

$$t^n e^{-at} u(t) \longleftrightarrow \frac{n!}{(s+a)^{n+1}} \quad \Re\{s\} > -a \quad (9.28)$$

or, equivalently,

$$\frac{t^n}{n!} e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^{n+1}} \quad \Re\{s\} > -a \quad (9.29)$$

and

$$\frac{t^{n-1}}{(n-1)!} e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^n} \quad \Re\{s\} > -a \quad (9.30)$$

9.7.4 Time Scaling

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$x(at) \longleftrightarrow \frac{1}{|a|} X\left(\frac{s}{a}\right) \quad \text{with ROC} = aR \quad (9.31)$$

Aside from the amplitude factor of $1/a$, scaling in time domain by a factor of a corresponds to scaling in the s -domain by a factor of $1/a$. Also, for any value of s in the ROC of $x(t)$, the value s/a will be in the ROC of $x(at)$; that is, the ROC associated with $x(at)$ is a compressed ($a > 1$) or expanded ($a < 1$) version of the ROC of $x(t)$.

Proof The Laplace transform of $x(at)$ is

$$\mathcal{L}[x(at)] = \int_{-\infty}^{\infty} x(at) e^{-st} dt$$

Case I: For a positive real constant ' a '

$$\mathcal{L}[x(at)] = \int_{-\infty}^{\infty} x(at) e^{-st} dt$$

A change of variables is performed by letting $\tau = at$, which also yields $d\tau = a dt$, $\tau \rightarrow -\infty$ as $t \rightarrow -\infty$, and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned}\mathcal{L}[x(at)] &= \frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-s/a\tau} d\tau \\ &= \frac{1}{a} X\left(\frac{s}{a}\right)\end{aligned}$$

Case II: For a negative real constant $-a$

$$\mathcal{F}[x(-at)] = \int_{-\infty}^{\infty} x(-at) e^{-st} dt$$

A change of variables is performed by letting $\tau = -at$, which also yields $d\tau = -a dt$, $\tau \rightarrow \infty$ as $t \rightarrow -\infty$, and $\tau \rightarrow -\infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned}\mathcal{L}[x(-at)] &= -\frac{1}{a} \int_{\infty}^{-\infty} x(\tau) e^{(s/a)\tau} d\tau \\ &= \frac{1}{a} \int_{-\infty}^{\infty} x(\tau) e^{-(-s/a)\tau} d\tau \\ &= \frac{1}{a} X\left(-\frac{s}{a}\right)\end{aligned}$$

Combining the two cases, we obtain

$$\mathcal{L}[x(at)] = \frac{1}{|a|} X\left(\frac{s}{a}\right)$$

Example 9.19 Determine the Laplace transform and ROC of the following signals:

(a) $g_1(t) = \delta(2t)$

(b) $g_2(t) = r(2t)$

Solution:

(a) We know that [Eq. (1.11)]

$$g_1(t) = \delta(2t) = \frac{1}{2}\delta(t)$$

Taking the Laplace transform, we obtain

$$G_1(s) = \mathcal{L}[\delta(2t)] = \mathcal{L}\left[\frac{1}{2}\delta(t)\right] = \frac{1}{2}\mathcal{L}[\delta(t)] = \frac{1}{2} \quad \text{ROC is the entire } s\text{-plane}$$

(b) **Method 1** Let $x(t) = r(t)$. Therefore,

$$g_2(t) = x(2t)$$

We know that [Eq. (9.14)]

$$X(s) = \mathcal{L}[r(t)] = \frac{1}{s^2} \quad \Re\{s\} > 0$$

Now, using the time-scaling property, we obtain

$$\begin{aligned} \mathcal{L}[g_2(t)] &= \mathcal{L}[x(2t)] = \frac{1}{2} X\left(\frac{s}{2}\right) \\ &= \frac{1}{2} \frac{1}{(s/2)^2} \end{aligned}$$

$$G_2(s) = \frac{2}{s^2} \quad \Re\{s\} > 0$$

Method 2 Given that

$$g_2(t) = r(2t) = 2tu(2t) = 2tu(t)$$

$$\mathcal{L}[g_2(t)] = 2\mathcal{L}[tu(t)]$$

$$G_2(s) = \frac{2}{s^2} \quad \Re\{s\} > 0$$

Example 9.20 Determine the Laplace transform of the following signals:

(a) $g_1(t) = \delta(2t - 3)$

(b) $g_2(t) = u(2t - 1)$

(c) $g_3(t) = r\left(\frac{1}{3}t - 2\right)$

Solution:

(a) We know that [Eq. (9.12)]

$$\delta(t) \longleftrightarrow 1$$

Using the time-shifting property, we obtain

$$\delta(t - 3) \longleftrightarrow e^{-3s}$$

Now, using the time-scaling property, we obtain

$$\delta(2t - 3) \longleftrightarrow \frac{1}{2} e^{-3/2s}$$

(b) We know that [Eq. (9.13)]

$$u(t) \longleftrightarrow \frac{1}{s}$$

Using the time-shifting property, we obtain

$$u(t-1) \longleftrightarrow \frac{1}{s} e^{-s}$$

Now, using the time-scaling property, we obtain

$$u(2t-1) \longleftrightarrow \frac{1}{2} \frac{1}{s/2} e^{-s/2}$$

$$u(2t-1) \longleftrightarrow \frac{1}{s} e^{-s/2}$$

(c) We know that [Eq. (9.14)]

$$r(t) \longleftrightarrow \frac{1}{s^2}$$

Using the time-shifting property, we obtain

$$r(t-2) \longleftrightarrow \frac{1}{s^2} e^{-2s}$$

Now, using the time-scaling property, we obtain

$$r\left(\frac{1}{3}t-2\right) \longleftrightarrow 3 \frac{1}{(3s)^2} e^{-6s}$$

$$r\left(\frac{1}{3}t-2\right) \longleftrightarrow \frac{1}{3s^2} e^{-6s}$$

9.7.5 Scaling in the s -Domain

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$\frac{1}{|a|} x\left(\frac{t}{a}\right) \longleftrightarrow X(as) \quad \text{with ROC} = \frac{R}{a} \quad (9.32)$$

Aside from the amplitude factor of $1/a$, scaling in s -domain by a factor of a corresponds to scaling in the time-domain by a factor of $1/a$. Also, for any value of s in the ROC of $x(t)$, the value as will be in the ROC of $x(t/a)$; that is, the ROC associated with $x(t/a)$ is a compressed ($a < 1$) or expanded ($a > 1$) version of the ROC of $x(t)$.

Proof The Laplace transform of $1/|a| x(t/a)$ is

$$\mathcal{L}\left[\frac{1}{|a|} x\left(\frac{t}{a}\right)\right] = \int_{-\infty}^{\infty} \frac{1}{|a|} x\left(\frac{t}{a}\right) e^{-st} dt$$

Case I For a positive real constant 'a',

$$\mathcal{L} \left[\frac{1}{a} x \left(\frac{t}{a} \right) \right] = \int_{-\infty}^{\infty} \frac{1}{a} x \left(\frac{t}{a} \right) e^{-st} dt$$

A change of variables is performed by letting $\tau = t/a$, which also yields $t = a\tau$, $dt = a d\tau$, $\tau \rightarrow -\infty$ as $t \rightarrow -\infty$, and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\mathcal{L} \left[\frac{1}{a} x \left(\frac{t}{a} \right) \right] = \int_{-\infty}^{\infty} x(\tau) e^{-as\tau} d\tau = X(as)$$

Case II For a negative real constant '-a',

$$\mathcal{F} \left[\frac{1}{-a} x \left(-\frac{t}{a} \right) \right] = \int_{-\infty}^{\infty} \frac{1}{-a} x \left(-\frac{t}{a} \right) e^{-st} dt$$

A change of variables is performed by letting $\tau = -t/a$, which also yields $t = -a\tau$, $dt = -a d\tau$, $\tau \rightarrow \infty$ as $t \rightarrow -\infty$, and $\tau \rightarrow -\infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned} \mathcal{L} \left[\frac{1}{-a} x \left(-\frac{t}{a} \right) \right] &= \int_{\infty}^{-\infty} x(\tau) e^{(as)\tau} d\tau \\ &= - \int_{-\infty}^{\infty} x(\tau) e^{(-as)\tau} d\tau \end{aligned}$$

$$\mathcal{L} \left[\frac{1}{-a} x \left(-\frac{t}{a} \right) \right] = -X(-as)$$

$$\mathcal{L} \left[\frac{1}{a} x \left(-\frac{t}{a} \right) \right] = X(-as)$$

Combining the two cases, we obtain

$$\mathcal{L} \left[\frac{1}{|a|} x \left(\frac{t}{a} \right) \right] = X(as)$$

9.7.6 Time Reversal

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$x(-t) \longleftrightarrow X(-s) \quad \text{with ROC} = -R \quad (9.33)$$

Proof Substituting $a = -1$ in Eq. (9.31), we have

$$\mathcal{L}[x(-t)] = \frac{1}{|-1|} X\left(\frac{s}{-1}\right)$$

$$\mathcal{L}[x(-t)] = X(-s)$$

Thus, time reversal of $x(t)$ results in a reversal of the ROC. An interesting consequence of the time-reversal property is that if $x(t)$ is even, then its Laplace transform is also even, i.e.,

$$\text{if } x(-t) = x(t) \text{ then } X(-s) = X(s) \quad (9.34)$$

Similarly, if $x(t)$ is odd, then so is its Laplace transform, that is

$$\text{if } x(-t) = -x(t), \text{ then } X(-s) = -X(s) \quad (9.35)$$

Example 9.21 Determine the Laplace transform of the following signals:

$$(a) \ g_2(t) = u(-2t - 1)$$

$$(b) \ g_3(t) = r(-3t - 2)$$

Solution:

(a) We know that [Eq. (9.13)]

$$u(t) \longleftrightarrow \frac{1}{s}$$

Using the time-shifting property, we obtain

$$u(t - 1) \longleftrightarrow \frac{1}{s} e^{-s}$$

Using the time-scaling property, we obtain

$$u(2t - 1) \longleftrightarrow \frac{1}{2} \frac{1}{s/2} e^{-s/2}$$

Now, using the time-reversal property, we obtain

$$u(-2t - 1) \longleftrightarrow \frac{1}{-s} e^{-s/2}$$

$$u(-2t - 1) \longleftrightarrow -\frac{1}{s} e^{s/2}$$

(b) We know that [Eq. (9.14)]

$$r(t) \longleftrightarrow \frac{1}{s^2}$$

Using the time-shifting property, we obtain

$$r(t-2) \longleftrightarrow \frac{1}{s^2} e^{-2s}$$

Using the time-scaling property, we obtain

$$r(3t-2) \longleftrightarrow \frac{1}{3} \frac{1}{(s/3)^2} e^{-2s/3}$$

$$r(3t-2) \longleftrightarrow \frac{3}{s^2} e^{-2s/3}$$

Now, using the time-reversal property, we obtain

$$r(-3t-2) \longleftrightarrow \frac{3}{(-s)^2} e^{-(-2s/3)}$$

$$r(-3t-2) \longleftrightarrow \frac{3}{s^2} e^{2s/3}$$

9.7.7 Differentiation in the Time Domain

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$\frac{dx(t)}{dt} \longleftrightarrow sX(s) \quad \text{with ROC containing } R \quad (9.36)$$

Proof From Eq. (9.4), we have

$$x(t) = \mathcal{L}^{-1}[X(s)] = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{st} ds$$

Differentiating both sides, we get

$$\frac{dx(t)}{dt} = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} [sX(s)] e^{st} ds$$

$$\frac{dx(t)}{dt} = \mathcal{L}^{-1}[sX(s)]$$

Therefore,

$$\frac{dx(t)}{dt} \longleftrightarrow sX(s)$$

Similarly, the differentiation property can be extended to yield

$$\frac{d^n x(t)}{dt^n} \longleftrightarrow s^n X(s) \quad (9.37)$$

Example 9.22 Using the bilateral time-shift and differentiation properties find the Laplace transform of

$$x(t) = \frac{d^2}{dt^2} \left(e^{-3(t-2)} u(t-2) \right)$$

Solution:

We know that

$$e^{-3t} u(t) \longleftrightarrow \frac{1}{s+3} \quad \Re\{s\} > -3$$

Using the time-shifting property [Eq. (9.24)], we obtain

$$e^{-3(t-2)} u(t-2) \longleftrightarrow \frac{1}{s+3} e^{-2s} \quad \Re\{s\} > -3$$

Now, using differentiation in the time-domain property [Eq. (9.37)], we obtain

$$\frac{d^2}{dt^2} e^{-3(t-2)} u(t-2) \longleftrightarrow \frac{s^2}{s+3} e^{-2s} \quad \Re\{s\} > -3$$

9.7.8 Differentiation in the s -Domain

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$tx(t) \longleftrightarrow -\frac{dX(s)}{ds} \quad \text{with ROC} = R \quad (9.38)$$

Proof By definition

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Differentiating both sides w.r.t. s , we get

$$\frac{dX(s)}{ds} = \int_{-\infty}^{\infty} [-tx(t)] e^{-st} dt$$

$$\frac{dX(s)}{ds} = \mathcal{L}[-tx(t)]$$

Therefore,

$$tx(t) \longleftrightarrow -\frac{dX(s)}{ds}$$

Similarly, differentiation in the s -domain property can be extended to yield

$$t^n x(t) \longleftrightarrow (-1)^n \frac{d^n X(s)}{ds^n} \quad (9.39)$$

■

Example 9.23 Determine the Laplace transform and ROC for the following signals:

- (a) $g_1(t) = t e^{-at} u(t)$
- (b) $g_2(t) = t \sin(\omega_0 t) u(t)$
- (c) $g_3(t) = t \cos(\omega_0 t) u(t)$

Solution

(a) Consider the given signal

$$g_1(t) = t e^{-at} u(t) = tx(t)$$

where $x(t) = e^{-at} u(t)$ and its Laplace transform is given by [Eq. (9.10)]

$$X(s) = \frac{1}{s+a} \quad \Re\{s\} > -a$$

Now, taking the Laplace transform of the given signal $g_1(t)$ and using differentiation in the s -domain property, we obtain

$$\begin{aligned} \mathcal{L}[g_1(t)] &= \mathcal{L}[tx(t)] \\ G_1(s) &= -\frac{dX(s)}{ds} = -\frac{d(1/s+a)}{ds} = \frac{1}{(s+a)^2} \end{aligned}$$

Therefore,

$$t e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^2} \quad \Re\{s\} > -a \quad (9.40)$$

(b) Consider the given signal

$$g_2(t) = t \sin(\omega_0 t) u(t) = tx(t)$$

where $x(t) = \sin(\omega_0 t) u(t)$ and its Laplace transform is given by [Eq. (9.15)]

$$X(s) = \frac{\omega_0}{s^2 + \omega_0^2} \quad \Re\{s\} > 0$$

Now, taking the Laplace transform of the given signal $g_2(t)$ and using differentiation in the s -domain property, we obtain

$$\mathcal{L}[g_2(t)] = \mathcal{L}[tx(t)]$$

$$G_2(s) = -\frac{dX(s)}{ds} = -\frac{d(\omega_0/s^2 + \omega_0^2)}{ds} = \frac{2\omega_0 s}{(s^2 + \omega_0^2)^2}$$

Therefore,

$$t \sin(\omega_0 t) u(t) \longleftrightarrow \frac{2\omega_0 s}{(s^2 + \omega_0^2)^2} \quad \Re\{s\} > 0 \quad (9.41)$$

(c) Consider the given signal

$$g_3(t) = t \cos(\omega_0 t) u(t) = tx(t)$$

where $x(t) = \cos(\omega_0 t) u(t)$ and its Laplace transform is given by [Eq. (9.16)]

$$X(s) = \frac{s}{s^2 + \omega_0^2} \quad \Re\{s\} > 0$$

Now, taking the Laplace transform of the given signal $g_3(t)$ and using differentiation in the s -domain property, we obtain

$$\mathcal{L}[g_3(t)] = \mathcal{L}[tx(t)]$$

$$G_3(s) = -\frac{dX(s)}{ds} = -\frac{d(s/s^2 + \omega_0^2)}{ds} = \frac{s^2 - \omega_0^2}{(s^2 + \omega_0^2)^2}$$

Therefore,

$$t \cos(\omega_0 t) u(t) \longleftrightarrow \frac{s^2 - \omega_0^2}{(s^2 + \omega_0^2)^2} \quad \Re\{s\} > 0 \quad (9.42)$$

9.7.9 Convolution Property

If

$$x_1(t) \longleftrightarrow X_1(s) \quad \text{with ROC} = R_1$$

and

$$x_2(t) \longleftrightarrow X_2(s) \quad \text{with ROC} = R_2$$

then

$$x_1(t) * x_2(t) \longleftrightarrow X_1(s)X_2(s) \quad \text{with ROC containing } R_1 \cap R_2 \quad (9.43)$$

The Laplace transform maps the convolution of two signals into the product of their Laplace transforms. The ROC of $X_1(s)X_2(s)$ includes the intersection of the ROCs of $X_1(s)$ and $X_2(s)$ and may be larger if pole-zero cancellation occurs in the product.

Proof The Laplace transform of $x_1(t) * x_2(t)$ is given by

$$\begin{aligned}\mathcal{L}[x_1(t) * x_2(t)] &= \int_{-\infty}^{\infty} [x_1(t) * x_2(t)] e^{-st} dt \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x_1(\tau) x_2(t - \tau) d\tau \right) e^{-st} dt\end{aligned}$$

Interchanging the order of integration and noting that $x_1(\tau)$ does not depend on t , we obtain

$$\mathcal{L}[x_1(t) * x_2(t)] = \int_{-\infty}^{\infty} x_1(\tau) \left(\int_{-\infty}^{\infty} x_2(t - \tau) e^{-st} dt \right) d\tau$$

Using the time-shifting property [Eq. (9.24)], the bracketed term is $X_2(s) e^{-s\tau}$. Substituting this into the above equation yields

$$\begin{aligned}\mathcal{L}[x_1(t) * x_2(t)] &= \int_{-\infty}^{\infty} x_1(\tau) (X_2(s) e^{-s\tau}) d\tau \\ &= X_2(s) \int_{-\infty}^{\infty} x_1(\tau) e^{-s\tau} d\tau\end{aligned}$$

$$\mathcal{L}[x_1(t) * x_2(t)] = X_2(s) X_1(s) = X_1(s) X_2(s)$$

Therefore,

$$x_1(t) * x_2(t) \longleftrightarrow X_1(s) X_2(s)$$

The convolution property states that convolution in the time domain corresponds to multiplication in the s -domain. ■

Example 9.24 Consider a signal $y(t)$ which is related to two signals $x_1(t)$ and $x_2(t)$ by

$$y(t) = x_1(t - 2) * x_2(-t + 3)$$

where

$$x_1(t) = e^{-2t} u(t) \quad \text{and} \quad x_2(t) = e^{-3t} u(t)$$

Use properties of the Laplace transform to determine the Laplace transform $Y(s)$ of $y(t)$.

Solution:

We know that [Eq. (9.10)]

$$x_1(t) = e^{-2t} u(t) \longleftrightarrow \frac{1}{s + 2} \quad \Re\{s\} > -2$$

Using the time-shifting property [Eq. (9.24)], we obtain

$$x_1(t-2) = e^{-2(t-2)}u(t-2) \longleftrightarrow \frac{e^{-2s}}{s+2} \quad \Re\{s\} > -2$$

Similarly, we obtain

$$x_2(t) = e^{-3t}u(t) \longleftrightarrow \frac{1}{s+3} \quad \Re\{s\} > -3$$

Using the time-shifting property [Eq. (9.24)], we obtain

$$x_1(t+3) = e^{-3(t+3)}u(t+3) \longleftrightarrow \frac{e^{3s}}{s+3} \quad \Re\{s\} > -3$$

Now, using the time-reversal property [Eq. (9.33)], we obtain

$$x_1(-t+3) = e^{-3(-t+3)}u(-t+3) \longleftrightarrow \frac{e^{-3s}}{-s+3} \quad \Re\{s\} < 3$$

Now, using the convolution property [Eq. (9.43)], we obtain

$$\begin{aligned} \mathcal{L}[y(t)] &= \mathcal{L}[x_1(t-2)]\mathcal{L}[x_2(-t+3)] \\ Y(s) &= \left(\frac{e^{-2s}}{s+2}\right)\left(\frac{e^{-3s}}{-s+3}\right) \quad -2 < \Re\{s\} < 3 \\ Y(s) &= \frac{e^{-5s}}{(s+2)(3-s)} \quad -2 < \Re\{s\} < 3 \end{aligned}$$

9.7.10 Multiplication Property

If

$$x_1(t) \longleftrightarrow X_1(s) \quad \text{with ROC} = R_1$$

and

$$x_2(t) \longleftrightarrow X_2(s) \quad \text{with ROC} = R_2$$

then

$$\begin{aligned} x_1(t)x_2(t) &\longleftrightarrow \frac{1}{2\pi j}[X_1(s) * X_2(s)] = \frac{1}{2\pi j} \int_{\sigma_1-j\infty}^{\sigma_1+j\infty} X_1(s_1)X_2(s-s_1) ds_1 \\ &\quad \text{with ROC containing } R_1 \cap R_2 \end{aligned} \quad (9.44)$$

The Laplace transform maps the multiplication of two signals into the convolution of their Laplace transforms.

Proof The Laplace transform of $x_1(t)x_2(t)$ is given by

$$\begin{aligned}\mathcal{L}[x_1(t)x_2(t)] &= \int_{-\infty}^{\infty} [x_1(t)x_2(t)] e^{-st} dt \\ &= \int_{-\infty}^{\infty} \left(\frac{1}{2\pi j} \int_{\sigma_1-j\infty}^{\sigma_1+j\infty} X_1(s_1) e^{s_1 t} ds_1 \right) x_2(t) e^{-st} dt\end{aligned}$$

Interchanging the order of integration and noting that $X_1(s_1)$ does not depend on t , we have

$$\mathcal{L}[x_1(t)x_2(t)] = \frac{1}{2\pi j} \int_{\sigma_1-j\infty}^{\sigma_1+j\infty} X_1(s_1) \left(\int_{-\infty}^{\infty} [x_2(t) e^{s_1 t}] e^{-st} dt \right) ds_1$$

Using shifting in the s -domain property [Eq. (9.23)], the bracketed term is $X_2(s - s_1)$. Substituting this into the above equation yields

$$\mathcal{L}[x_1(t)x_2(t)] = \frac{1}{2\pi j} \int_{\sigma_1-j\infty}^{\sigma_1+j\infty} X_1(s_1) X_2(s - s_1) ds_1 = \frac{1}{2\pi j} [X_1(s) * X_2(s)]$$

Therefore,

$$x_1(t)x_2(t) \longleftrightarrow \frac{1}{2\pi j} [X_1(s) * X_2(s)]$$

9.7.11 Integration in the Time Domain

If

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{s} X(s) \quad \text{with ROC containing } R \cap \{\Re\{s\} > 0\} \quad (9.45)$$

Proof If a signal $x(t)$ is convolved with a unit step function $u(t)$, we get

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t - \tau) d\tau$$

Since

$$u(t - \tau) = \begin{cases} 1, & \tau < t \\ 0, & \tau > t \end{cases}$$

we have

$$x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$$

The convolution of a signal with a unit step function is the same as the cumulative integral of the signal. Now we can prove the integration property of the Laplace transform,

$$\int_{-\infty}^t x(\tau) d\tau = x(t) * u(t)$$

$$\mathcal{L} \left[\int_{-\infty}^t x(\tau) d\tau \right] = \mathcal{L}[x(t) * u(t)]$$

Using the convolution property [Eq. (9.43)] and Eq. (9.13), we obtain

$$\mathcal{L} \left[\int_{-\infty}^t x(\tau) d\tau \right] = X(s)U(s) = \frac{1}{s}X(s)$$

Therefore,

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{1}{s}X(s)$$

Example 9.25 Determine the bilateral Laplace transform and the corresponding ROC for each of the following signals:

$$(a) \quad x(t) = e^{-t} \frac{d}{dt} (e^{-(t+1)} u(t+1))$$

$$(b) \quad x(t) = \int_{-\infty}^t e^{2\tau} \sin(\tau) u(-\tau) d\tau$$

Solution:

(a) We know that

$$e^{-t} u(t) \longleftrightarrow \frac{1}{s+1} \quad \Re\{s\} > -1$$

Using the time-shifting property [Eq. (9.24)], we obtain

$$e^{-(t+1)} u(t+1) \longleftrightarrow \frac{1}{s+1} e^s \quad \Re\{s\} > -1$$

Now, using differentiation in the time-domain property [Eq. (9.36)], we obtain

$$\frac{d}{dt} \left(e^{-(t+1)} u(t+1) \right) \longleftrightarrow \frac{s}{s+1} e^s \quad \Re\{s\} > -1$$

Now, using shifting in the s -domain property [Eq. (9.25)], we obtain

$$e^{-t} \frac{d}{dt} \left(e^{-(t+1)} u(t+1) \right) \longleftrightarrow \frac{s+1}{s+2} e^{s+1} \quad \Re\{s\} > -2$$

(b) We know that

$$\sin(t)u(-t) \longleftrightarrow \frac{-1}{s^2+1} \quad \Re\{s\} < 0$$

Using shifting in the s -domain property [Eq. (9.25)], we obtain

$$e^{2t} \sin(t)u(-t) \longleftrightarrow \frac{-1}{(s-2)^2+1} \quad \Re\{s\} < 2$$

Now, using integration in the time-domain property [Eq. (9.45)], we obtain

$$x(t) * u(t) = \int_{-\infty}^t e^{2\tau} \sin(\tau)u(-\tau) d\tau \longleftrightarrow \frac{-1}{s[(s-2)^2+1]} \quad 0 < \Re\{s\} < 2$$

9.7.12 Conjugation Property

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If

$$x(t) \longleftrightarrow X(s)$$

then

$$x^*(t) \longleftrightarrow X^*(s^*) \quad (9.46)$$

Proof The Laplace transform of $x^*(t)$ is

$$\mathcal{L}[x^*(t)] = \int_{-\infty}^{\infty} x^*(t) e^{-st} dt = \left[\int_{-\infty}^{\infty} x(t) e^{-s^*t} dt \right]^* = [X(s^*)]^* = X^*(s^*)$$

Therefore, if $x(t)$ is real, that is, $x(t) = x^*(t)$, then

$$X(s) = X^*(s^*) \quad (9.47)$$

$$X^*(s) = X(s^*) \quad (9.48)$$

Consequently, if $x(t)$ is real and if $X(s)$ has a pole or zero at $s = s_0$, then $X(s)$ also has a pole or zero at the complex conjugate point $s = s_0^*$. ■

9.8 LAPLACE TRANSFORM OF CAUSAL PERIODIC SIGNALS

The Laplace transform of a causal periodic signal can be determined from the knowledge of the Laplace transform of its first cycle (period). Consider a causal periodic signal $x(t)$ with period T_0 as shown in Fig. 9.14.

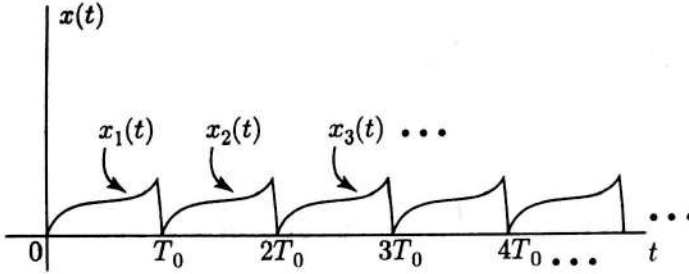


Fig. 9.14 Causal periodic signal

Let $x_1(t)$, $x_2(t)$, $x_3(t)$, ..., be the signals representing the 1st, 2nd, 3rd, ..., cycles of a causal periodic signal $x(t)$. Therefore, the causal periodic signal $x(t)$ can be written as

$$\begin{aligned} x(t) &= x_1(t) + x_2(t) + x_3(t) + x_4(t) + \dots \\ &= x_1(t) + x_1(t - T_0) + x_1(t - 2T_0) + x_1(t - 3T_0) + \dots \end{aligned}$$

Assume $x_1(t) \longleftrightarrow X_1(s)$. Using the time-shifting property [Eq. (9.24)], the Laplace transform of the above equation becomes

$$\begin{aligned} X(s) &= X_1(s) + X_1(s) e^{-sT_0} + X_1(s) e^{-s2T_0} + X_1(s) e^{-s3T_0} + \dots \\ &= X_1(s) [1 + e^{-sT_0} + e^{-s2T_0} + e^{-s3T_0} + \dots] \\ &= X_1(s) \sum_{n=0}^{\infty} e^{-snT_0} \\ &= X_1(s) \sum_{n=0}^{\infty} (e^{-sT_0})^n \\ X(s) &= X_1(s) \frac{1}{1 - e^{-sT_0}} \end{aligned}$$

Therefore, the Laplace transform of a periodic signal $x(t)$ is given by

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}} \quad (9.49)$$

Example 9.26 Find the laplace transform of the square wave shown in Fig. 9.15(a).

Solution:

We know that the Laplace transform of a causal periodic signal is given by [Eq. (9.49)]

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

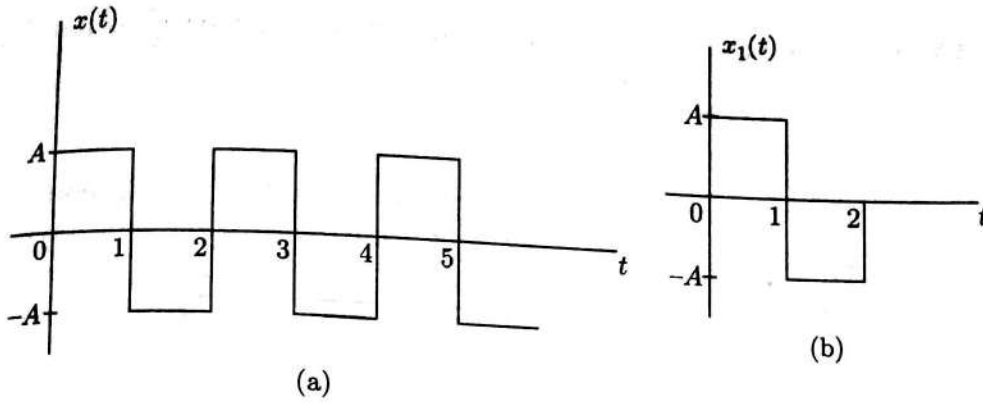


Fig. 9.15 (a) Causal periodic square wave $x(t)$ and (b) first cycle of $x(t)$

where $X_1(s)$ is the Laplace transform of the first cycle $x_1(t)$ of the causal periodic signal $x(t)$. The given square wave is a causal periodic signal with period $T_0 = 2$. The first cycle $x_1(t)$ of $x(t)$ is as shown in Fig. 9.15(b). Using the step function $u(t)$, $x_1(t)$ can be written as

$$x_1(t) = Au(t) - 2Au(t-1) + Au(t-2)$$

Taking the Laplace transform of the above equation, we obtain

$$\mathcal{L}[x_1(t)] = A\mathcal{L}[u(t)] - 2A\mathcal{L}[u(t-1)] + A\mathcal{L}[u(t-2)]$$

$$X_1(s) = \frac{A}{s} - \frac{2A}{s} e^{-s} + \frac{A}{s} e^{-2s} = \frac{A}{s} [1 - 2e^{-s} + e^{-2s}] = \frac{A}{s} (1 - e^{-s})^2$$

Substituting $X_1(s)$ and $T_0 = 2$ in the expression of $X(s)$, we get

$$\begin{aligned} X(s) &= \frac{X_1(s)}{1 - e^{-sT_0}} = \frac{A(1 - e^{-s})^2}{s(1 - e^{-2s})} \\ &= \frac{A}{s} \frac{(1 - e^{-s})^2}{(1 + e^{-s})(1 - e^{-s})} \\ &= \frac{A}{s} \frac{1 - e^{-s}}{1 + e^{-s}} \\ &= \frac{A e^{-s/2} (e^{s/2} - e^{-s/2})}{s e^{-s/2} (e^{s/2} + e^{-s/2})} \end{aligned}$$

$$X(s) = \frac{A}{s} \left(\frac{e^{s/2} - e^{-s/2}}{e^{s/2} + e^{-s/2}} \right)$$

$$X(s) = \frac{A}{s} \tanh\left(\frac{s}{2}\right)$$

Example 9.27 Find the Laplace transform of the saw-tooth wave shown in Fig. 9.16(a).

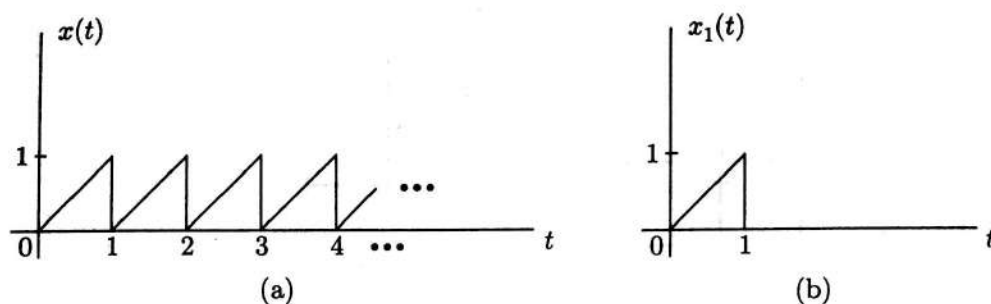


Fig. 9.16 (a) Causal periodic saw-tooth wave $x(t)$ and (b) first cycle of $x(t)$

Solution

We know that the Laplace transform of a causal periodic signal is given by [Eq. (9.49)]

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

where $X_1(s)$ is the Laplace transform of the first cycle $x_1(t)$ of the causal periodic signal $x(t)$. The given saw-tooth wave is a causal periodic signal with period $T_0 = 1$. The first cycle $x_1(t)$ of $x(t)$ is as shown in Fig. 9.16(b). Using the step function $u(t)$ and ramp function $r(t)$, $x_1(t)$ can be written as

$$x_1(t) = r(t) - r(t-1) - u(t-1)$$

Taking the Laplace transform of the above equation, we obtain

$$\mathcal{L}[x_1(t)] = \mathcal{L}[r(t)] - \mathcal{L}[r(t-1)] - \mathcal{L}[u(t-1)]$$

$$X_1(s) = \frac{1}{s^2} - \frac{1}{s^2} e^{-s} - \frac{1}{s} e^{-s}$$

$$X_1(s) = \frac{1}{s^2} [1 - e^{-s} - s e^{-s}]$$

Substituting $X_1(s)$ and $T_0 = 1$ in the expression of $X(s)$, we get

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}} = \frac{1}{s^2} \frac{1 - e^{-s} - s e^{-s}}{1 - e^{-s}}$$

Example 9.28 Find the Laplace transform of the triangular wave shown in Fig. 9.17(a).

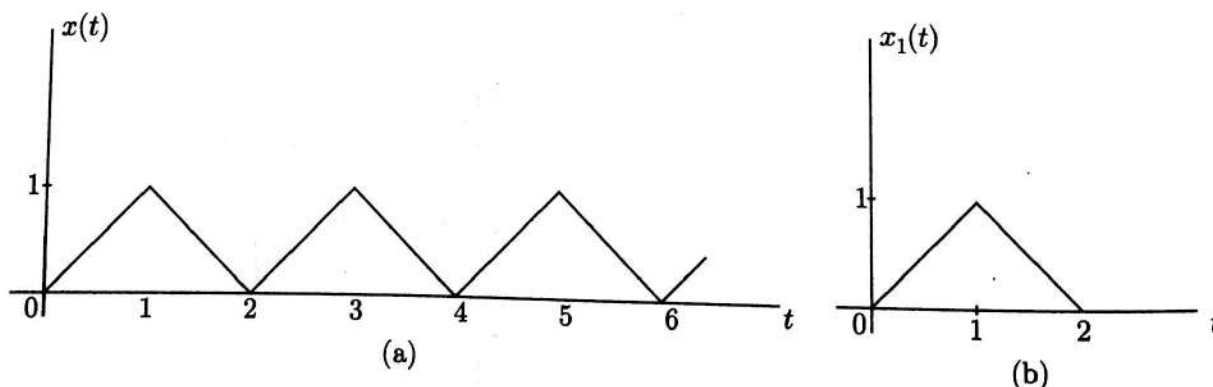


Fig. 9.17 (a) Causal periodic triangular wave $x(t)$ and (b) first cycle of $x(t)$

Solution

We know that the Laplace transform of a causal periodic signal is given by [Eq. (9.49)]

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

where $X_1(s)$ is the Laplace transform of the first cycle $x_1(t)$ of the causal periodic signal $x(t)$. The given saw-tooth wave is a causal periodic signal with period $T_0 = 2$. The first cycle $x_1(t)$ of $x(t)$ is as shown in Fig. 9.17(b). Using the ramp function $r(t)$, $x_1(t)$ can be written as

$$x_1(t) = r(t) - 2r(t-1) + r(t-2)$$

Taking the Laplace transform of the above equation, we obtain

$$\begin{aligned} \mathcal{L}[x_1(t)] &= \mathcal{L}[r(t)] - 2\mathcal{L}[r(t-1)] + \mathcal{L}[r(t-2)] \\ X_1(s) &= \frac{1}{s^2} - \frac{2}{s^2} e^{-s} + \frac{1}{s^2} e^{-2s} = \frac{1}{s^2} [1 - 2e^{-s} + e^{-2s}] = \frac{1}{s^2} (1 - e^{-s})^2 \end{aligned}$$

Substituting $X_1(s)$ and $T_0 = 2$ in the expression of $X(s)$, we get

$$\begin{aligned} X(s) &= \frac{X_1(s)}{1 - e^{-sT_0}} = \frac{1}{s^2} \frac{(1 - e^{-s})^2}{1 - e^{-2s}} \\ &= \frac{1}{s^2} \frac{(1 - e^{-s})^2}{(1 + e^{-s})(1 - e^{-s})} \\ &= \frac{1}{s^2} \frac{1 - e^{-s}}{1 + e^{-s}} \\ &= \frac{1}{s^2} \frac{e^{-s/2}(e^{s/2} - e^{-s/2})}{e^{-s/2}(e^{s/2} + e^{-s/2})} \\ X(s) &= \frac{1}{s^2} \left(\frac{e^{s/2} - e^{-s/2}}{e^{s/2} + e^{-s/2}} \right) \\ X(s) &= \frac{1}{s^2} \tanh\left(\frac{s}{2}\right) \end{aligned}$$

9 ANALYSIS AND CHARACTERIZATION OF LTI SYSTEMS USING THE LAPLACE TRANSFORM

If $x(t)$ and $y(t)$ are the input and output of an LTI system with impulse response $h(t)$ [Fig. 9.18], then

$$y(t) = x(t) * h(t)$$

Application of the time convolution property to the above equation yields

$$\mathcal{L}[y(t)] = \mathcal{L}[x(t)] * \mathcal{L}[h(t)]$$

$$Y(s) = X(s)H(s) \quad (9.50)$$

or

$$H(s) = \frac{Y(s)}{X(s)} \quad (9.51)$$

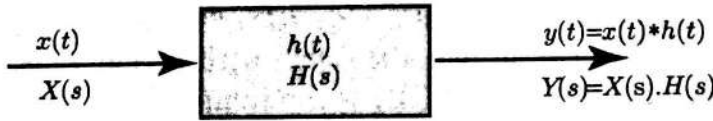


Fig. 9.18 LTI system

$H(s)$ is commonly referred to as the *system function* or *transfer function*. The transfer function $H(s)$ of an LTI system is equal to the ratio of the Laplace transform of the output signal to the Laplace transform of the input signal when all initial conditions are zero. For $s = j\omega$, $H(s)$ is the frequency response of the LTI system.

9.9.1 The Transfer Function and Differential-equation System Description

The transfer function may be related directly to the differential-equation description of an LTI system by using the bilateral Laplace transform. The relationship between the input and output of an N th-order LTI system is described by the differential equation

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Taking the Laplace transform of the above equation, we obtain

$$\sum_{k=0}^N a_k s^k Y(s) = \sum_{k=0}^M b_k s^k X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{\sum_{k=0}^M b_k s^k}{\sum_{k=0}^N a_k s^k}$$

$H(s)$ is a ratio of polynomials in s and is thus termed a rational transfer function.

9.9.2 Impulse Response and Step Response

The impulse response $h(t)$ is defined as the output of an LTI system due to a unit impulse signal input $\delta(t)$ applied at time $t = 0$. Also,

$$h(t) \longleftrightarrow H(s)$$

where $H(s)$ is the system function or transfer function.

The step response $s(t)$ is defined as the output of an LTI system due to a unit step input signal, i.e., $x(t) = u(t)$. The step response of an LTI system with impulse response $h(t)$ is given by

$$s(t) = h(t) * x(t)$$

$$= h(t) * u(t)$$

$$= \int_{-\infty}^{\infty} h(\tau) u(t - \tau) d\tau$$

$$s(t) = \int_{-\infty}^t h(\tau) d\tau$$

Therefore,

$$\mathcal{L}[s(t)] = \mathcal{L} \left[\int_{-\infty}^t h(\tau) d\tau \right]$$

$$S(s) = \frac{H(s)}{s} \quad (9.52)$$

Conversely,

$$h(t) = \frac{ds(t)}{dt}$$

$$\mathcal{L}[h(t)] = \mathcal{L} \left[\frac{ds(t)}{dt} \right]$$

$$H(s) = s S(s) \quad (9.53)$$

Example 9.29 Find the transfer function and the impulse response of a causal LTI system described by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} - 2x(t)$$

Solution

Given that

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{dx(t)}{dt} - 2x(t)$$

Taking the Laplace transform of the above equation, we obtain

$$s^2 Y(s) + 2sY(s) + Y(s) = s^2 X(s) - 2X(s)$$

$$Y(s)(s^2 + 2s + 1) = X(s)(s - 2)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s - 2}{s^2 + 2s + 1}$$

$$H(s) = \frac{s - 2}{(s + 1)^2} = \frac{s - 3 + 1}{(s + 1)^2} = \frac{s + 1}{(s + 1)^2} - \frac{3}{(s + 1)^2}$$

$$H(s) = \frac{1}{s + 1} - \frac{3}{(s + 1)^2}$$

We know that

$$e^{-t}u(t) \longleftrightarrow \frac{1}{s+1}$$

$$3t e^{-t}u(t) \longleftrightarrow \frac{3}{(s+1)^2}$$

Therefore, the impulse response

$$h(t) = e^{-t}u(t) - 3t e^{-t}u(t)$$

Example 9.30 The unit step response of an LTI system is $s(t) = 2 e^{-t}u(t)$. Determine its system function and the impulse response.

Solution

Given that the unit step response

$$s(t) = 2 e^{-t}u(t)$$

$$\mathcal{L}[s(t)] = \mathcal{L}[2 e^{-t}u(t)]$$

$$S(s) = \frac{2}{s+1}$$

The relationship between $\mathcal{L}[h(t)] = H(s)$ and $\mathcal{L}[s(t)] = S(s)$ is given by

$$H(s) = s S(s) = \frac{2s}{s+1} = \frac{2s+2-2}{s+1} = \frac{2(s+1)}{s+1} - \frac{2}{s+1}$$

$$H(s) = 2 - \frac{2}{s+1}$$

The inverse transform of this equation yields

$$h(t) = 2\delta(t) - 2 e^{-t}u(t)$$

Example 9.31 An LTI system has a unit step response given by $s(t) = (1 - e^{-t} - t e^{-t})u(t)$. For a certain input $x(t)$, the output is observed to be equal to $y(t) = (2 - 3 e^{-t} + e^{-3t})u(t)$. What is $x(t)$?

Solution

Given that

$$y(t) = (2 - 3 e^{-t} + e^{-3t})u(t)$$

$$Y(s) = \frac{2}{s} - \frac{3}{s+1} + \frac{1}{s+3}$$

$$Y(s) = \frac{2(s^2 + 4s + 3) - 3s(s+3) + s(s+1)}{s(s+1)(s+3)} = \frac{6}{s(s+1)(s+3)}$$

Consider the given unit step response

$$s(t) = (1 - e^{-t} - t e^{-t})u(t)$$

$$s(t) = u(t) - e^{-t}u(t) - t e^{-t}u(t)$$

The Laplace transform of this equation yields

$$S(s) = \frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2}$$

We know that [Eq. (9.53)]

$$H(s) = s S(s) = s \left(\frac{1}{s} - \frac{1}{s+1} - \frac{1}{(s+1)^2} \right)$$

$$\begin{aligned} H(s) &= 1 - \frac{s}{s+1} - \frac{s}{(s+1)^2} \\ &= \frac{(s+1)^2 - s(s+1) - s}{(s+1)^2} \\ &= \frac{s^2 + 2s + 1 - s^2 - s - s}{(s+1)^2} \end{aligned}$$

$$H(s) = \frac{1}{(s+1)^2} = \frac{Y(s)}{X(s)}$$

Therefore,

$$\begin{aligned} X(s) &= \frac{Y(s)}{H(s)} = \frac{6/(s(s+1)(s+3))}{1/((s+1)^2)} \\ &= \frac{6(s+1)}{s(s+3)} \\ X(s) &= \frac{2}{s} + \frac{4}{s+3} \end{aligned}$$

The inverse transform of this equation yields

$$x(t) = 2u(t) + 4e^{-3t}u(t)$$

9.9.3 Causality

For a causal LTI system, the impulse response $h(t) = 0$ for $t < 0$ and thus is right-sided. We know that if a signal is right-sided and of infinite duration, then the ROC is the region in the s -plane to the right of the rightmost pole. Consequently, *the ROC associated with*

the system function for a causal system is a right-half plane. However, the converse of this statement is not necessarily true. If $H(s)$ is rational, then we can determine whether the system is causal simply by checking to see if its ROC is a right-half plane, that is, for a system with a rational system function, causality of the system is equivalent to the ROC being the right-half plane to the right of the rightmost pole.

In an exactly analogous manner, we can deal with the concept of anticausality. For an anticausal LTI system, the impulse response $h(t) = 0$ for $t > 0$ and is thus left-sided. We know that if a signal is left-sided and of infinite duration, then the ROC is the region in the s -plane to the left of the leftmost pole. Consequently, the ROC associated with the system function for an anticausal system is a left-half plane. However, the converse of this statement is not necessarily true.

Example 9.32 For the following system functions, check whether the corresponding LTI system is causal, anticausal, or noncausal.

- (a) $H_1(s) = \frac{1}{s^2 + 5s + 6} \quad \Re\{s\} > -2$
- (b) $H_2(s) = \frac{1}{s^2 + 5s + 6} \quad \Re\{s\} < -3$
- (c) $H_3(s) = \frac{1}{s^2 + 5s + 6} \quad -2 < \Re\{s\} < -3$
- (d) $H_4(s) = \frac{e^{2s}}{s + 1} \quad \Re\{s\} > -1$

Solution

(a) Consider the given system function

$$H_1(s) = \frac{1}{s^2 + 5s + 6} = \frac{1}{(s + 2)(s + 3)} \quad \Re\{s\} > -2$$

Since the given system function $H_1(s)$ is rational and its ROC [shown in Fig. 9.19(a)] is to the right of the rightmost pole, the corresponding LTI system is causal. Also it can be verified by its impulse response. Given that

$$\begin{aligned} H_1(s) &= \frac{1}{(s + 2)(s + 3)} \\ &= \frac{1}{s + 2} - \frac{1}{s + 3} \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$h_1(t) = e^{-2t}u(t) - e^{-3t}u(t)$$

Since $h_1(t) = 0$ for $t < 0$, this system is causal.

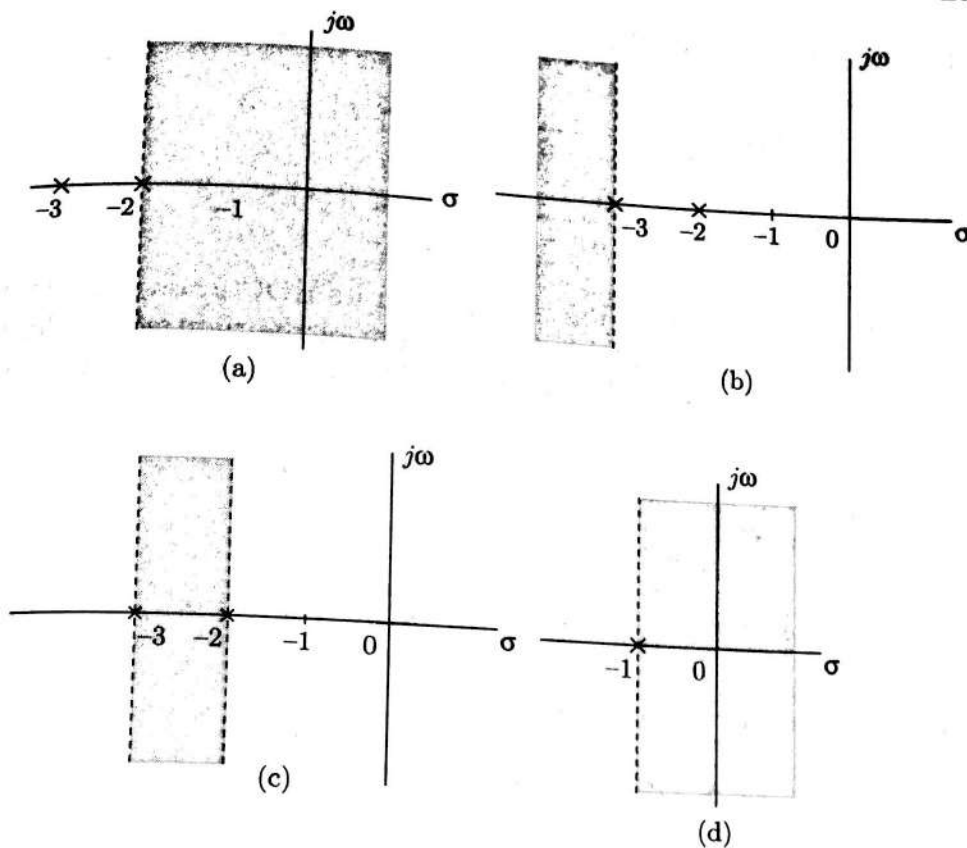


Fig. 9.19 (a) ROC for $H_1(s)$, (b) ROC for $H_2(s)$, (c) ROC for $H_3(s)$, and (d) ROC for $H_4(s)$

(b) Consider the given system function

$$H_2(s) = \frac{1}{s^2 + 5s + 6} \quad \Re\{s\} < -3$$

$$H_2(s) = \frac{1}{(s+2)(s+3)} \quad \Re\{s\} < -3$$

Since the given system function $H_2(s)$ is rational and its ROC [shown in Fig. 9.19(b)] is to the left of the leftmost pole, the corresponding LTI system is anticausal. Also it can be verified by its impulse response. Given that

$$\begin{aligned} H_2(s) &= \frac{1}{(s+2)(s+3)} \\ &= \frac{1}{s+2} - \frac{1}{s+3} \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$h_2(t) = -e^{-2t}u(-t) + e^{-3t}u(-t)$$

Since $h_2(t) = 0$ for $t > 0$, this system is anticausal.

(c) Consider the given system function

$$H_3(s) = \frac{1}{s^2 + 5s + 6} \quad -2 < \Re\{s\} < -3$$

$$H_3(s) = \frac{1}{(s+2)(s+3)} \quad -2 < \Re\{s\} < -3$$

Since the given system function $H_3(s)$ is rational and its ROC [shown in Fig. 9.19(c)] is neither to the left of the leftmost pole nor to the right of the rightmost pole, but it is a strip, the corresponding LTI system is noncausal. Also it can be verified by its impulse response. Given that

$$H_3(s) = \frac{1}{(s+2)(s+3)}$$

$$H_3(s) = \frac{1}{s+2} - \frac{1}{s+3} \quad -2 < \Re\{s\} < -3$$

Taking the inverse Laplace transform, we obtain

$$h_3(t) = e^{-2t}u(t) + e^{-3t}u(-t)$$

Since $h_3(t)$ is two-sided, this system is noncausal.

(d) Consider the given system function

$$H_4(s) = \frac{e^{2s}}{s+1} \quad \Re\{s\} > -1$$

Since the given system function $H_4(s)$ is not rational, we cannot determine whether the system is causal by checking its ROC. For this system, the ROC [shown in Fig. 9.19(a)] is to the right of the rightmost pole; therefore, the impulse response must be right-sided. We know that

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a$$

$$e^{-t}u(t) \longleftrightarrow \frac{1}{s+1} \quad \Re\{s\} > -1$$

$$e^{-(t+2)}u(t+2) \longleftrightarrow \frac{e^{2s}}{s+1} \quad \Re\{s\} > -1$$

Therefore,

$$h_4(t) = e^{-(t+2)}u(t+2)$$

Since $h_4(t)$ is two-sided, this system is noncausal.

9.9.4 Stability

An LTI system is stable if the impulse response $h(t)$ is *absolutely integrable*, that is,

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty$$

This implies that the Fourier transform exists, and thus the ROC includes the $j\omega$ -axis in the s -plane. Thus, *an LTI system is stable if and only if the ROC of its system function $H(s)$ includes the $j\omega$ -axis.*

9.9.5 Stability of a Causal LTI System

Consider a causal LTI system with a rational system function $H(s)$. Since the system is causal, the ROC is to the right of the rightmost pole. Consequently, for this system to be stable (i.e., for the ROC to include the $j\omega$ -axis), the rightmost pole of $H(s)$ must be to the left of the $j\omega$ -axis. That is, *a causal system with rational system function $H(s)$ is stable if and only if all the poles of $H(s)$ lie in the left-half of the s -plane.*

Example 9.33 For the following system functions, check whether the corresponding LTI system is causal and stable.

$$(a) H_1(s) = \frac{1}{s^2 - s - 6} \quad \Re\{s\} > 3$$

$$(b) H_2(s) = \frac{1}{s^2 - s - 6} \quad \Re\{s\} < -2$$

$$(c) H_3(s) = \frac{1}{s^2 - s - 6} \quad -2 < \Re\{s\} < 3$$

Solution

(a) Consider the given system function

$$H_1(s) = \frac{1}{s^2 - s - 6} = \frac{1}{(s - 3)(s + 2)} \quad \Re\{s\} > 3$$

Since the given system function $H_1(s)$ is rational and its ROC [shown in Fig. 9.20(a)] is to the right of the rightmost pole, the corresponding LTI system is causal. Also, note that the ROC does not include the $j\omega$ -axis, and, consequently, the corresponding system is unstable. This can be verified by its impulse response. Given that

$$\begin{aligned} H_1(s) &= \frac{1}{(s - 3)(s + 2)} \\ &= \frac{1/5}{s - 3} - \frac{1/5}{s + 2} \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$h_1(t) = \frac{1}{5} [e^{3t}u(t) - e^{-2t}u(t)]$$

Note that $h_1(t) = 0$ for $t < 0$, and $\int_{-\infty}^{\infty} h_1(t) dt = \infty$, i.e., the impulse response is not absolutely integrable, so this system is causal and unstable.

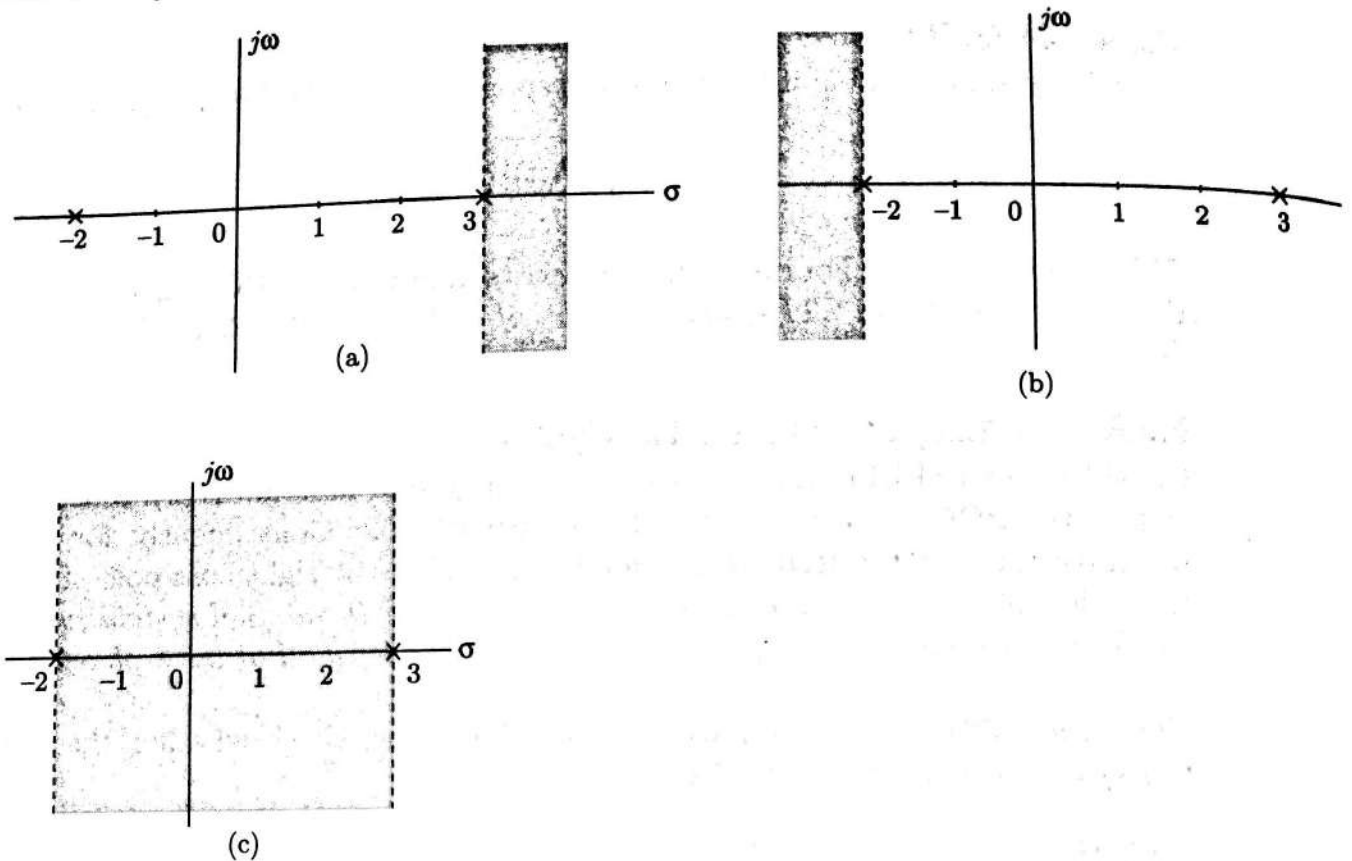


Fig. 9.20 ROC for (a) causal and unstable system $H_1(s)$, (b) anticausal and unstable system $H_2(s)$ and (c) noncausal and stable system $H_3(s)$.

(b) Consider the given system function

$$H_2(s) = \frac{1}{s^2 - s - 6} \quad \Re\{s\} < -2$$

$$H_2(s) = \frac{1}{(s-3)(s+2)} \quad \Re\{s\} < -2$$

Since the given system function $H_2(s)$ is rational and its ROC [shown in Fig. 9.20(b)] is to the left of the leftmost pole, the corresponding LTI system is anticausal. Also, note that the ROC does not include the $j\omega$ -axis, and, consequently, the corresponding system is unstable. This can be verified by its impulse response. Given that

$$\begin{aligned} H_2(s) &= \frac{1}{(s-3)(s+2)} \quad \Re\{s\} < -2 \\ &= \frac{1/5}{s-3} - \frac{1/5}{s+2} \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$h_2(t) = \frac{1}{5} [-e^{3t}u(-t) + e^{-2t}u(-t)]$$

Note that $h_2(t) = 0$ for $t > 0$, and $\int_{-\infty}^{\infty} h_2(t) dt = \infty$, i.e., the impulse response is not absolutely integrable, so this system is anticausal and unstable.

(c) Consider the given system function

$$H_3(s) = \frac{1}{s^2 - s - 6} \quad -2 < \Re\{s\} < 3$$

$$H_3(s) = \frac{1}{(s-3)(s+2)} \quad -2 < \Re\{s\} < 3$$

Since the given system function $H_3(s)$ is rational and its ROC [shown in Fig. 9.20(c)] is a strip, the corresponding LTI system is noncausal. Also, note that the ROC includes the $j\omega$ -axis, and, consequently, the corresponding system is stable. This can be verified by its impulse response. Given that

$$\begin{aligned} H_3(s) &= \frac{1}{(s-3)(s+2)} \quad -2 < \Re\{s\} < 3 \\ &= \frac{1/5}{s-3} - \frac{1/5}{s+2} \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$h_3(t) = \frac{1}{5}[-e^{3t}u(-t) - e^{-2t}u(t)]$$

Note that $h_3(t)$ is two-sided, and $\int_{-\infty}^{\infty} h_3(t) dt < \infty$, i.e., the impulse response is absolutely integrable, so this system is noncausal and stable.

Example 9.34 For the following system functions, check whether the corresponding LTI system is causal and stable.

$$H(s) = \frac{1}{s^2 + 5s + 6} \quad \Re\{s\} > -2$$

Solution

Consider the given system function

$$H(s) = \frac{1}{s^2 + 5s + 6} = \frac{1}{(s+2)(s+3)} \quad \Re\{s\} > -2$$

Since the given system function $H(s)$ is rational and its ROC [shown in Fig. 9.21] is to the right of the rightmost pole, the corresponding LTI system is causal. Also, note that the ROC includes the $j\omega$ -axis, and, consequently, the corresponding system is stable. This can be verified by its impulse response. Given that

$$\begin{aligned} H(s) &= \frac{1}{(s+2)(s+3)} \quad \Re\{s\} > -2 \\ &= \frac{1}{s+2} - \frac{1}{s+3} \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$h(t) = e^{-2t}u(t) - e^{-3t}u(t)$$

Note that $h(t) = 0$ for $t < 0$, and $\int_{-\infty}^{\infty} h(t) dt < \infty$, i.e., the impulse response is absolutely integrable, so this system is causal and stable.

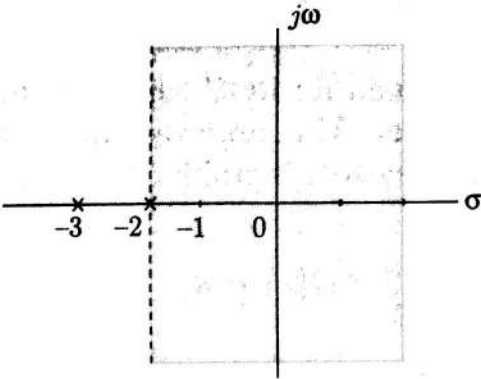


Fig. 9.21 ROC for Example 9.34

Example 9.35 The transfer function of the system is given by

$$H(s) = \frac{2}{s + 3} + \frac{1}{s - 2}$$

Determine the impulse response if the system is (a) stable and (b) causal. State whether the system will be stable and causal simultaneously.

Solution

Given that

$$H(s) = \frac{2}{s + 3} + \frac{1}{s - 2}$$

The system has poles at -3 and 2 .

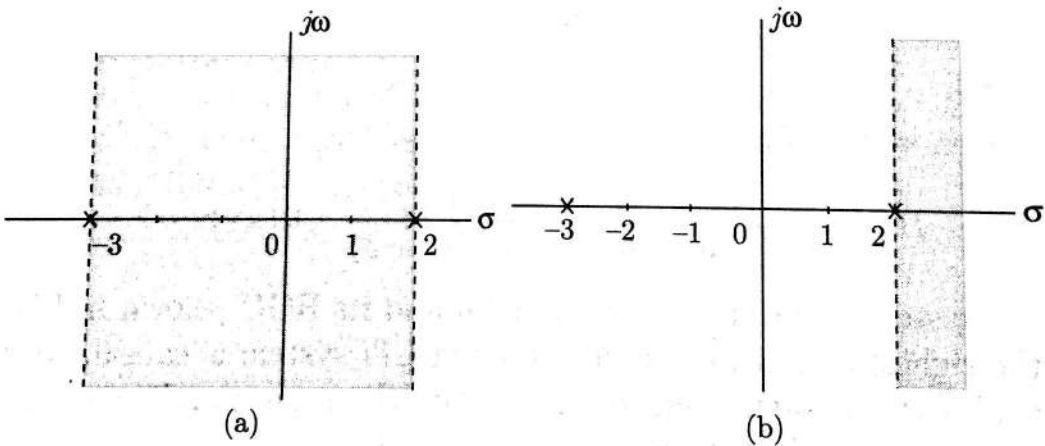


Fig. 9.22 ROC for the system function $H(s)$ of Example 9.35; (a) stable system, (b) causal system

(a) For this system to be stable, the ROC must include the $j\omega$ -axis. To include the $j\omega$ -axis, its ROC must lie between -3 and 2 , i.e., $-3 < \Re\{s\} < 2$ [shown in Fig. 9.22(a)]. The pole of the first term is at -3 . The ROC lies to the right of this pole, so this pole corresponds to causal (right-sided) signal. Therefore,

$$2 e^{-3t} u(t) \longleftrightarrow \frac{2}{s + 3}$$

The second term has a pole at $s = 2$. Here the ROC is to the left of this pole, so this pole corresponds to the anticausal (left-sided) signal. Therefore,

$$-e^{2t}u(-t) \longleftrightarrow \frac{1}{s-2}$$

and hence we obtain

$$h(t) = 2e^{-3t}u(t) - e^{2t}u(-t)$$

(b) For this system to be causal, the ROC lies to the right of the rightmost pole, i.e., $\Re\{s\} > 2$ [shown in Fig. 9.22(b)]. The ROC, $\Re\{s\} > 2$, is to the right of the rightmost pole, so all the poles correspond to causal (right-sided) signals. Therefore,

$$\begin{aligned} 2e^{-3t}u(t) &\longleftrightarrow \frac{2}{s+3} \\ e^{2t}u(t) &\longleftrightarrow \frac{1}{s-2} \end{aligned}$$

and hence we obtain

$$h(t) = 2e^{-3t}u(t) + e^{2t}u(t)$$

For a system to be stable and causal simultaneously, all the poles must lie in the left half of the s -plane. Since this system has one pole ($s = 2$) in the right half of the s -plane, it cannot be stable and causal simultaneously.

Example 9.36 Find the inverse Laplace transform of

$$H(s) = \frac{4s^2 + 15s + 8}{(s+2)^2(s-1)}$$

assuming that (a) $h(t)$ is causal and (b) the Fourier transform of $h(t)$ exists, i.e., $h(t)$ is absolutely integrable.

Solution

Given that

$$H(s) = \frac{4s^2 + 15s + 8}{(s+2)^2(s-1)}$$

Using partial fraction expansion, we obtain

$$H(s) = \frac{1}{s+2} + \frac{2}{(s+2)^2} + \frac{3}{s-1}$$

The system has a double pole at -2 and a single pole at 1 .

(a) For this system to be causal, the ROC lies to the right of the rightmost pole, i.e., $\Re\{s\} > 1$ [shown in Fig. 9.23(a)]. Since the ROC, $\Re\{s\} > 1$, is to the right of the rightmost

pole, all the poles correspond to causal (right-sided) signals. Therefore,

$$\begin{aligned} e^{-2t}u(t) &\longleftrightarrow \frac{1}{s+2} \\ 2t e^{-2t}u(t) &\longleftrightarrow \frac{1}{(s+2)^2} \\ 3e^t u(t) &\longleftrightarrow \frac{3}{s-1} \end{aligned}$$

and hence we obtain

$$h(t) = e^{-2t}u(t) + 2t e^{-2t}u(t) + 3 e^t u(t)$$

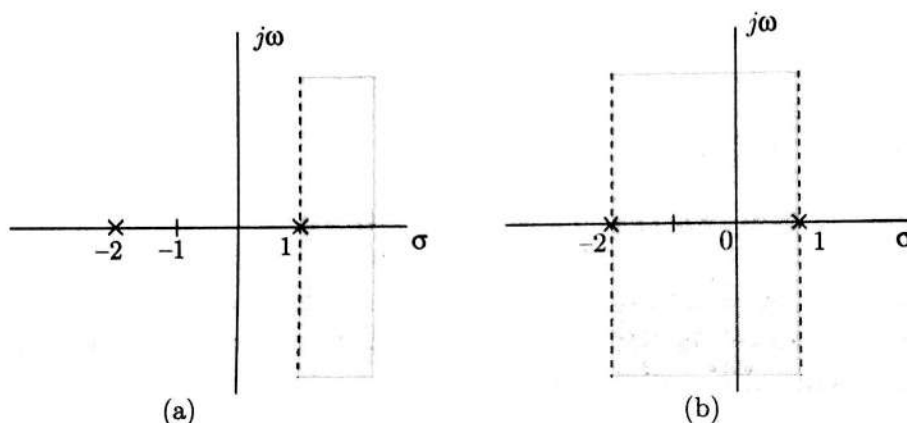


Fig. 9.23

(b) For $h(t)$ to be Fourier transformable, i.e., for $h(t)$ to be absolutely summable (stable system), the ROC must include the $j\omega$ -axis. To include the $j\omega$ -axis, its ROC is in the region $-2 < \Re\{s\} < 1$ [shown in Fig. 9.23(b)]. The pole of the first and second term is at -2 . The ROC lies to the right of this pole, so this pole corresponds to a causal (right-sided) signal. Therefore,

$$\begin{aligned} e^{-2t}u(t) &\longleftrightarrow \frac{1}{s+2} \\ 2t e^{-2t}u(t) &\longleftrightarrow \frac{1}{(s+2)^2} \end{aligned}$$

The third term has a pole at $s = 1$. Here the ROC is to the left of this pole, so this pole corresponds to an anticausal (left-sided) signal. Therefore,

$$-3 e^t u(-t) \longleftrightarrow \frac{3}{s-1}$$

and hence we obtain

$$h(t) = e^{-2t}u(t) + 2t e^{-2t}u(t) - 3 e^t u(-t)$$

Example 9.37 Consider a continuous-time LTI system for which the input $x(t)$ and output $y(t)$ are related by the differential equation

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Let $X(s)$ and $Y(s)$ denote the Laplace transforms of $x(t)$ and $y(t)$, respectively, and let $H(s)$ denote the Laplace transform of $h(t)$, the system impulse response.

- (a) Determine $H(s)$ as a ratio of two polynomials in s . Sketch the pole-zero pattern of $H(s)$.
- (b) Determine $h(t)$ for each of the following cases:
 1. The system is stable.
 2. The system is causal.
 3. The system is neither stable nor causal.

Solution

(a) Consider the given differential equation

$$\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$$

Taking the Laplace transform of the above equation, we obtain

$$s^2 Y(s) - sY(s) - 2Y(s) = X(s)$$

$$Y(s)[s^2 - s - 2] = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2 - s - 2}$$

$$H(s) = \frac{1}{(s-2)(s+1)}$$

The system has poles at -1 and 2 . The pole-zero pattern is shown in Fig. 9.24(a).

(b) Using partial fraction expansion

$$H(s) = \frac{1/3}{s-2} - \frac{1/3}{s+1}$$

1. For this system to be stable, the ROC must include the $j\omega$ -axis. To include the $j\omega$ -axis, its ROC must be in the region $-1 < \Re\{s\} < 2$ [shown in Fig. 9.24(b)].

The first term has a pole at $s = 2$. Here the ROC is to the left of this pole, so this pole corresponds to the anti-causal (left-sided) signal. Therefore,

$$-\frac{1}{3} e^{2t} u(-t) \longleftrightarrow \frac{1/3}{s-2}$$

The pole of the second term is at -1 . The ROC lies to the right of this pole, so this pole corresponds to a causal (right-sided) signal. Therefore,

$$\frac{1}{3} e^{-t} u(t) \longleftrightarrow \frac{1/3}{s+1}$$

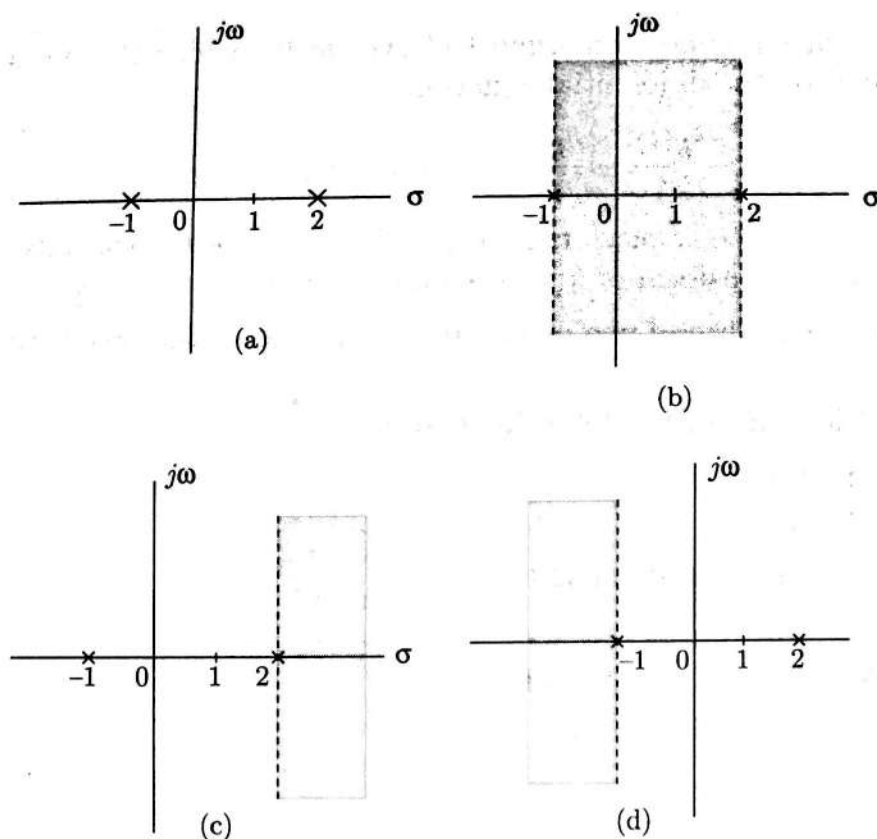


Fig. 9.24 (a) Pole-zero pattern of $H(s)$ and ROC for (b) stable system, (c) causal system, and (d) neither stable nor causal system of Example 9.37

and hence we obtain

$$h(t) = -\frac{1}{3} e^{2t} u(-t) - \frac{1}{3} e^{-t} u(t)$$

2. For this system to be causal, the ROC lies to the right of the rightmost pole, i.e., $\Re\{s\} > 2$ [shown in Fig. 9.24(c)].

Since the ROC, $\Re\{s\} > 2$, is to the right of the rightmost pole, all the poles correspond to causal (right-sided) signals. Therefore,

$$\begin{aligned} \frac{1}{3} e^{2t} u(t) &\longleftrightarrow \frac{1/3}{s-2} \\ \frac{1}{3} e^{-t} u(t) &\longleftrightarrow \frac{1/3}{s+1} \end{aligned}$$

and hence we obtain

$$h(t) = \frac{1}{3} e^{2t} u(t) - \frac{1}{3} e^{-t} u(t)$$

3. For this system to be neither stable nor causal, the ROC must not include the $j\omega$ -axis and also not be to the left of the leftmost pole. So for this case, the ROC will be $\Re\{s\} < -1$ [shown in Fig. 9.24(d)]. Since the ROC, $\Re\{s\} < -1$, is to the left of the

leftmost pole, all the poles correspond to anti-causal (left-sided) signals. Therefore,

$$\begin{aligned} -\frac{1}{3} e^{2t} u(-t) &\longleftrightarrow \frac{1/3}{s-2} \\ -\frac{1}{3} e^{-t} u(-t) &\longleftrightarrow \frac{1/3}{s+1} \end{aligned}$$

and hence we obtain

$$h(t) = -\frac{1}{3} e^{2t} u(-t) + \frac{1}{3} e^{-t} u(-t)$$

Example 9.38 Find the response $y(t)$ of a noncausal system with the transfer function

$$H(s) = \frac{-1}{s-1} \quad \Re\{s\} < 1$$

to the input $x(t) = e^{-2t} u(t)$.

Solution

Given that

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$$x(t) = e^{-2t} u(t)$$

$$\mathcal{L}[x(t)] = X(s) = \frac{1}{s+2} \quad \Re\{s\} > -2$$

We know that [Eq. (9.50)]

$$Y(s) = X(s)H(s) = \frac{-1}{(s-1)(s+2)}$$

The ROC of $Y(s)$ is in the region $-2 < \Re\{s\} < 1$. Using partial fraction expansion, we obtain

$$Y(s) = \frac{-1/3}{s-1} + \frac{1/3}{s+2} \quad -2 < \Re\{s\} < 1$$

Taking the inverse Laplace transform, we obtain

$$y(t) = \frac{1}{3} [e^t u(-t) + e^{-2t} u(t)]$$

Example 9.39 Find the response $y(t)$ of an LTI system with the transfer function

$$H(s) = \frac{1}{s+5} \quad \Re\{s\} > -5$$

and the input

$$x(t) = e^{-t} u(t) + e^{-2t} u(-t)$$

Solution

Consider the given input

$$x(t) = e^{-t}u(t) + e^{-2t}u(-t) = x_1(t) + x_2(t)$$

We know that

$$x_1(t) = e^{-t}u(t) \longleftrightarrow X_1(s) = \frac{1}{s+1} \quad \Re\{s\} > -1$$

$$x_2(t) = e^{-2t}u(-t) \longleftrightarrow X_2(s) = -\frac{1}{s+2} \quad \Re\{s\} < -2$$

Although the Laplace transform of each of the individual terms have an ROC, there is no common ROC, and thus $x(t)$ has no Laplace transform.

In this case, we must determine separately the system response to each of the two input components, $x_1(t) = e^{-t}u(t)$ and $x_2(t) = e^{-2t}u(-t)$. If $y_1(t)$ and $y_2(t)$ are the system responses to $x_1(t)$ and $x_2(t)$, respectively, then

$$Y_1(s) = X_1(s)H(s) = \frac{1}{(s+1)(s+5)} \quad \Re\{s\} > -1$$

$$Y_1(s) = \frac{1/4}{s+1} - \frac{1/4}{s+5} \quad \Re\{s\} > -1$$

$$y_1(t) = \frac{1}{4}[e^{-t}u(t) + e^{-5t}u(t)]$$

and

$$Y_2(s) = X_2(s)H(s) = \frac{-1}{(s+2)(s+5)} \quad -5 < \Re\{s\} < -2$$

$$Y_2(s) = \frac{-1/3}{s+2} + \frac{1/3}{s+5} \quad -5 < \Re\{s\} < -2$$

$$y_2(t) = \frac{1}{3}[e^{-2t}u(-t) + e^{-5t}u(t)]$$

Therefore, the response $y(t)$ is

$$y(t) = y_1(t) + y_2(t)$$

$$= \frac{1}{4}[e^{-t}u(t) + e^{-5t}u(t)] + \frac{1}{3}[e^{-2t}u(-t) + e^{-5t}u(t)]$$

$$y(t) = \frac{1}{4}e^{-t}u(t) + \frac{1}{12}e^{-5t}u(t) + \frac{1}{3}e^{-2t}u(-t)$$

9.10 UNILATERAL (ONE-SIDED) LAPLACE TRANSFORM

The unilateral Laplace transform of a signal $x(t)$ is defined as

$$\mathcal{L}_u[x(t)] = X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt \quad (9.54)$$

where the lower limit of integration, 0^- , implies that we do include discontinuities and impulses that occur at $t = 0$ in the interval of integration. Hence $X(s)$ depends on $x(t)$ for $t \geq 0$. The evaluation of the inverse unilateral Laplace transform is also the same as for bilateral Laplace transforms. We shall denote the relationship between $x(t)$ and $X(s)$ as

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s) \quad (9.55)$$

The difference in the definitions of the unilateral and bilateral Laplace transforms lies in the lower limit of integral. The bilateral Laplace transform in Eq. (9.3) involves an integration from $-\infty$ to $+\infty$, while the unilateral Laplace transform in Eq. (9.54) involves an integration from 0^- to $+\infty$. Consequently, two signals that differ for $t < 0$, but that are identical for $t \geq 0$, will have different bilateral Laplace transforms but identical unilateral Laplace transforms. Similarly, any signal that is identically zero for $t < 0$ will have identical bilateral and unilateral transforms.

The unilateral Laplace transform is used in analysing causal systems and, particularly, systems specified by linear constant-coefficient differential equations with nonzero initial conditions (i.e., systems that are not initially at rest).

For a unilateral Laplace transform, there is a unique inverse transform of $X(s)$; consequently, there is no need to specify the ROC explicitly (ROC for a unilateral Laplace transform must always be a right-half plane). For this reason, we shall generally ignore any mention of the ROC for a unilateral Laplace transform.

Example 9.40 Find (a) bilateral Laplace transform and (b) unilateral Laplace transform of the signal

$$x(t) = e^{-a(t+1)}u(t+1)$$

Solution

(a) We know that

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a$$

Using the time-shifting property, we get

$$e^{-a(t+1)}u(t+1) \longleftrightarrow \frac{e^s}{s+a} \quad \Re\{s\} > -a$$

Therefore,

$$\mathcal{L}[x(t)] = X(s) = \frac{e^s}{s+a} \quad \Re\{s\} > -a$$

(b) From Eq. (9.54), we have

$$\begin{aligned}
 \mathcal{L}_u[x(t)] = X(s) &= \int_{0^-}^{\infty} x(t) e^{-st} dt = \int_{0^-}^{\infty} e^{-a(t+1)} u(t+1) e^{-st} dt \\
 &= \int_{0^-}^{\infty} e^{-a(t+1)} e^{-st} dt \\
 &= e^{-a} \int_{0^-}^{\infty} e^{-(s+a)t} dt \\
 \mathcal{L}_u[x(t)] = X(s) &= \frac{e^{-a}}{s+a} \quad \Re\{s\} > -a
 \end{aligned}$$

Example 9.41 Find the inverse Laplace transform of

$$X(s) = \frac{s+3+5e^{-2s}}{(s+1)(s+2)}$$

Solution

Given that

$$\begin{aligned}
 X(s) &= \frac{s+3+5e^{-2s}}{(s+1)(s+2)} = \frac{s+3}{(s+1)(s+2)} + \frac{5e^{-2s}}{(s+1)(s+2)} \\
 &= X_1(s) + X_2(s) e^{-2s}
 \end{aligned}$$

where

$$X_1(s) = \frac{s+3}{(s+1)(s+2)} = \frac{2}{s+1} - \frac{1}{s+2}$$

and

$$X_2(s) = \frac{5}{(s+1)(s+2)} = \frac{5}{s+1} - \frac{5}{s+2}$$

Taking the inverse Laplace transform of $X_1(s)$ and $X_2(s)$, we obtain

$$x_1(t) = [2e^{-t} - e^{-2t}]u(t)$$

$$x_2(t) = [5e^{-t} - 5e^{-2t}]u(t)$$

Also, because

$$X(s) = X_1(s) + X_2(s) e^{-2s}$$

We can write

$$\begin{aligned} x(t) &= x_1(t) + x_2(t-2) \\ &= [2e^{-t} - e^{-2t}]u(t) + [5e^{-(t-2)} - 5e^{-2(t-2)}]u(t-2) \end{aligned}$$

Example 9.42 Determine the unilateral Laplace transform of each of the following signals:

- (a) $x(t) = e^{-2t}u(t+1)$
 (b) $x(t) = \delta(t+1) + \delta(t) + e^{-2(t+3)}u(t+1)$

Solution

(a) From Eq. (9.54), we have

$$\begin{aligned} \mathcal{L}_u[x(t)] &= X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt = \int_{0^-}^{\infty} e^{-2t}u(t+1) e^{-st} dt \\ &= \int_{0^-}^{\infty} e^{-2t} e^{-st} dt = \int_{0^-}^{\infty} e^{-(s+2)t} dt \\ &= -\frac{e^{-(s+2)t}}{s+2} \Big|_{0^-}^{\infty} = -\frac{1}{s+2}[0-1] = \frac{1}{s+2} \\ \mathcal{L}_u[x(t)] &= X(s) = \frac{1}{s+2} \end{aligned}$$

(b) From Eq. (9.54), we have

$$\begin{aligned} \mathcal{L}_u[x(t)] &= X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt \\ &= \int_{0^-}^{\infty} [\delta(t+1) + \delta(t) + e^{-2(t+3)}u(t+1)] e^{-st} dt \\ &= \int_{0^-}^{\infty} \delta(t+1) e^{-st} dt + \int_{0^-}^{\infty} \delta(t) e^{-st} dt + \int_{0^-}^{\infty} e^{-2(t+3)}u(t+1) e^{-st} dt \\ &= 0 + e^{-st} \Big|_{t=0} + \int_{0^-}^{\infty} e^{-2(t+3)} e^{-st} dt \end{aligned}$$

$$\begin{aligned}
&= 1 + e^{-6} \int_{0^-}^{\infty} e^{-(s+2)t} dt \\
&= 1 - \frac{e^{-(s+2)t}}{s+2} \Big|_{0^-}^{\infty} \\
&= 1 - \frac{1}{s+2} [0 - 1] \\
&= 1 + \frac{1}{s+2} \\
\mathcal{L}_u[x(t)] = X(s) &= \frac{s+3}{s+2}
\end{aligned}$$

9.11 RELATIONSHIP BETWEEN BILATERAL AND UNILATERAL LAPLACE TRANSFORMS

The bilateral Laplace transform can be evaluated using the unilateral Laplace transform in two steps:

1. Split $x(t)$ into its causal and anticausal components $x_1(t)$ and $x_2(t)$, respectively.

$$x_1(t) = x(t)u(t)$$

$$x_2(t) = x(t)u(-t)$$

2. The signals $x_1(t)$ and $x_2(-t)$ are both causal. Determine $\mathcal{L}_u[x_1(t)]$ and add it with $\mathcal{L}_u[x_2(-t)]$, with s replaced by $-s$, i.e.,

$$X(s) = \mathcal{L}_u[x_1(t)] + \mathcal{L}_u[x_2(-t)] \Big|_{s \rightarrow -s}$$

or, equivalently,

$$\mathcal{L}[x(t)] = \mathcal{L}_u[x(t)u(t)] + \mathcal{L}_u[x(-t)u(t)] \Big|_{s \rightarrow -s} \quad (9.56)$$

Proof We separate $x(t)$ into two components $x_1(t)$ and $x_2(t)$, representing the causal component and the anticausal component of $x(t)$, respectively:

$$x_1(t) = x(t)u(t)$$

$$x_2(t) = x(t)u(-t)$$

The bilateral Laplace transform of $x(t)$ is given by

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t) e^{-st} dt \\ &= \int_{-\infty}^{0^-} x_2(t) e^{-st} dt + \int_{0^-}^{\infty} x_1(t) e^{-st} dt \\ X(s) &= X_2(s) + X_1(s) \end{aligned}$$

Now, consider $X_2(s)$, given by

$$X_2(s) = \int_{-\infty}^{0^-} x_2(t) e^{-st} dt$$

Using the substitution $t = -\tau$ yields

$$X_2(s) = \int_{0^+}^{\infty} x_2(-\tau) e^{s\tau} d\tau$$

If $x(t)$ has any singularities at the origin, they are included in $x_1(t)$. Consequently, $x_2(t) = 0$ at $t = 0$. Hence,

$$\begin{aligned} X_2(s) &= \int_{0^-}^{\infty} x_2(-\tau) e^{s\tau} d\tau \\ X_2(-s) &= \int_{0^-}^{\infty} x_2(-\tau) e^{-s\tau} d\tau \\ X_2(-s) &= \int_{0^-}^{\infty} x_2(-t) e^{-st} dt \\ &= \mathcal{L}_u[x_2(-t)] \end{aligned}$$

Therefore,

$$X_2(s) = X_2(-s) \Big|_{s \rightarrow -s} = \mathcal{L}_u[x_2(-t)] \Big|_{s \rightarrow -s}$$

Also, because

$$\begin{aligned} X(s) &= X_1(s) + X_2(s) \\ X(s) &= \mathcal{L}_u[x_1(t)] + \mathcal{L}_u[x_2(-t)] \Big|_{s \rightarrow -s} \end{aligned}$$

9.12 PROPERTIES OF UNILATERAL LAPLACE TRANSFORM

The unilateral and bilateral Laplace transforms have many properties in common, although there are important differences. The properties of linearity, scaling, s -domain shifting, conjugation, convolution, and differentiation in the s -domain properties are identical for the bilateral and unilateral Laplace transforms. The derivation of each of these properties is identical to that of its bilateral counterpart.

9.12.1 Linearity

If

$$x_1(t) \xleftrightarrow{\mathcal{L}_u} X_1(s) \quad \text{and} \quad x_2(t) \xleftrightarrow{\mathcal{L}_u} X_2(s)$$

then

$$ax_1(t) + bx_2(t) \xleftrightarrow{\mathcal{L}_u} aX_1(s) + bX_2(s) \quad (9.57)$$

9.12.2 Time Scaling

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$x(at) \xleftrightarrow{\mathcal{L}_u} \frac{1}{a} X\left(\frac{s}{a}\right) \quad a > 0 \quad (9.58)$$

9.12.3 Shifting in the s -Domain

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$e^{s_0 t} x(t) \xleftrightarrow{\mathcal{L}_u} X(s - s_0) \quad (9.59)$$

9.12.4 Conjugation

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$x^*(t) \xleftrightarrow{\mathcal{L}_u} X^*(s^*) \quad (9.60)$$

9.12.5 Differentiation in s -Domain

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$tx(t) \xleftrightarrow{\mathcal{L}_u} -\frac{dX(s)}{ds} \quad (9.61)$$

Example 9.43 Find the inverse Laplace transform of

$$X(s) = \ln \left(1 + \frac{\omega^2}{s^2} \right)$$

Solution

Given that

$$X(s) = \ln \left(1 + \frac{\omega^2}{s^2} \right)$$

Differentiating w.r.t. s , we get

$$\begin{aligned} \frac{dX(s)}{ds} &= \frac{1}{1 + (\omega^2/s^2)} (-2) \frac{\omega^2}{s^3} \\ \frac{dX(s)}{ds} &= -\frac{2\omega^2}{s(s^2 + \omega^2)} \\ -\frac{dX(s)}{ds} &= \frac{2(\omega^2 + s^2 - s^2)}{s(s^2 + \omega^2)} \\ &= \frac{2(\omega^2 + s^2)}{s(s^2 + \omega^2)} - \frac{2s^2}{s(s^2 + \omega^2)} \\ -\frac{dX(s)}{ds} &= \frac{2}{s} - 2\frac{s}{s^2 + \omega^2} \end{aligned}$$

Using the differentiation property [Eq. (9.61)], we obtain

$$\mathcal{L}_u^{-1} \left[-\frac{dX(s)}{ds} \right] = tx(t) = \mathcal{L}_u^{-1} \left[\frac{2}{s} - 2\frac{s}{s^2 + \omega^2} \right]$$

$$tx(t) = 2u(t) - 2\cos(\omega t)u(t)$$

$$x(t) = \frac{2}{t}[1 - \cos(\omega t)]u(t)$$

9.12.6 Convolution

If $x_1(t) = x_2(t) = 0$, for all $t < 0$, and if

$$x_1(t) \xleftrightarrow{\mathcal{L}_u} X_1(s) \quad \text{and} \quad x_2(t) \xleftrightarrow{\mathcal{L}_u} X_2(s)$$

then

$$x_1(t) * x_2(t) \xleftrightarrow{\mathcal{L}_u} X_1(s)X_2(s) \quad (9.62)$$

It is important to note that the convolution property for unilateral transform applies only if the signals $x_1(t)$ and $x_2(t)$ are both zero for $t < 0$. That is, while we have seen that the bilateral Laplace transform of $x_1(t) * x_2(t)$ always equals to the product of the bilateral transforms of $x_1(t)$ and $x_2(t)$, the unilateral transform of $x_1(t) * x_2(t)$ in general does not equal to the product of the unilateral transforms if either $x_1(t)$ or $x_2(t)$ is nonzero for $t < 0$.

Example 9.44 Use the convolution property of the Laplace transform to determine

$$y(t) = e^{at}u(t) * e^{bt}u(t)$$

Solution

Given that

$$y(t) = e^{at}u(t) * e^{bt}u(t)$$

Using the convolution property of the Laplace transform [Eq. (9.60)], we obtain

$$Y(s) = \mathcal{L}_u[e^{at}u(t)]\mathcal{L}_u[e^{bt}u(t)]$$

$$Y(s) = \frac{1}{s-a} \frac{1}{s-b}$$

Using partial fraction expansion, we obtain

$$Y(s) = \frac{1}{a-b} \left[\frac{1}{s-a} - \frac{1}{s-b} \right]$$

Taking the inverse Laplace transform of this equation, we obtain

$$y(t) = \frac{1}{a-b} [e^{at}u(t) - e^{bt}u(t)]$$

9.12.7 Multiplication

If $x_1(t) = x_2(t) = 0$, for all $t < 0$, and if

$$x_1(t) \xleftrightarrow{\mathcal{L}_u} X_1(s) \quad \text{and} \quad x_2(t) \xleftrightarrow{\mathcal{L}_u} X_2(s)$$

then

$$x_1(t)x_2(t) \xleftrightarrow{\mathcal{L}_u} \frac{1}{2\pi j} [X_1(s) * X_2(s)] = \frac{1}{2\pi j} \int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} X_1(s_1)X_2(s - s_1) ds_1 \quad (9.63)$$

9.12.8 Integration in the s -Domain

If $x(t)$ satisfies the condition of existence $\int_{0^-}^{\infty} |x(t) e^{-\sigma t}| dt < \infty$, and the limit of $x(t)/t$, as t approaches 0 from the right, exists, and

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$\frac{x(t)}{t} \xleftrightarrow{\mathcal{L}_u} \int_s^{\infty} X(s) ds \quad (9.64)$$

Proof By definition

$$\mathcal{L}_u[x(t)] = X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt$$

Integrating both sides w.r.t. s , and taking the limit from s to ∞ , we get

$$\begin{aligned} \int_s^{\infty} X(s) ds &= \int_s^{\infty} \left(\int_{0^-}^{\infty} x(t) e^{-st} dt \right) ds \\ &= \int_{0^-}^{\infty} x(t) \left(\int_s^{\infty} e^{-st} ds \right) dt \\ &= \int_{0^-}^{\infty} x(t) \left(\frac{e^{-st}}{t} \right) dt \\ \int_s^{\infty} X(s) ds &= \int_{0^-}^{\infty} \left[\frac{x(t)}{t} \right] e^{-st} dt \\ \int_s^{\infty} X(s) ds &= \mathcal{L}_u \left[\frac{x(t)}{t} \right] \end{aligned}$$

Therefore,

$$\frac{x(t)}{t} \xleftrightarrow{\mathcal{L}_u} \int_s^{\infty} X(s) ds$$

Example 9.45 Find the Laplace transform of the signal

$$g(t) = \frac{1}{t} \sin(\omega t) u(t)$$

Solution

Given that

$$g(t) = \frac{1}{t} \sin(\omega t) u(t) = \frac{x(t)}{t}$$

where $x(t) = \sin(\omega t) u(t)$ and its Laplace transform is given by

$$X(s) = \frac{\omega}{s^2 + \omega^2}$$

Also,

$$g(t) = \frac{x(t)}{t}$$

Now, using integration in the s -domain property [Eq. (9.64)], we obtain

$$\mathcal{L}_u[g(t)] = G(s) = \mathcal{L}_u \left[\frac{x(t)}{t} \right] = \int_s^\infty X(s) ds = \int_s^\infty \frac{\omega}{s^2 + \omega^2} ds$$

$$G(s) = \tan^{-1} \left(\frac{s}{\omega} \right) \Big|_s^\infty$$

$$G(s) = \tan^{-1}(\infty) - \tan^{-1} \left(\frac{s}{\omega} \right)$$

$$G(s) = \frac{\pi}{2} - \tan^{-1} \left(\frac{s}{\omega} \right) \quad (9.65)$$

$$\tan^{-1} \left(\frac{s}{\omega} \right) = \frac{\pi}{2} - G(s)$$

$$\frac{s}{\omega} = \tan \left(\frac{\pi}{2} - G(s) \right)$$

$$\frac{s}{\omega} = \cot(G(s))$$

$$\frac{1}{\cot(G(s))} = \frac{1}{\frac{s}{\omega}}$$

$$\tan(G(s)) = \frac{\omega}{s}$$

$$G(s) = \tan^{-1} \left(\frac{\omega}{s} \right)$$

Therefore,

$$\frac{1}{t} \sin(\omega t) u(t) \longleftrightarrow \tan^{-1} \left(\frac{\omega}{s} \right) \quad (9.66)$$

Example 9.46 Show that

$$\frac{e^{-at} - e^{-bt}}{t} u(t) \xleftrightarrow{\mathcal{L}_u} \ln \left(\frac{s+b}{s+a} \right)$$

Solution

Let

$$g(t) = \frac{e^{-at} - e^{-bt}}{t} u(t) = \frac{x(t)}{t}$$

where $x(t) = [e^{-at} - e^{-bt}]u(t)$ and its Laplace transform is given by

$$X(s) = \frac{1}{s+a} - \frac{1}{s+b}$$

Also,

$$g(t) = \frac{x(t)}{t}$$

Now, using integration in the s -domain property [Eq. (9.64)], we obtain

$$\mathcal{L}_u[g(t)] = G(s) = \mathcal{L}_u \left[\frac{x(t)}{t} \right] = \int_s^\infty X(s) ds = \lim_{N \rightarrow \infty} \int_s^N X(s) ds$$

$$\begin{aligned} G(s) &= \lim_{N \rightarrow \infty} \int_s^N \left(\frac{1}{s+a} - \frac{1}{s+b} \right) ds \\ &= \lim_{N \rightarrow \infty} \left[\ln(s+a) \Big|_s^N - \ln(s+b) \Big|_s^N \right] \\ &= \lim_{N \rightarrow \infty} [\ln(N+a) - \ln(s+a) - \ln(N+b) + \ln(s+b)] \\ &= \lim_{N \rightarrow \infty} \left[\ln \left(\frac{N+a}{N+b} \right) + \ln \left(\frac{s+b}{s+a} \right) \right] \\ &= \lim_{N \rightarrow \infty} \left[\ln \left(\frac{1 + \frac{a}{N}}{1 + \frac{b}{N}} \right) \right] + \ln \left(\frac{s+b}{s+a} \right) \end{aligned}$$

$$G(s) = \ln(1) + \ln \left(\frac{s+b}{s+a} \right)$$

$$G(s) = \ln \left(\frac{s+b}{s+a} \right)$$

Therefore,

$$\frac{e^{-at} - e^{-bt}}{t} u(t) \longleftrightarrow \ln \left(\frac{s+b}{s+a} \right) \quad (9.67)$$

Example 9.47 Show that

$$\frac{e^{bt} - e^{at}}{t} u(t) \xleftrightarrow{\mathcal{L}_u} \ln \left(\frac{s-a}{s-b} \right)$$

Solution

Let

$$g(t) = \frac{e^{bt} - e^{at}}{t} u(t) = \frac{x(t)}{t}$$

where $x(t) = [e^{bt} - e^{at}]u(t)$ and its Laplace transform is given by

$$X(s) = \frac{1}{s-b} - \frac{1}{s-a}$$

Also,

$$g(t) = \frac{x(t)}{t}$$

Now, using integration in the s -domain property [Eq. (9.64)], we obtain

$$\mathcal{L}_u[g(t)] = G(s) = \mathcal{L}_u \left[\frac{x(t)}{t} \right] = \int_s^\infty X(s) ds = \lim_{N \rightarrow \infty} \int_s^N X(s) ds$$

$$G(s) = \lim_{N \rightarrow \infty} \int_s^N \left(\frac{1}{s-b} - \frac{1}{s-a} \right) ds$$

$$= \lim_{N \rightarrow \infty} \left[\ln(s-b) \Big|_s^N - \ln(s-a) \Big|_s^N \right]$$

$$G(s) = \lim_{N \rightarrow \infty} [\ln(N-b) - \ln(s-b) - \ln(N-a) + \ln(s-a)]$$

$$= \lim_{N \rightarrow \infty} \left[\ln \left(\frac{N-b}{N-a} \right) + \ln \left(\frac{s-a}{s-b} \right) \right]$$

$$= \lim_{N \rightarrow \infty} \left[\ln \left(\frac{1 - \frac{b}{N}}{1 - \frac{a}{N}} \right) \right] + \ln \left(\frac{s-a}{s-b} \right)$$

$$G(s) = \ln(1) + \ln \left(\frac{s-a}{s-b} \right)$$

$$G(s) = \ln \left(\frac{s-a}{s-b} \right)$$

Therefore,

$$\frac{e^{bt} - e^{at}}{t} u(t) \xleftrightarrow{\mathcal{L}_u} \ln \left(\frac{s-a}{s-b} \right) \quad (9.68)$$

9.12.9 Time Shifting

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$x(t - t_0) \xleftrightarrow{\mathcal{L}_u} X(s) e^{-st_0} \quad (9.69)$$

Observe that $x(t)$ starts at $t = 0$, and, therefore, $x(t - t_0)$ starts at $t = t_0$. So, this property may be restated as follows. If

$$x(t)u(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$x(t - t_0)u(t - t_0) \xleftrightarrow{\mathcal{L}_u} X(s) e^{-st_0} \quad (9.70)$$

Proof By definition

$$\mathcal{L}_u [x(t - t_0)u(t - t_0)] = \int_{0^-}^{\infty} x(t - t_0)u(t - t_0) e^{-st} dt$$

A change of variables is performed by letting $\tau = t - t_0$, which also yields $d\tau = dt$, $\tau \rightarrow -t_0$ as $t \rightarrow 0^-$, and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\mathcal{L}[x(t - t_0)u(t - t_0)] = \int_{-t_0}^{\infty} x(\tau)u(\tau) e^{-s(\tau+t_0)} d\tau$$

Since

$$u(\tau) = \begin{cases} 1, & \tau > 0 \\ 0, & \tau < 0 \end{cases}$$

we obtain

$$\begin{aligned} \mathcal{L}[x(t - t_0)u(t - t_0)] &= \int_0^{\infty} x(\tau) e^{-s(\tau+t_0)} d\tau \\ &= e^{-st_0} \int_0^{\infty} x(\tau) e^{-s\tau} d\tau \end{aligned}$$

$$\mathcal{L}[x(t - t_0)u(t - t_0)] = X(s) e^{-st_0}$$

9.12.10 Differentiation in the Time Domain

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$\frac{dx(t)}{dt} \xleftrightarrow{\mathcal{L}_u} sX(s) - x(0^-) \quad (9.71)$$

Repeated application of this property yields

$$\frac{d^2x(t)}{dt^2} \xleftrightarrow{\mathcal{L}_u} s^2X(s) - sx(0^-) - \dot{x}(0^-) \quad (9.72)$$

where $\dot{x}(0^-) = dx(t)/dt|_{t=0^-}$. In general,

$$\frac{d^nx(t)}{dt^n} \xleftrightarrow{\mathcal{L}_u} s^nX(s) - \sum_{k=1}^n s^{n-k}x^{(k-1)}(0^-) \quad (9.73)$$

where $x^{(r)}(0^-) = \frac{dx^r(t)}{dt^r}|_{t=0^-}$.

Proof From Eq. (9.54), we have

$$\mathcal{L}_u \left[\frac{dx(t)}{dt} \right] = \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

Integrating by parts, we obtain

$$\mathcal{L}_u \left[\frac{dx(t)}{dt} \right] = x(t) e^{-st} \Big|_{0^-}^{\infty} + s \int_{0^-}^{\infty} x(t) e^{-st} dt$$

For the Laplace transform $X(s)$ to exist, it is necessary that $x(t)e^{-st} \rightarrow 0$ as $t \rightarrow \infty$ for $\Re\{s\} > 0$. Thus,

$$\mathcal{L}_u \left[\frac{dx(t)}{dt} \right] = [0 - x(0^-)] + sX(s)$$

$$\mathcal{L}_u \left[\frac{dx(t)}{dt} \right] = sX(s) - x(0^-)$$

9.12.11 Integration in the Time Domain

If

$$x(t) \xleftrightarrow{\mathcal{L}_u} X(s)$$

then

$$\int_{0^-}^t x(\tau) d\tau \xleftrightarrow{\mathcal{L}_u} \frac{X(s)}{s} \quad \text{for } x(t) = 0, \quad t < 0 \quad (9.74)$$

and

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{L}_u} \frac{X(s)}{s} + \frac{\int_{-\infty}^{0^-} x(\tau) d\tau}{s} \quad (9.75)$$

Proof We know that if $x(t) = 0$ for $t < 0$, then

$$\int_{0^-}^{\infty} x(\tau) d\tau = x(t) * u(t)$$

Taking the unilateral Laplace transform on both sides of the above equation, we obtain

$$\begin{aligned} \mathcal{L}_u \left[\int_{0^-}^{\infty} x(\tau) d\tau \right] &= \mathcal{L}_u [x(t) * u(t)] \\ &= \frac{X(s)}{s} \end{aligned}$$

Therefore,

$$\int_{0^-}^{\infty} x(\tau) d\tau \xleftrightarrow{\mathcal{L}_u} \frac{X(s)}{s}$$

To prove Eq. (9.75), observe that

$$\int_{-\infty}^t x(\tau) d\tau = \int_{-\infty}^{0^-} x(\tau) d\tau + \int_{0^-}^t x(\tau) d\tau$$

Note that the first term on the RHS is a constant for $t \geq 0$. Taking the Laplace transform of the above equation, we obtain

$$\begin{aligned} \mathcal{L}_u \left[\int_{-\infty}^t x(\tau) d\tau \right] &= \mathcal{L}_u \left[\int_{-\infty}^{0^-} x(\tau) d\tau \right] + \mathcal{L}_u \left[\int_{0^-}^t x(\tau) d\tau \right] \\ &= \frac{\int_{-\infty}^{0^-} x(\tau) d\tau}{s} + \frac{X(s)}{s} \end{aligned}$$

Therefore,

$$\int_{-\infty}^t x(\tau) d\tau \xleftrightarrow{\mathcal{L}_u} \frac{X(s)}{s} + \frac{\int_{-\infty}^{0^-} x(\tau) d\tau}{s}$$

9.12.12 Initial-value Theorem

The initial-value theorem allows us to determine the initial value $x(0^+)$ of $x(t)$ directly from $X(s)$. The theorem states that if $x(t) = 0$, for $t < 0$, and if it contains no impulses or higher-order singularities at the origin, then

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s) \quad (9.76)$$

The initial-value theorem does not apply to rational functions $X(s)$ in which the order of the numerator polynomial is greater than or equal to that of the denominator polynomial.

Proof Method 1 Since $x(t) = 0$ for $t < 0$, $x(t) = x(t)u(t)$. Now, expanding $x(t)$ as a Taylor series at $t = 0^+$, we obtain

$$x(t) = \left[x(0^+) + x^{(1)}(0^+)t + \cdots + x^{(n)}(0^+)\frac{t^n}{n!} + \cdots \right] u(t)$$

where $x^{(n)}(0^+) = d^n x(t)/dt^n|_{t=0^+}$. Taking the Laplace transform of both sides yields

$$X(s) = \frac{x(0^+)}{s} + \frac{x^{(1)}(0^+)}{s^2} + \cdots + \frac{x^{(n)}(0^+)}{s^{n+1}} + \cdots \quad (9.77)$$

Multiplying both sides of Eq. (9.77) by s and taking the limit as $s \rightarrow \infty$, we obtain

$$\lim_{s \rightarrow \infty} sX(s) = x(0^+) + \lim_{s \rightarrow \infty} \frac{x^{(1)}(0^+)}{s} + \cdots + \lim_{s \rightarrow \infty} \frac{x^{(n)}(0^+)}{s^n} + \cdots$$

$$\lim_{s \rightarrow \infty} sX(s) = x(0^+)$$

As a generalization, multiplying both sides by s^{n+1} and taking the limit as $s \rightarrow \infty$ yields

$$x^{(n)}(0^+) = \lim_{s \rightarrow \infty} [s^{n+1}X(s) - s^n x(0^+) - s^{n-1}x^{(1)}(0^+) - \cdots - sx^{(n-1)}(0^+)] \quad (9.78)$$

This more general form of the initial-value theorem is simplified if $x^{(n)}(0^+) = 0$ for $n < N$. In that case,

$$x^{(N)}(0^+) = \frac{dx^N(t)}{dt^N}\bigg|_{t=0^+} = \lim_{s \rightarrow \infty} s^{N+1}X(s) \quad (9.79)$$

This property is useful, since it allows us to compute the initial value of the signal $x(t)$ and its derivative directly from the Laplace transform $X(s)$ without having to find the inverse $x(t)$.

Method 2 Taking the unilateral Laplace transform of $dx(t)/dt$, we obtain

$$\mathcal{L}_u \left[\frac{dx(t)}{dt} \right] = \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

Using differentiation in the time-domain property [Eq. (9.71)], we obtain

$$\begin{aligned} sX(s) - x(0^-) &= \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt \\ &= \int_{0^-}^{0^+} \frac{dx(t)}{dt} e^{-st} dt + \int_{0^+}^{\infty} \frac{dx(t)}{dt} e^{-st} dt \\ &= x(t) \Big|_{0^-}^{0^+} + \int_{0^+}^{\infty} \frac{dx(t)}{dt} e^{-st} dt \end{aligned}$$

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$$sX(s) - x(0^-) = x(0^+) - x(0^-) + \int_{0^+}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$sX(s) = x(0^+) + \int_{0^+}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$\lim_{s \rightarrow \infty} sX(s) = x(0^+) + \lim_{s \rightarrow \infty} \int_{0^+}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$\lim_{s \rightarrow \infty} sX(s) = x(0^+)$$

Example 9.48 Use the initial-value theorem to find the initial value of the signal corresponding to the Laplace transform

$$X(s) = \frac{s+1}{s(s+2)}$$

Verify that the answer obtained is correct.

Solution

From the initial-value theorem [Eq. (9.76)], we have

$$\begin{aligned} x(0^+) &= \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} s \left(\frac{s+1}{s(s+2)} \right) \\ &= \lim_{s \rightarrow \infty} \left(\frac{s+1}{s+2} \right) \end{aligned}$$

$$x(0^+) = \lim_{s \rightarrow \infty} \left(\frac{1 + (1/s)}{1 + (2/s)} \right)$$

$$x(0^+) = 1$$

Using partial fraction expansion, the Laplace transform can be written as

$$X(s) = \frac{1/2}{s} + \frac{1/2}{s+2}$$

The inverse transform of this equation yields

$$x(t) = \frac{1}{2}u(t) + \frac{1}{2}e^{-2t}u(t)$$

Therefore,

$$x(t)|_{t=0^-} = x(0^+) = \frac{1}{2} + \frac{1}{2} = 1$$

9.12.13 Final-value Theorem

The final-value theorem allows us to determine the final value $x(\infty)$ of $x(t)$ directly from $X(s)$. The theorem states that if $x(t) = 0$, for $t < 0$, and if it contains no impulses or higher-order singularities at the origin, then

$$\lim_{t \rightarrow \infty} x(t) = x(\infty) = \lim_{s \rightarrow 0} sX(s) \quad (9.80)$$

The final-value theorem is applicable only if the poles of $sX(s)$ are in the left half of the s -plane, with at most a single pole at $s = 0$. If $X(s)$ has a pole in the right half of the s -plane, $x(t)$ contains an exponentially growing term and $x(\infty)$ does not exist. If there is a pole on the imaginary axis, then $x(t)$ contains an oscillating term and $x(\infty)$ does not exist. However, if there is a pole at the origin, then $x(t)$ contains a constant term, and hence $x(\infty)$ exists and is a constant.

Proof Taking the unilateral Laplace transform of $dx(t)/dt$, we obtain

$$\mathcal{L}_u \left[\frac{dx(t)}{dt} \right] = \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

Using differentiation in the time-domain property [Eq. (9.71)], we obtain

$$sX(s) - x(0^-) = \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt$$

$$\begin{aligned} \lim_{s \rightarrow 0} [sX(s) - x(0^-)] &= \lim_{s \rightarrow 0} \int_{0^-}^{\infty} \frac{dx(t)}{dt} e^{-st} dt \\ &= x(t) \Big|_{0^-}^{\infty} \end{aligned}$$

$$\lim_{s \rightarrow 0} sX(s) - x(0^-) = x(\infty) - x(0^-)$$

$$\lim_{s \rightarrow 0} sX(s) = x(\infty)$$

■

Example 9.49 The final value of $x(t) = [2 + e^{-3t}]u(t)$ is obviously $x(\infty) = 2$. Show that this final value can be found with the final-value theorem.

Solution

Given that

$$x(t) = [2 + e^{-3t}]u(t)$$

The Laplace transform of the above equation yields

$$X(s) = \frac{2}{s} + \frac{1}{s+3}$$

Using the final-value theorem [Eq. (9.80)], we have

$$\lim_{t \rightarrow \infty} x(t) = x(\infty) = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} s \left[\frac{2}{s} + \frac{1}{s+3} \right]$$

$$x(\infty) = \lim_{s \rightarrow 0} \left[2 + \frac{s}{s+3} \right]$$

$$x(\infty) = 2$$

Example 9.50 Determine the initial and final values of $x(t)$ if its Laplace transform is given by

$$X(s) = \frac{10(2s+3)}{s(s^2+2s+5)}$$

Solution

From the initial-value theorem [Eq. (9.76)], we have

$$x(0^+) = \lim_{s \rightarrow \infty} sX(s) = \lim_{s \rightarrow \infty} s \frac{10(2s+3)}{s(s^2+2s+5)} = \lim_{s \rightarrow \infty} \frac{10(2s+3)}{s^2+2s+5} = 0$$

From the final-value theorem [Eq. (9.80)], we have

$$\lim_{t \rightarrow \infty} x(t) = x(\infty) = \lim_{s \rightarrow 0} sX(s) = \lim_{s \rightarrow 0} s \left[\frac{10(2s+3)}{s(s^2+2s+5)} \right] = \lim_{s \rightarrow 0} \left[\frac{10(2s+3)}{s^2+2s+5} \right] = 6$$

Example 9.51 Find the final value of the response $y(t)$ of a system whose transfer function is

$$H(s) = \frac{s+3}{s^2+4s+5}$$

when the system is excited by

(a) a unit step, i.e., $x(t) = u(t)$

(b) a unit impulse, i.e., $x(t) = \delta(t)$

Solution

(a) Given that $x(t) = u(t)$, so $X(s) = 1/s$. We know that [Eq. (9.50)]

$$Y(s) = X(s)H(s) = \frac{1}{s} \frac{s+3}{s^2+4s+5}$$

Since all the poles of $sY(s)$ lie in the left half of the s -plane, the final-value theorem is applicable. From the final-value theorem [Eq. (9.80)], we have

$$\lim_{t \rightarrow \infty} y(t) = y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s \left[\frac{1}{s} \frac{s+3}{s^2+4s+5} \right]$$

$$y(\infty) = \lim_{s \rightarrow 0} \left[\frac{s+3}{s^2+4s+5} \right]$$

$$y(\infty) = \frac{3}{5}$$

(b) Given that $x(t) = \delta(t)$, so $X(s) = 1$. We know that [Eq. (9.50)]

$$Y(s) = X(s)H(s) = \frac{s+3}{s^2+4s+5}$$

Since all the poles of $sY(s)$ lie in the left half of the s -plane, the final-value theorem is applicable. From the final-value theorem [Eq. (9.80)], we have

$$\begin{aligned} \lim_{t \rightarrow \infty} y(t) = y(\infty) &= \lim_{s \rightarrow 0} sY(s) \\ &= \lim_{s \rightarrow 0} s \left[\frac{s+3}{s^2+4s+5} \right] \\ y(\infty) &= 0 \end{aligned}$$

9.13 SOLUTION OF DIFFERENTIAL AND INTEGRO-DIFFERENTIAL EQUATIONS

One of the most common applications of the Laplace transform is to solve linear constant-coefficient differential equations. Such equations are used to model continuous-time LTI systems. Solving of these equations depends on the differentiation property of the Laplace transform. Because $dy^n(t)/dt^n \longleftrightarrow s^n Y(s)$, the Laplace transform of a differential equation is an algebraic equation that can be readily solved for $Y(s)$. The procedure is straightforward and systematic, and we summarize it in the following three steps:

1. For a given set of initial conditions, take the Laplace transform of both sides of the differential equation to obtain an algebraic equation in $Y(s)$.
2. Solve the algebraic equation for $Y(s)$.
3. Take the inverse Laplace transform to obtain $y(t)$.

Example 9.52 Solve the second-order linear differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

for the initial conditions $y(0^-) = 2$ and $dy(t)/dt|_{t=0^-} = \dot{y}(0^-) = 1$ and the input $x(t) = e^{-4t}u(t)$.

Solution

If $x(t) = e^{-4t}u(t)$, then $x(0^-) = 0$ and

$$X(s) = \frac{1}{s+4}$$

Now, consider the given differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

Taking the Laplace transform of the above equation, we obtain

$$[s^2Y(s) - sy(0^-) - \dot{y}(0^-)] + 5[sY(s) - y(0^-)] + 6Y(s) = [sX(s) - x(0^-)] + X(s)$$

$$[s^2Y(s) - 2s - 1] + 5[sY(s) - 2] + 6Y(s) = [sX(s) - 0] + X(s)$$

$$(s^2 + 5s + 6)Y(s) - (2s + 11) = X(s)(s + 1)$$

$$(s^2 + 5s + 6)Y(s) = (2s + 11) + \frac{s + 1}{s + 4}$$

$$(s + 2)(s + 3)Y(s) = \frac{2s^2 + 20s + 45}{s + 4}$$

$$Y(s) = \frac{2s^2 + 20s + 45}{(s + 2)(s + 3)(s + 4)}$$

Expanding the RHS into partial fractions yields

$$Y(s) = \frac{13/2}{s + 2} - \frac{3}{s + 3} - \frac{3/2}{s + 4}$$

Taking the inverse Laplace transform of the above equation, we obtain

$$y(t) = \left[\frac{13}{2} e^{-2t} - 3 e^{-3t} - \frac{3}{2} e^{-4t} \right] u(t)$$

9.13.1 Zero-input Response and Zero-state Response

The Laplace transform method for solving differential equations offers a clear separation between the *zero-input response* of the system to initial conditions and the *zero-state response* of the system associated with the input. The *zero-input response* represents the response of the system when the input is zero, i.e., $x(t) = 0$, and thus it is the result of internal system conditions (such as energy storage, initial conditions) alone. In contrast, the

zero-state response represents the system response to the external input $x(t)$ when the initial conditions are zero. The total response is defined as

$$\text{Total response} = \text{Zero-input response} + \text{Zero-state response} \quad (9.81)$$

Natural Response and Forced Response: In total response, the response due to the poles of the system is called *natural response* of the system. The response due to the poles of the input is called the *forced response* of the system.

Example 9.53 Use the unilateral Laplace transform to determine the output of a system represented by the differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + 6x(t)$$

in response to the input $x(t) = u(t)$. Assume that the initial conditions on the system are $y(0^-) = 1$ and $\dot{y}(0^-) = 2$. Identify the zero-state response $y_{zs}(t)$, of the system, and the zero-input response $y_{zi}(t)$.

Solution

If $x(t) = u(t)$, then $x(0^-) = 0$ and $X(s) = 1/s$. Now, consider the given differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + 6x(t)$$

Taking the Laplace transform of the above equation, we obtain

$$[s^2Y(s) - sy(0^-) - \dot{y}(0^-)] + 5[sY(s) - y(0^-)] + 6Y(s) = [sX(s) - x(0^-)] + 6X(s)$$

$$[s^2Y(s) - s - 2] + 5[sY(s) - 1] + 6Y(s) = [sX(s) - 0] + 6X(s)$$

$$(s^2 + 5s + 6)Y(s) - (s + 7) = X(s)(s + 6)$$

$$(s^2 + 5s + 6)Y(s) = \underbrace{(s + 7)}_{\text{initial condition terms}} + \underbrace{X(s)(s + 6)}_{\text{input terms}}$$

$$Y(s) = \underbrace{\frac{s + 7}{s^2 + 5s + 6}}_{\text{zero-input component}} + \underbrace{\frac{s + 6}{s(s^2 + 5s + 6)}}_{\text{zero-state component}}$$

$$Y(s) = \frac{s + 7}{(s + 2)(s + 3)} + \frac{s + 6}{s(s + 2)(s + 3)}$$

Using partial fraction expansion, we obtain

$$Y(s) = \left[\frac{5}{s + 2} - \frac{4}{s + 3} \right] + \left[\frac{1}{s} - \frac{2}{s + 2} + \frac{1}{s + 3} \right]$$

The inverse transform of this equation yields

$$y(t) = [5e^{-2t}u(t) - 4e^{-3t}u(t)] + [u(t) - 2e^{-2t}u(t) + e^{-3t}u(t)]$$

$$y(t) = \underbrace{[5e^{-2t} - 4e^{-3t}]u(t)}_{\text{Zero-input response or natural response}} + \underbrace{[1 - 2e^{-2t} + e^{-3t}]u(t)}_{\text{Zero-state response or forced response}}$$

$$y(t) = y_{zi}(t) + y_{zs}(t)$$

where the zero-input response is

$$y_{zi}(t) = [5e^{-2t} - 4e^{-3t}]u(t)$$

and the zero-state response is

$$y_{zs}(t) = [1 - 2e^{-2t} + e^{-3t}]u(t)$$

Example 9.54 Determine the forced response and natural response of an LTI system described by the following differential equation with specified input and initial conditions:

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = -4x(t) - 3\frac{dx(t)}{dt} \quad x(t) = e^{-t}u(t)$$

$$y(0^-) = -1 \quad \dot{y}(0^-) = 5$$

Solution

Consider the given differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = -4x(t) - 3\frac{dx(t)}{dt}$$

Taking the bilateral Laplace transform of the above equation to get the system function

$$H(s) = \frac{Y(s)}{X(s)}$$

$$s^2Y(s) + 5sY(s) + 6Y(s) = -4X(s) - 3sX(s)$$

$$(s^2 + 5s + 6)Y(s) + s = -X(s)(4 + 3s)$$

$$H(s) = \frac{Y(s)}{X(s)} = -\frac{4 + 3s}{s^2 + 5s + 6} = -\frac{4 + 3s}{(s + 2)(s + 3)}$$

The system function has poles at $p_1 = -2$, and $p_2 = -3$. The response due to these poles is known as natural response.

If $x(t) = e^{-t}u(t)$, then $x(0^-) = 0$ and

$$X(s) = \frac{1}{s + 1}$$

Now, consider the given differential equation

$$\frac{d^2y(t)}{dt^2} + 5\frac{dy(t)}{dt} + 6y(t) = -4x(t) - 3\frac{dx(t)}{dt}$$

Taking the Laplace transform of the above equation, we obtain

$$[s^2Y(s) - sy(0^-) - \dot{y}(0^-)] + 5[sY(s) - y(0^-)] + 6Y(s) = -4X(s) - 3[sX(s) - x(0^-)]$$

$$[s^2Y(s) + s - 5] + 5[sY(s) + 1] + 6Y(s) = -4X(s) - 3[sX(s) - 0]$$

$$(s^2 + 5s + 6)Y(s) + s = -X(s)(4 + 3s)$$

$$(s^2 + 5s + 6)Y(s) = \underbrace{-s}_{\text{Initial condition terms}} + \underbrace{-X(s)(4 + 3s)}_{\text{Input terms}}$$

$$Y(s) = \underbrace{-\frac{s}{s^2 + 5s + 6}}_{\text{Zero-input response}} + \underbrace{-\frac{4 + 3s}{(s + 1)(s^2 + 5s + 6)}}_{\text{Zero-state response}}$$

$$Y(s) = -\frac{s}{(s + 2)(s + 3)} - \frac{4 + 3s}{(s + 1)(s + 2)(s + 3)}$$

Using partial fraction expansion, we obtain

$$Y(s) = \left[\frac{2}{s + 2} - \frac{3}{s + 3} \right] + \left[-\frac{1/2}{s + 1} - \frac{2}{s + 2} + \frac{5/2}{s + 3} \right]$$

The inverse transform of this equation yields

$$y(t) = [2e^{-2t}u(t) - 3e^{-3t}u(t)] + \left[-\frac{1}{2}e^{-t}u(t) - 2e^{-2t}u(t) + \frac{5}{2}e^{-3t}u(t) \right]$$

$$y(t) = \underbrace{[2e^{-2t} - 3e^{-3t}]u(t)}_{\text{Natural response}} + \underbrace{\left[-\frac{1}{2}e^{-t} - 2e^{-2t} + \frac{5}{2}e^{-3t} \right]u(t)}_{\text{Forced response}}$$

$$y(t) = y_n(t) + y_f(t)$$

where the natural response is

$$y_n(t) = [2e^{-2t} - 3e^{-3t}]u(t)$$

and the forced response is

$$y_f(t) = \left[-\frac{1}{2}e^{-t} - 2e^{-2t} + \frac{5}{2}e^{-3t} \right]u(t)$$

Example 9.55 Consider the system S characterized by the differential equation

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t)$$

- Determine the zero-state response of this system for the input $x(t) = e^{-4t}u(t)$.
- Determine the zero-input response of the system for $t > 0^-$, given that

$$y(0^-) = 1, \quad \dot{y}(0^-) = -1, \quad \ddot{y}(0^-) = 1$$

- (c) Determine the total response of the system when the input is $x(t) = e^{-4t}u(t)$ and the initial conditions are the same as those specified in part (b).

Solution

(a) The zero-state response represents the system response to the external input $x(t)$ when the initial conditions are zero.

Given that $x(t) = e^{-4t}u(t)$, its Laplace transform $X(s) = 1/(s + 4)$. Now, consider the given differential equation

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t)$$

Taking the Laplace transform of the given differential equation, we obtain

$$s^3Y(s) + 6s^2Y(s) + 11sY(s) + 6Y(s) = X(s)$$

$$Y(s)[s^3 + 6s^2 + 11s + 6] = \frac{1}{s + 4}$$

$$Y(s) = \frac{1}{(s + 4)(s^3 + 6s^2 + 11s + 6)}$$

$$Y(s) = \frac{1}{(s + 1)(s + 2)(s + 3)(s + 4)}$$

Using partial fraction expansion, we obtain

$$Y(s) = \frac{1/6}{s + 1} + \frac{1/2}{s + 2} - \frac{1/2}{s + 3} - \frac{1/6}{s + 4}$$

Taking the inverse Laplace transform, we get the zero-state response

$$y_{zs}(t) = \frac{1}{6} e^{-t}u(t) + \frac{1}{2} e^{-2t}u(t) - \frac{1}{2} e^{-3t}u(t) - \frac{1}{6} e^{-4t}u(t)$$

(b) The zero-input response represents the system output when the input is zero, i.e., $x(t) = 0$ or, equivalently, $X(s) = 0$. Now, consider the given differential equation

$$\frac{d^3y(t)}{dt^3} + 6\frac{d^2y(t)}{dt^2} + 11\frac{dy(t)}{dt} + 6y(t) = x(t)$$

Taking the Laplace transform of the given differential equation, we obtain

$$[s^3Y(s) - s^2y(0^-) - s\dot{y}(0^-) - \ddot{y}(0^-)] + 6[s^2Y(s) - sy(0^-) - \dot{y}(0^-)]$$

$$+ 11[sY(s) - y(0^-)] + 6Y(s) = X(s)$$

$$[s^3Y(s) - s^2 + s - 1] + 6[s^2Y(s) - s + 1] + 11[sY(s) - 1] + 6Y(s) = 0$$

$$Y(s)[s^3 + 6s^2 + 11s + 6] - s^2 - 5s - 6 = 0$$

$$Y(s)[s^3 + 6s^2 + 11s + 6] = s^2 + 5s + 6$$

$$Y(s) = \frac{s^2 + 5s + 6}{s^3 + 6s^2 + 11s + 6}$$

$$Y(s) = \frac{(s+2)(s+3)}{(s+1)(s+2)(s+3)}$$

$$Y(s) = \frac{1}{s+1}$$

Taking the inverse Laplace transform, we get the zero-input response

$$y_n(t) = e^{-t}u(t)$$

(c) The total response is the sum of the zero-state and zero-input responses.

$$y(t) = y_f(t) + y_n(t) = \frac{1}{6} e^{-t}u(t) + \frac{1}{2} e^{-2t}u(t) - \frac{1}{2} e^{-3t}u(t) - \frac{1}{6} e^{-4t}u(t) + e^{-t}u(t)$$

$$y(t) = \frac{7}{6} e^{-t}u(t) + \frac{1}{2} e^{-2t}u(t) - \frac{1}{2} e^{-3t}u(t) - \frac{1}{6} e^{-4t}u(t)$$

9.14 BLOCK DIAGRAM REPRESENTATION

As we saw earlier, the Laplace transform is a useful tool for computing the system transfer function if the system is described by its differential equation or if the output is expressed explicitly in terms of the input. The situation changes considerably in cases where a large number of components or elements are interconnected to form the complete system. In such cases, it is convenient to represent the system by suitably interconnected subsystems, each of which can be separately and easily analysed. A linear system can be characterized by its transfer function $H(s)$. Figure 9.25(a) shows a block diagram of a system with a transfer function $H(s)$ and its input and output $X(s)$ and $Y(s)$, respectively. Subsystems may be interconnected by using cascade (series), parallel, and feedback interconnections, the three elementary types.

9.14.1 Cascade Interconnection

When the systems are connected in series, as depicted in Fig. 9.25(b), then the transfer function of the overall system is the product of the individual transfer functions. This result can also be proved by observing that in Fig. 9.25(b)

$$H(s) = \frac{Y(s)}{X(s)} = \frac{W(s)}{X(s)} \frac{Y(s)}{W(s)} = H_1(s)H_2(s)$$

We can extend this result to any number of systems connected in cascade. It follows from this discussion that the systems connected in cascade can be interchanged without affecting the overall transfer function. This commutation property of LTI systems follows from the commutative and associative properties of convolution (see Section 2.3).

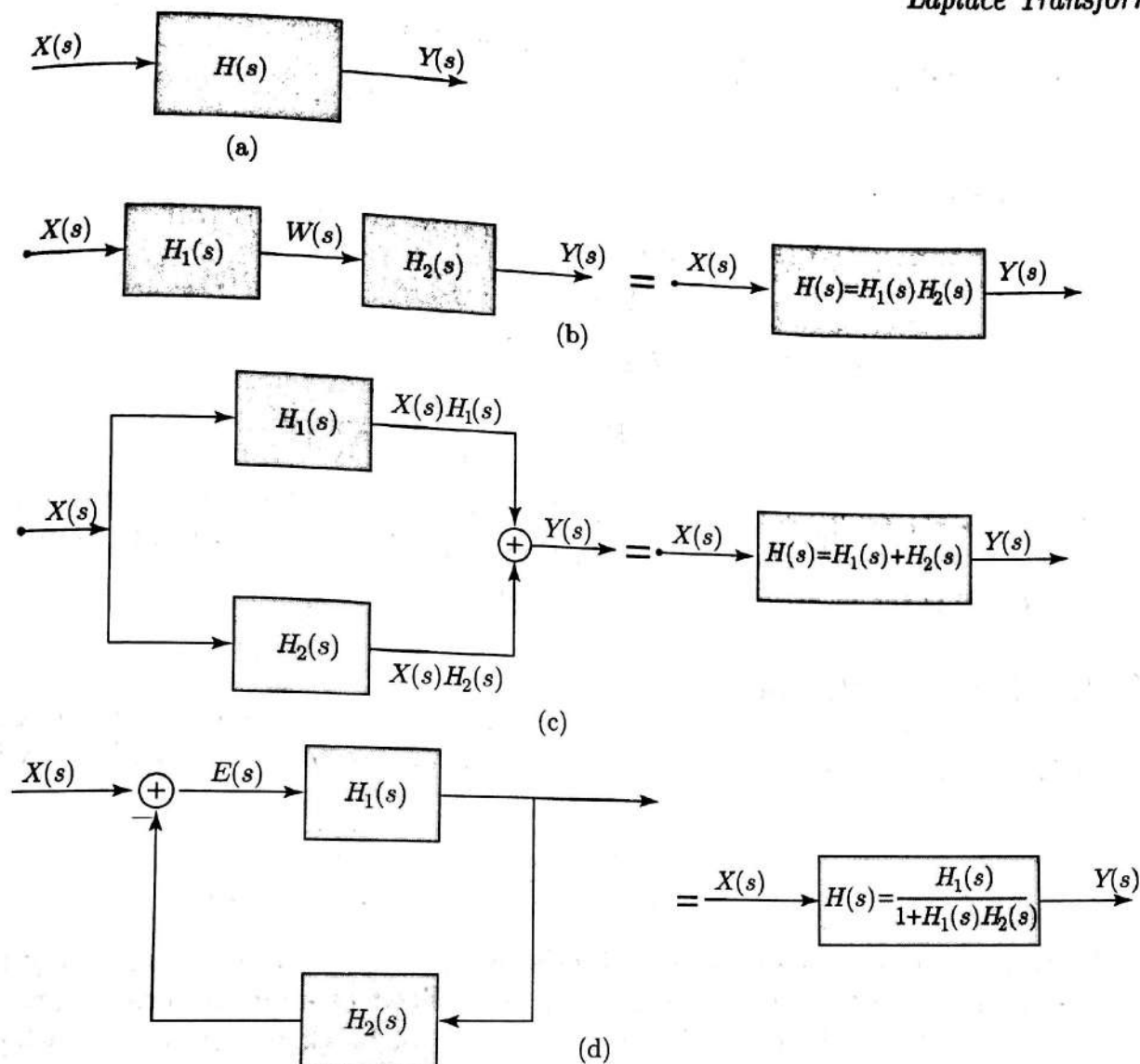


Fig. 9.25 Elementary connection of block diagrams

9.14.2 Parallel Interconnection

When the systems are connected in parallel, as depicted in Fig. 9.25(c), then the transfer function of the overall system is the sum of the individual transfer functions. This result can also be proved by observing that in Fig. 9.25(c)

$$Y(s) = X(s)H_1(s) + X(s)H_2(s)$$

$$Y(s) = X(s)[H_1(s) + H_2(s)]$$

$$\frac{Y(s)}{X(s)} = H(s) = H_1(s) + H_2(s)$$

We can extend this result to any number of systems connected in parallel.

9.14.3 Feedback Interconnection

When the output is fed back to the input, as shown in Fig. 9.25(d), the overall transfer function $H(s) = Y(s)/X(s)$ can be computed as follows. The input to the adder are $X(s)$

and $-H_2(s)Y(s)$. Therefore, $E(s)$, the output of the adder, is

$$E(s) = X(s) - H_2(s)Y(s)$$

and the output is given by

$$Y(s) = H_1(s)E(s) = H_1(s)[X(s) - H_2(s)Y(s)]$$

$$Y(s) = H_1(s)X(s) - H_1(s)H_2(s)Y(s)$$

$$Y(s)[1 + H_1(s)H_2(s)] = H_1(s)X(s)$$

$$\frac{Y(s)}{X(s)} = H(s) = \frac{H_1(s)}{1 + H_1(s)H_2(s)} \quad (9.82)$$

Therefore, the feedback loop can be replaced by a single block with the transfer function shown in Eq. (9.82). The transfer function $H_1(s)$ is the forward path transfer function and $H_2(s)$ is the feedback path transfer function.

9.15 SYSTEM REALIZATION

We now develop a systematic method for realization (or implementation) of an arbitrary N th-order transfer function. The most general transfer function with $M = N$ is given by

$$H(s) = \frac{b_0 s^N + b_1 s^{N-1} + \cdots + b_{N-1} s + b_N}{s^N + a_1 s^{N-1} + \cdots + a_{N-1} s + a_N}$$

Since realization is basically a synthesis problem, there is no unique way of realizing a system. A given transfer function can be realized in many different ways. A transfer function $H(s)$ can be conveniently represented in block diagram from using the basic building blocks representing the integrators or differentiators, the multiplier, the adder, and the pick-off node as shown in Fig. 9.26.

We avoid use of differentiators because

1. a differentiator represents an unstable system because a bounded input like the step input $u(t)$ results in an unbounded output $\delta(t)$.

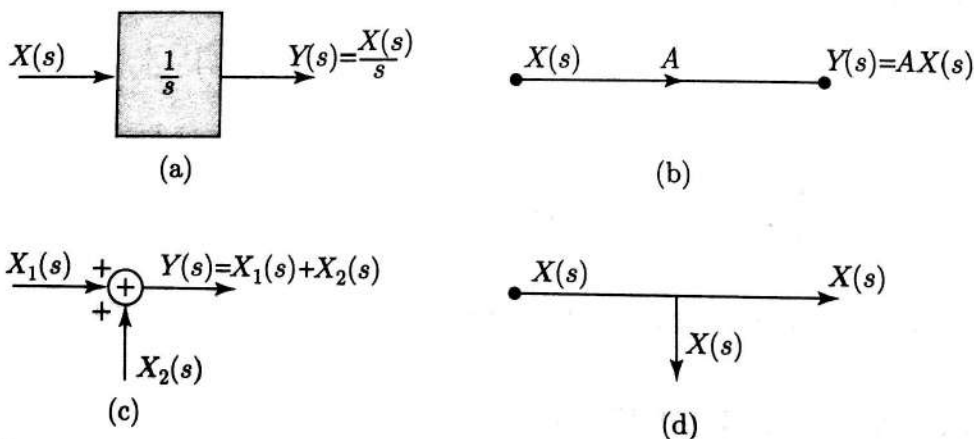


Fig. 9.26 (a) Basic building block representing (a) integrator, (b) multiplier, (c) adder, and (d) pick-off node

2. the noise is enhanced by a differentiator. Noise is a wideband signal containing components of all frequencies from 0 to a very high frequency approaching ∞ . Hence, noise contains a significant amount of rapidly varying components and we know that the derivative of any rapidly varying signal is high.

There are several advantages in representing the system in block diagram form:

1. It is easy to write down the computational algorithm by inspection.
2. It is easy to analyse the block diagram to determine the explicit relation between the output and the input.
3. It is easy to manipulate a block diagram to derive other equivalent block diagrams yielding different computational algorithms.
4. It is easy to determine the hardware requirements.
5. It is easier to develop block diagram representations from the transfer function directly leading to a variety of equivalent representations.

9.15.1 Direct Form I Realization

An N th-order transfer function is characterized by $2N + 1$ unique coefficients and, in general, requires $2N + 1$ multipliers and $2N$ two-input adders for implementation. Structures in which the multiplier coefficients are precisely the coefficients of the transfer function are called *direct form* structures.

Consider for simplicity a third-order system characterized by a transfer function

$$H(s) = \frac{b_0 s^3 + b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$$

$$H(s) = \frac{b_0 + (b_1/s) + (b_2/s^2) + (b_3/s^3)}{1 + (a_1/s) + (a_2/s^2) + (a_3/s^3)}$$

We can express $H(s)$ as

$$H(s) = \underbrace{\left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} \right)}_{H_1(s)} \underbrace{\left(\frac{1}{1 + a_1/s + (a_2/s^2) + (a_3/s^3)} \right)}_{H_2(s)}$$

We can realize $H(s)$ as a cascade of transfer function $H_1(s)$ followed by $H_2(s)$, as depicted in Fig. 9.27(a), where

$$H_1(s) = \frac{W(s)}{X(s)} = b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} \quad (9.83)$$

$$W(s) = \left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} \right) X(s) \quad (9.84)$$

$$\text{and} \quad H_2(s) = \frac{Y(s)}{W(s)} = \frac{1}{1 + (a_1/s) + (a_2/s^2) + (a_3/s^3)} \quad (9.85)$$

$$Y(s) = W(s) - \left(\frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3} \right) Y(s) \quad (9.86)$$

We shall first realize $H_1(s)$ given by Eq. (9.83). Equation (9.84) shows that the output $W(s)$ can be synthesized by adding the input $b_0X(s)$ to $b_1X(s)/s$, $b_2X(s)/s^2$, and $b_3X(s)/s^3$. Because the transfer function of an integrator is $1/s$, the signals $X(s)/s$, $X(s)/s^2$, and $X(s)/s^3$ can be obtained by successive integration of the input $x(t)$. The left-half section of Fig. 9.27(b) shows the realization of $H_1(s)$.

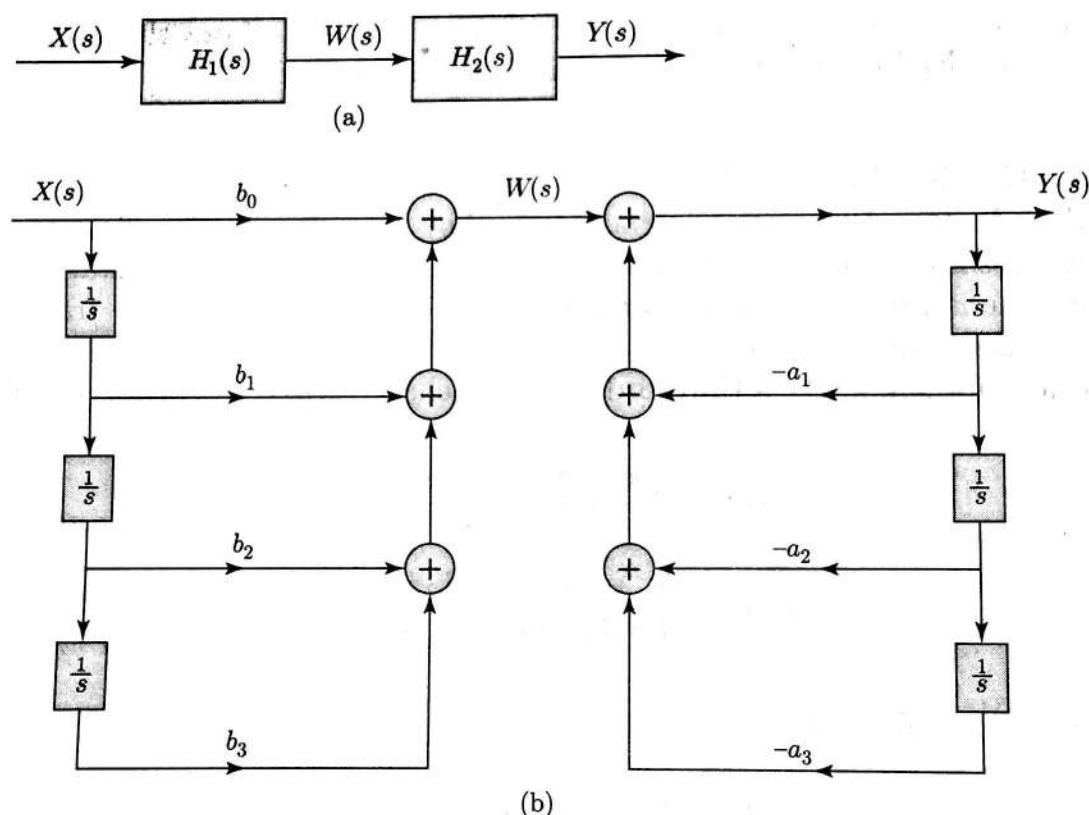


Fig. 9.27 (a) Realization of a function in two steps and (b) direct form I realization of a third-order continuous-time LTI system

We next consider the realization of $H_2(s)$ given by Eq. (9.85). Equation (9.86) shows that the output $Y(s)$ can be synthesized by subtracting $a_1Y(s)/s$, $a_2Y(s)/s^2$, and $a_3Y(s)/s^3$ from $W(s)$. To obtain signals $Y(s)/s$, $Y(s)/s^2$, and $Y(s)/s^3$, we assume that we already have the desired output $Y(s)$. Successive integration of $Y(s)$ yields the needed signals $Y(s)/s$, $Y(s)/s^2$, and $Y(s)/s^3$. The right-half section of Fig. 9.27(b) shows the realization of $H_2(s)$. We can generalize this procedure, known as the *direct form I* realization, for any value of N . This procedure requires $2N$ integrators to realize an N th-order transfer function. This realization is *noncanonic* since it employs six integrators to implement a third-order transfer function. A realization is *canonic* if the number of integrators

used in the realization is equal to the order of the transfer function realized. Thus, canonic realization has no redundant integrators.

9.15.2 Direct Form II (or Canonic) Realization

In the direct form I, we realize $H(s)$ by implementing $H_1(s)$ followed by $H_2(s)$, as shown in Fig. 9.27(a). We can also realize $H(s)$, as shown in Fig. 9.28(a), where $H_2(s)$ is followed by $H_1(s)$. This procedure is known as the *direct form II* realization.

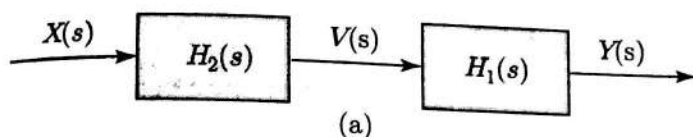


Fig. 9.28(a)

The direct form I realization [Fig. 9.27(b)] implements zeros first [the left-half section represented by $H_1(s)$] followed by realization of poles [the right-half section represented by $H_2(s)$] of $H(s)$. In contrast, the direct form II realization implements poles first followed by zeros.

Again consider for simplicity a third-order system characterized by a transfer function

$$H(s) = \frac{b_0 s^3 + b_1 s^2 + b_2 s + b_3}{s^3 + a_1 s^2 + a_2 s + a_3}$$

$$H(s) = \frac{b_0 + (b_1/s) + (b_2/s^2) + (b_3/s^3)}{1 + (a_1/s) + (a_2/s^2) + (a_3/s^3)}$$

We can express $H(s)$ as

$$H(s) = \underbrace{\left(\frac{1}{1 + (a_1/s) + (a_2/s^2) + (a_3/s^3)} \right)}_{H_2(s)} \underbrace{\left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} \right)}_{H_1(s)}$$

We can realize $H(s)$ as a cascade of transfer function $H_2(s)$ followed by $H_1(s)$, as depicted in Fig. 9.28(a), where

$$H_2(s) = \frac{V(s)}{X(s)} = \frac{1}{1 + (a_1/s) + (a_2/s^2) + (a_3/s^3)} \quad (9.87)$$

$$V(s) = X(s) - \left(\frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3} \right) V(s) \quad (9.88)$$

and

$$H_1(s) = \frac{Y(s)}{V(s)} = b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} \quad (9.89)$$

$$Y(s) = \left(b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} + \frac{b_3}{s^3} \right) V(s) \quad (9.90)$$

The left-half section of Fig. 9.28(b) shows the realization of $H_2(s)$ and the right-half section shows the realization of $H_1(s)$.

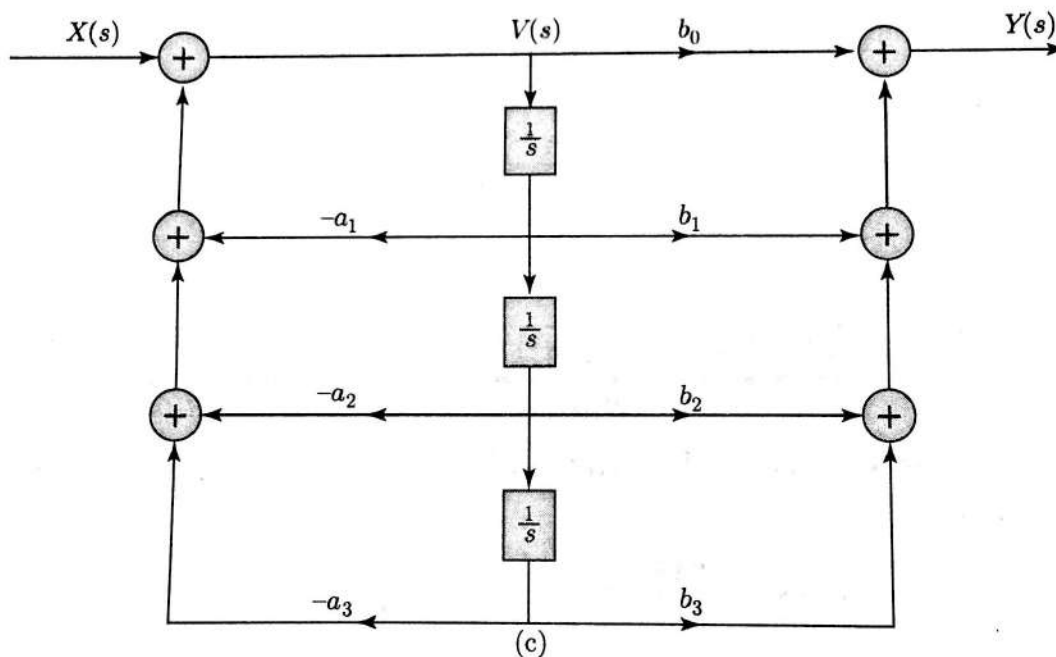
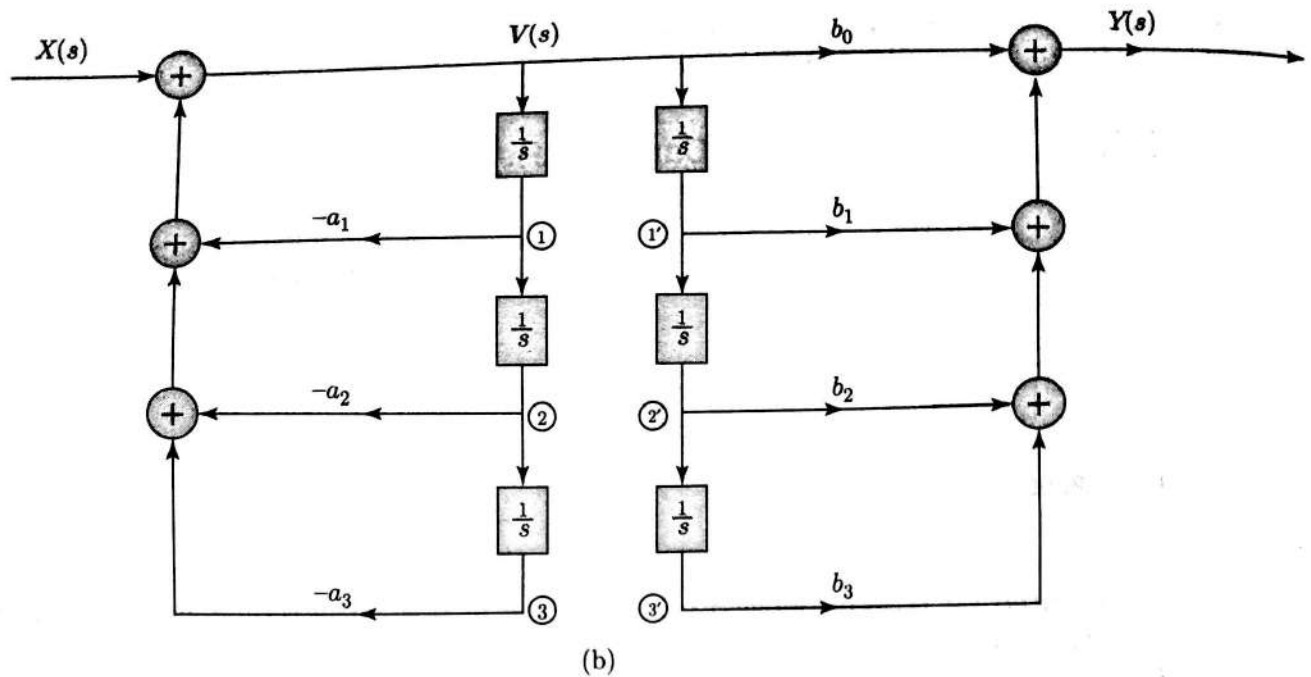


Fig. 9.28 Direct form II realization of a third-order LTI system

We observe that in Fig. 9.28(b), the signal variables at nodes 1 and 1' are the same, and hence the two top integrators can be shared. Likewise, the signal variables at nodes 2 and 2' are the same, which permits the sharing of the two middle integrators. Following the same argument, we can share the integrators, leading to the final structure shown in Fig. 9.28(c).

This implementation halves the number of integrators to N , and is thus more efficient in hardware utilization. This is the direct form II realization. This realization is *canonic* since it employs N integrators to implement an N th-order transfer function.

Example 9.56 Find the canonic realization of the following transfer functions:

$$(a) H(s) = \frac{5}{s+7}$$

$$(b) H(s) = \frac{s}{s+7}$$

$$(c) H(s) = \frac{s+5}{s+7}$$

$$(d) H(s) = \frac{4s+28}{s^2+6s+5}$$

Solution

(a) Given that

$$H(s) = \frac{Y(s)}{X(s)} = \frac{V(s)}{X(s)} \frac{Y(s)}{V(s)} = \frac{5}{s+7} = \frac{\frac{5}{s}}{1+7/s} = \underbrace{\left(\frac{1}{1+7/s} \right)}_{H_2(s)} \underbrace{\left(\frac{5}{s} \right)}_{H_1(s)}$$

where

$$H_2(s) = \frac{V(s)}{X(s)} = \frac{1}{1+7/s}$$

$$V(s) = X(s) - 7 \frac{V(s)}{s}$$

and

$$H_1(s) = \frac{Y(s)}{V(s)} = \frac{5}{s}$$

$$Y(s) = 5 \frac{V(s)}{s}$$

The given transfer function $H(s) = 5/(s+7)$ is of the first order; therefore, we need only one integrator for its realization, as shown in Fig. 9.29(a).

(b) Given that

$$H(s) = \frac{s}{s+7}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{1+7/s}$$

$$Y(s) = X(s) - 7 \frac{Y(s)}{s}$$

The given transfer function $H(s) = s/(s+7)$ is of the first order; therefore, we need only one integrator for its realization, as shown in Fig. 9.29(b).

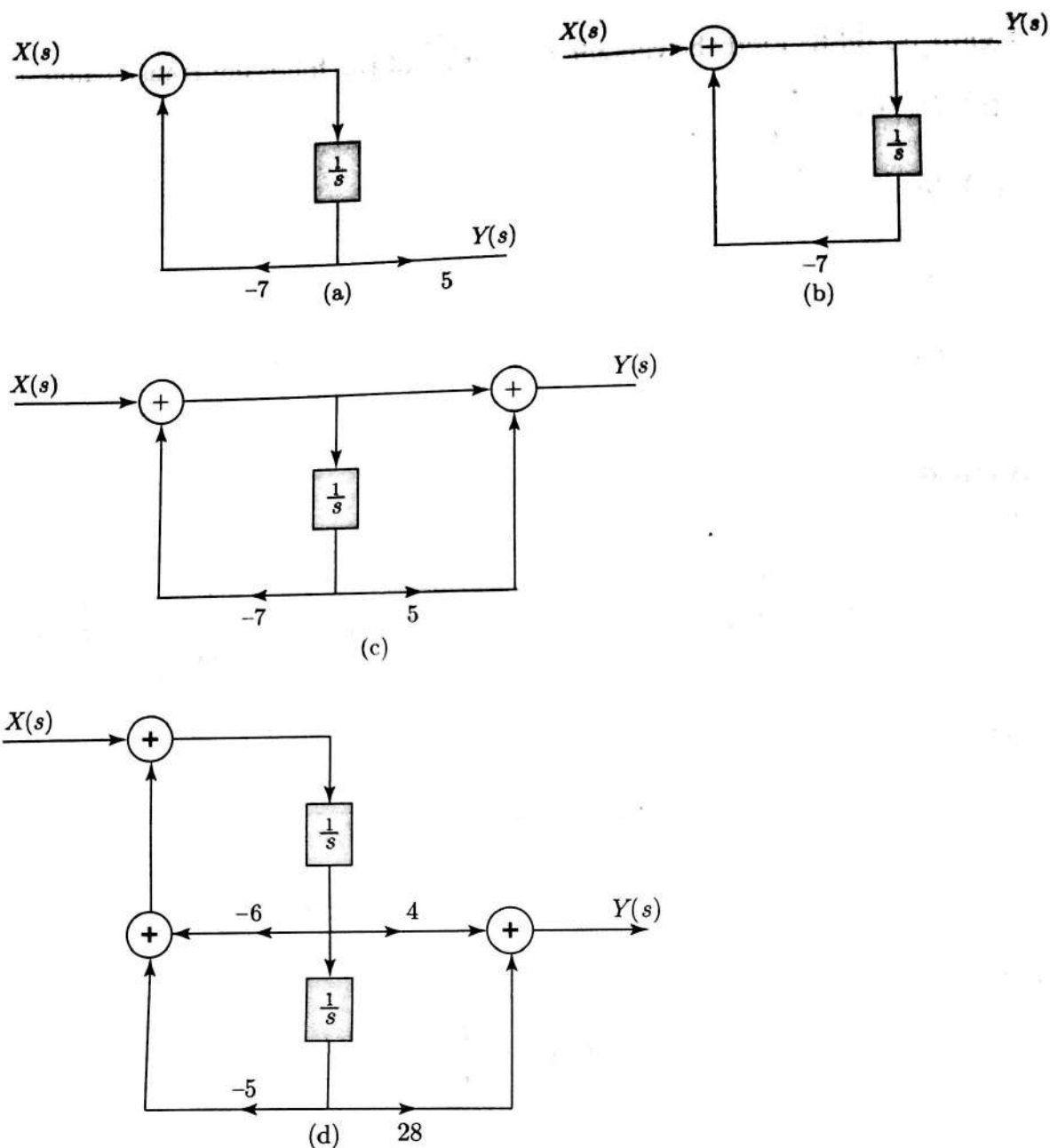


Fig. 9.29

(c) Given that

$$H(s) = \frac{Y(s)}{X(s)} = \frac{V(s) Y(s)}{X(s) V(s)} = \frac{s+5}{s+7} = \frac{1+\frac{5}{s}}{1+\frac{7}{s}} = \underbrace{\left(\frac{1}{1+\frac{7}{s}}\right)}_{H_2(s)} \underbrace{\left(1+\frac{5}{s}\right)}_{H_1(s)}$$

where

$$H_2(s) = \frac{V(s)}{X(s)} = \frac{1}{1+\frac{7}{s}}$$

$$V(s) = X(s) - 7 \frac{V(s)}{s}$$

and

$$H_1(s) = \frac{Y(s)}{V(s)} = 1 + \frac{5}{s}$$

$$Y(s) = V(s) + 5 \frac{V(s)}{s}$$

The given transfer function $H(s) = (s+5)/(s+7)$ is of the first order; therefore, we need only one integrator for its realization, as shown in Fig. 9.29(c).

(d) Given that

$$\begin{aligned} H(s) = \frac{Y(s)}{X(s)} &= \frac{V(s)}{X(s)} \frac{Y(s)}{V(s)} = \frac{4s+28}{s^2+6s+5} = \frac{\frac{4}{s} + (\frac{28}{s^2})}{1 + (\frac{6}{s}) + (\frac{5}{s^2})} \\ &= \underbrace{\left(\frac{1}{1 + (\frac{6}{s}) + (\frac{5}{s^2})} \right)}_{H_2(s)} \underbrace{\left(\frac{4}{s} + \frac{28}{s^2} \right)}_{H_1(s)} \end{aligned}$$

where

$$H_2(s) = \frac{V(s)}{X(s)} = \frac{1}{1 + (\frac{6}{s}) + (\frac{5}{s^2})}$$

$$V(s) = X(s) - 6 \frac{V(s)}{s} - 5 \frac{V(s)}{s^2}$$

and

$$H_1(s) = \frac{Y(s)}{V(s)} = \frac{4}{s} + \frac{28}{s^2}$$

$$Y(s) = 4 \frac{V(s)}{s} + 28 \frac{V(s)}{s^2}$$

The given transfer function $H(s) = \frac{4s+28}{s^2+6s+5}$ is of the second order; therefore, we need only two integrators for its realization, as shown in Fig. 9.29(d).

9.15.3 Cascade and Parallel Realization

By expressing the numerator polynomial $N(s)$ and the denominator polynomial $D(s)$ of the transfer function $H(s)$ as a product of polynomials of lower degree, a digital filter is often realized as a cascade of low-order systems. Consider, for example, $H(s) = N(s)/D(s)$ expressed as

$$H(s) = \frac{N_1(s)N_2(s)N_3(s)}{D_1(s)D_2(s)D_3(s)}$$

Various different cascade realizations of $H(s)$ can be obtained by different pole-zero polynomial pairings. Some examples of such realizations are shown in Fig. 9.30(a). Additional cascade realizations are obtained by simply changing the ordering of the systems. Figure 9.30(b) illustrates examples of different structures obtained by changing the ordering of the systems.

A system can be realized in a parallel form by making use of partial fraction expansion of the transfer function. From a practical viewpoint, parallel and cascade forms are preferable because parallel and certain cascade forms are numerically less sensitive than a canonic direct form to small parameter variations in the system.

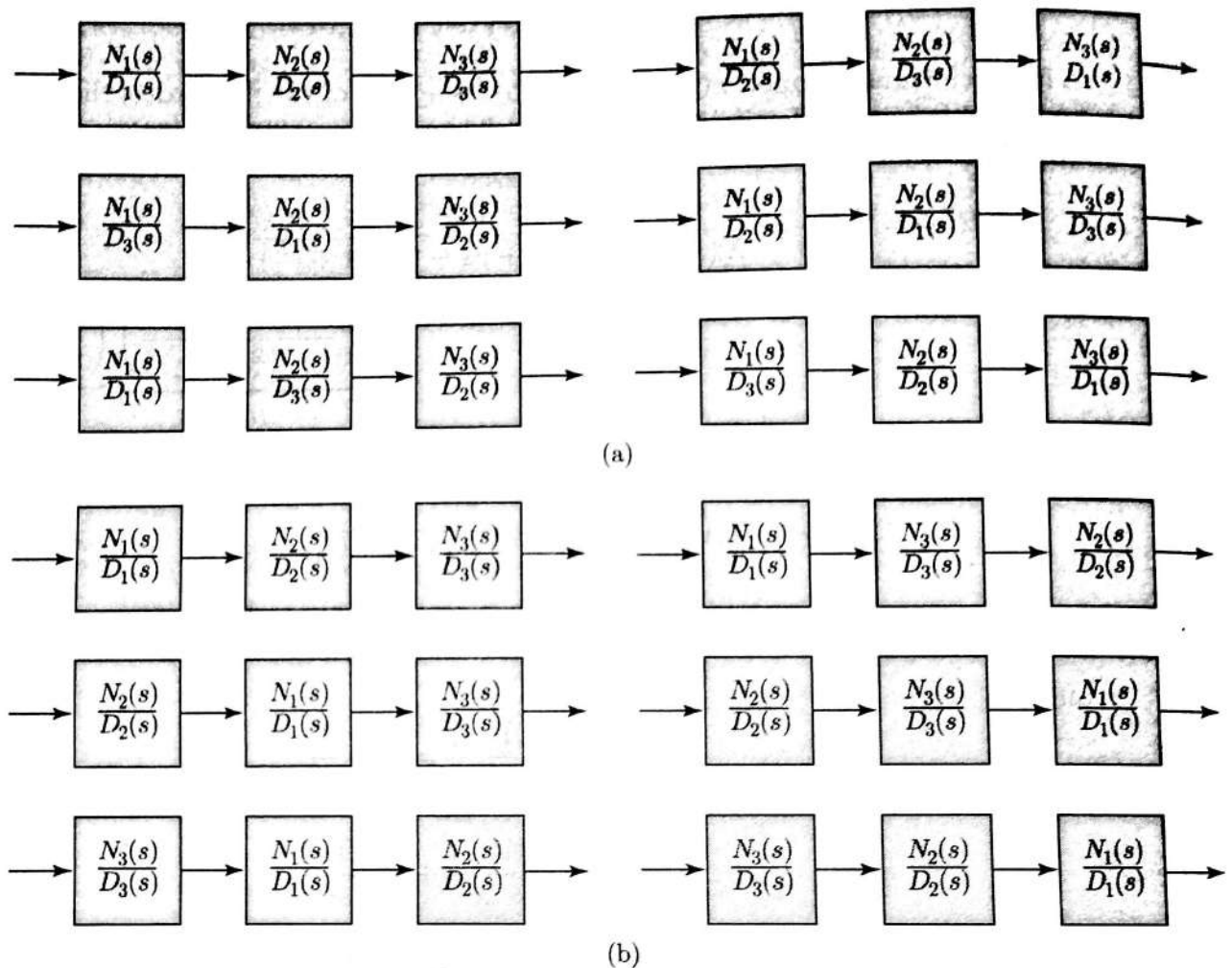


Fig. 9.30 Different cascade realizations obtained by (a) different pole-zero pairings and (b) changing the ordering of the systems

Example 9.57 Find the canonic, cascade, and parallel realizations of

$$H(s) = \frac{s+3}{s^2+7s+10}$$

Solution

Canonic realization Given that

$$\begin{aligned} H(s) &= \frac{Y(s)}{X(s)} = \frac{V(s)Y(s)}{X(s)V(s)} = \frac{s+3}{s^2+7s+10} = \frac{(1/s) + (3/s^2)}{1 + (7/s) + (10/s^2)} \\ &= \underbrace{\left(\frac{1}{1 + (7/s) + (10/s^2)} \right)}_{H_2(s)} \underbrace{\left(\frac{1}{s} + \frac{3}{s^2} \right)}_{H_1(s)} \end{aligned}$$

where

$$H_2(s) = \frac{V(s)}{X(s)} = \frac{1}{1 + (7/s) + (10/s^2)}$$

$$V(s) = X(s) - 7\frac{V(s)}{s} - 10\frac{V(s)}{s^2}$$

and

$$H_1(s) = \frac{Y(s)}{V(s)} = \frac{1}{s} + \frac{3}{s^2}$$

$$Y(s) = \frac{V(s)}{s} + 3\frac{V(s)}{s^2}$$

The given transfer function $H(s) = (s+3)/(s^2+7s+10)$ is of the second order; therefore, we need only two integrators for its realization, as shown in Fig. 9.31(a).

Cascade realization Given that

$$H(s) = \frac{s+3}{s^2+7s+10} = \frac{s+3}{(s+2)(s+5)} = \left(\frac{s+3}{s+2}\right) \left(\frac{1}{s+5}\right) = H_1(s)H_2(s)$$

where

$$H_1(s) = \frac{s+3}{s+2} = \frac{1+3/s}{1+2/s}$$

and

$$H_2(s) = \frac{1}{s+5} = \frac{1/s}{1+5/s}$$

We can realize $H(s)$ as a cascade of $H_1(s)$ and $H_2(s)$. Both $H_1(s)$ and $H_2(s)$ are first-order transfer functions and can be implemented by using canonic realization, as shown in Fig. 9.31(b).

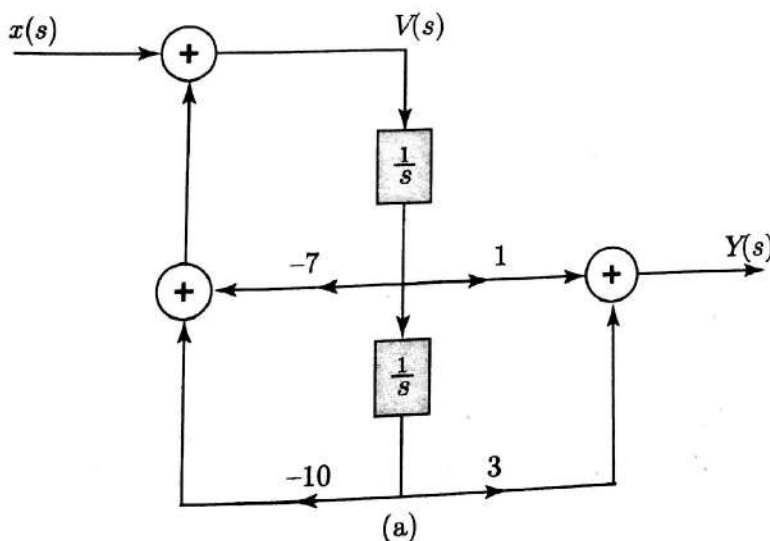


Fig. 9.31(a)

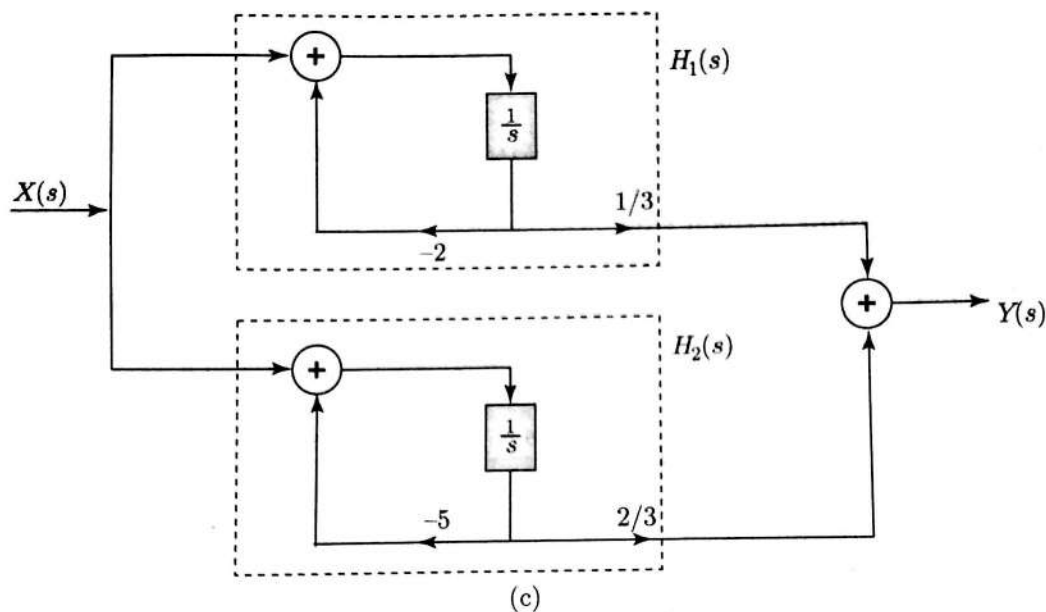
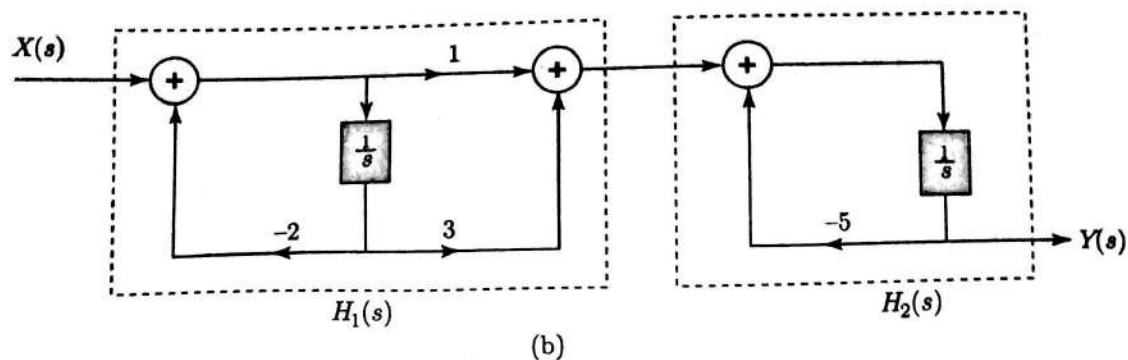


Fig. 9.31(b), (c)

Parallel realization We can express $H(s)$ as a sum of partial fractions as

$$H(s) = \frac{s+3}{(s+2)(s+5)} = \frac{1/3}{s+2} + \frac{2/3}{s+5} = H_1(s) + H_2(s)$$

where

$$H_1(s) = \frac{1/3}{s+2} = \frac{(1/3)(1/s)}{1+2/s}$$

and

$$H_2(s) = \frac{2/3}{s+5} = \frac{(2/3)(1/s)}{1+5/s}$$

We can realize $H(s)$ as a sum of $H_1(s)$ and $H_2(s)$. Both $H_1(s)$ and $H_2(s)$ are first-order transfer functions and can be implemented by using canonic realization, as shown in Fig. 9.31(c).

9.15.4 Transposed Realization

Two realizations are said to be *equivalent* if they have the same transfer function. A simple way to generate an equivalent structure from a given realization is via the *transpose operation*, which is as follows:

1. Reverse all paths.
2. Replace pick-off nodes by adders, and vice versa.
3. Interchange the input and output nodes.

Figure 9.32 shows the transposed version of the canonic direct form realization shown in Fig. 9.27(c) found according to the rules just listed.

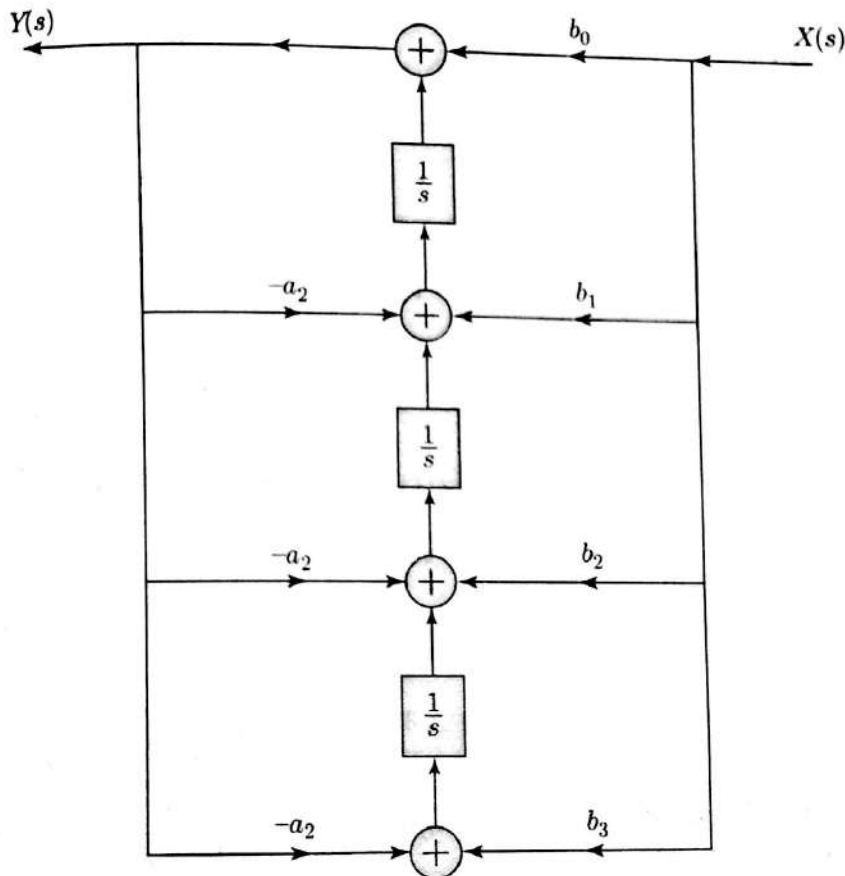


Fig. 9.32 Transposed version of the canonic direct form-II realization.

SOLVED EXAMPLES

Example 9.58 Determine the Laplace transform and the associated ROC for each of the following functions of time:

- (a) $x(t) = e^{-2t}u(t) + e^{-3t}u(t)$
- (b) $x(t) = e^{-4t}u(t) + e^{-5t} \sin(5t)u(t)$
- (c) $g(t) = t e^{-2|t|}$
- (d) $x(t) = |t| e^{-2|t|}$
- (e) $x(t) = |t| e^{2t}u(-t)$
- (f) $x(t) = \delta(t) + u(t)$
- (g) $x(t) = \delta(3t) + u(3t)$

Solution

(a) Consider the given signal

$$x(t) = e^{-2t}u(t) + e^{-3t}u(t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[e^{-2t}u(t) + e^{-3t}u(t)]$$

$$X(s) = \mathcal{L}[e^{-2t}u(t)] + \mathcal{L}[e^{-3t}u(t)]$$

From Eq. (9.10), we have

$$e^{-at}u(t) \longleftrightarrow \frac{1}{s+a} \quad \Re\{s\} > -a$$

Therefore,

$$e^{-2t}u(t) \longleftrightarrow \frac{1}{s+2} \quad \Re\{s\} > -2$$

$$e^{-3t}u(t) \longleftrightarrow \frac{1}{s+3} \quad \Re\{s\} > -3$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > -2$, and thus, we obtain

$$X(s) = \frac{1}{s+2} + \frac{1}{s+3} \quad \Re\{s\} > -2$$

$$X(s) = \frac{2s+5}{s^2+5s+6} \quad \Re\{s\} > -2$$

(b) Consider the given signal

$$x(t) = e^{-4t}u(t) + e^{-5t} \sin(5t)u(t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[e^{-4t}u(t) + e^{-5t} \sin(5t)u(t)]$$

$$X(s) = \mathcal{L}[e^{-4t}u(t)] + \mathcal{L}[e^{-5t} \sin(5t)u(t)]$$

From Eq. (9.10), we obtain

$$e^{-4t}u(t) \longleftrightarrow \frac{1}{s+4} \quad \Re\{s\} > -4$$

and from Eq. (9.27), we have

$$e^{-at} \sin(\omega_0 t)u(t) \longleftrightarrow \frac{\omega_0}{(s+a)^2 + \omega_0^2} \quad \Re\{s\} > -a$$

Therefore,

$$e^{-5t} \sin(5t)u(t) \longleftrightarrow \frac{5}{(s+5)^2 + 25} \quad \Re\{s\} > -5$$

The set of values of $\Re\{s\}$ for which the Laplace transforms of both terms converge is $\Re\{s\} > -4$, and thus, we obtain

$$X(s) = \frac{1}{s+4} + \frac{5}{(s+5)^2 + 25} \quad \Re\{s\} > -4$$

$$X(s) = \frac{s^2 + 15s + 70}{(s+4)(s^2 + 10s + 50)} \quad \Re\{s\} > -4$$

(c) Consider the given signal

$$g(t) = t e^{-2|t|} = tx(t)$$

where

$$x(t) = e^{-2|t|}$$

From Eq. (9.19), we have

$$e^{-a|t|} \longleftrightarrow \frac{-2a}{s^2 - a^2} \quad -a < \Re\{s\} < a$$

Therefore,

$$\underbrace{e^{-2|t|}}_{x(t)} \longleftrightarrow \underbrace{\frac{-4}{s^2 - 4}}_{X(s)} \quad -2 < \Re\{s\} < 2$$

Now, taking the Laplace transform of the given signal $g(t)$ and using differentiation in the s -domain property, we obtain

$$\mathcal{L}[g(t)] = \mathcal{L}[tx(t)]$$

$$G(s) = -\frac{dX(s)}{ds}$$

$$= -\frac{d(-4/s^2 - 4)}{ds}$$

$$G(s) = -\frac{8s}{(s^2 - 4)^2}$$

(d) Given that

$$x(t) = |t| e^{-2|t|}$$

$$= -t e^{2t}u(-t) + t e^{-2t}u(t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[-t e^{2t}u(-t)] + \mathcal{L}[t e^{-2t}u(t)]$$

We know that

$$-e^{2t}u(-t) \longleftrightarrow \frac{1}{s-2} \quad \Re\{s\} < 2$$

$$t[-e^{2t}u(-t)] \longleftrightarrow -\frac{d}{ds} \left(\frac{1}{s-2} \right)$$

$$t[-e^{2t}u(-t)] \longleftrightarrow \frac{1}{(s-2)^2} \quad \Re\{s\} < 2$$

and

$$e^{-2t}u(t) \longleftrightarrow \frac{1}{s+2} \quad \Re\{s\} > -2$$

$$t[e^{-2t}u(t)] \longleftrightarrow -\frac{d}{ds} \left(\frac{1}{s+2} \right)$$

$$t[e^{-2t}u(t)] \longleftrightarrow \frac{1}{(s+2)^2} \quad \Re\{s\} > -2$$

Therefore,

$$X(s) = \frac{1}{(s-2)^2} + \frac{1}{(s+2)^2} \quad -2 < \Re\{s\} < 2$$

$$X(s) = \frac{2s^2 + 8}{(s^2 - 4)^2} \quad -2 < \Re\{s\} < 2$$

(e) Given that

$$x(t) = |t| e^{2t} u(-t) \\ = -t e^{2t} u(-t)$$

We know that

$$-e^{2t} u(-t) \longleftrightarrow \frac{1}{s-2} \quad \Re\{s\} < 2$$

$$t[-e^{2t} u(-t)] \longleftrightarrow -\frac{d}{ds} \left(\frac{1}{s-2} \right)$$

$$t[-e^{2t} u(-t)] \longleftrightarrow \frac{1}{(s-2)^2} \quad \Re\{s\} < 2$$

Therefore,

$$X(s) = \frac{1}{(s-2)^2} \quad \Re\{s\} < 2$$

(f) Given that

$$x(t) = \delta(t) + u(t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[\delta(t)] + \mathcal{L}[u(t)]$$

$$X(s) = 1 + \frac{1}{s} \quad \Re\{s\} > 0$$

$$X(s) = \frac{s+1}{s} \quad \Re\{s\} > 0$$

(g) Given that

$$x(t) = \delta(3t) + u(3t)$$

$$= \frac{1}{3} \delta(t) + u(t)$$

$$\mathcal{L}[x(t)] = \frac{1}{3} \mathcal{L}[\delta(t)] + \mathcal{L}[u(t)]$$

$$X(s) = \frac{1}{3} + \frac{1}{s} \quad \Re\{s\} > 0$$

$$X(s) = \frac{s+3}{3s} \quad \Re\{s\} > 0$$

Example 9.59 Determine the Laplace transform of the following signals:

(a) $x(t) = \cos^3(3t)u(t)$

(b) $x(t) = (1 + 0.5 \sin(t)) \sin(1000t)u(t)$

Solution

(a) Given that

$$x(t) = \cos^3(3t)u(t)$$

$$x(t) = \frac{1}{4} \cos(9t)u(t) + \frac{3}{4} \cos(3t)u(t)$$

Since $\cos^3(x) = 1/4 \cos(3x) + 3/4 \cos(x)$, we have

$$\mathcal{L}[x(t)] = \mathcal{L}\left[\frac{1}{4} \cos(9t)u(t)\right] + \mathcal{L}\left[\frac{3}{4} \cos(3t)u(t)\right]$$

$$X(s) = \frac{1}{4} \frac{s}{s^2 + 81} + \frac{3}{4} \frac{s}{s^2 + 9}$$

$$X(s) = \frac{1}{4} \left(\frac{s}{s^2 + 81} + \frac{3s}{s^2 + 9} \right)$$

(b) Given that

$$x(t) = [1 + 0.5 \sin(t)] \sin(1000t)u(t)$$

$$= \sin(1000t)u(t)$$

$$+ 0.5 \sin(t) \sin(1000t)u(t)$$

$$x(t) = \sin(1000t)u(t) + 0.25 \cos(999t)u(t)$$

$$- 0.25 \cos(1001t)u(t)$$

$$\mathcal{L}[x(t)] = \mathcal{L}[\sin(1000t)u(t)]$$

$$+ 0.25 \mathcal{L}[\cos(999t)u(t)]$$

$$- 0.25 \mathcal{L}[\cos(1001t)u(t)]$$

$$X(s) = \frac{1000}{s^2 + 10^6} + 0.25 \frac{s}{s^2 + (999)^2}$$

$$- 0.25 \frac{s}{s^2 + (1001)^2}$$

Example 9.60 Find the unilateral Laplace transform of

$$x(t) = \frac{1}{\sqrt{t}} u(t) = \begin{cases} \frac{1}{\sqrt{t}}, & t > 0 \\ 0, & t < 0 \end{cases}$$

Solution

Given that

$$x(t) = \frac{1}{\sqrt{t}} u(t) = t^{-1/2} u(t)$$

We know that [Eq. (9.54)]

$$\begin{aligned} X(s) &= \int_{0^-}^{\infty} x(t) e^{-st} dt \\ &= \int_{0^-}^{\infty} t^{-1/2} u(t) e^{-st} dt \\ X(s) &= \int_0^{\infty} t^{-1/2} e^{-st} dt \end{aligned}$$

A change of variables is performed by letting $\tau = st$, which also yields $d\tau = dt$, $\tau \rightarrow 0$ as $t \rightarrow 0$, and $\tau \rightarrow \infty$ as $t \rightarrow \infty$. Therefore,

$$\begin{aligned} X(s) &= \frac{1}{\sqrt{s}} \int_0^{\infty} \tau^{-1/2} e^{-\tau} d\tau \\ &= \frac{1}{\sqrt{s}} \Gamma\left(\frac{1}{2}\right) \\ X(s) &= \sqrt{\frac{\pi}{s}} \end{aligned}$$

Example 9.61 An absolutely integrable signal $x(t)$ is known to have a pole at $s = 2$. Answer the following questions:

- Could $x(t)$ be of finite duration?
- Could $x(t)$ be left sided?
- Could $x(t)$ be right sided?
- Could $x(t)$ be two sided?

Solution

(a) No. We know that for a finite duration signal, the ROC is the entire s -plane. Therefore, there can be no poles in finite s -plane for a finite duration signal. Since, the given signal $x(t)$ has a pole at $s = 2$, clearly, in this problem this is not the case.

(b) Yes. Since the signal is absolutely integrable, the ROC must include the $j\omega$ -axis. Furthermore, $X(s)$ has a pole at $s = 2$. Therefore, one valid ROC for the signal would be $\Re\{s\} < 2$. Since the ROC lies left to a pole, it would correspond to a left-sided signal.

(c) No. Since the signal is absolutely integrable, the ROC must include the $j\omega$ -axis. Furthermore, $X(s)$

has a pole at $s = 2$. For a right-sided signal ROC is of the form $\Re\{s\} > a$ for some constant a and ROC does not contain any pole. Therefore, $x(t)$ cannot be right-sided signal.

(d) Yes. Since the signal is absolutely integrable, the ROC must include the $j\omega$ -axis. Furthermore, $X(s)$ has a pole at $s = 2$. Therefore, a valid ROC for the signal could be a strip in the region $a < \Re\{s\} < 2$ such that $a < 0$ and we know that this would correspond to a two-sided signal.

Example 9.62 How many signals have a Laplace transform that may be expressed as

$$X(s) = \frac{s-1}{(s+2)(s+3)(s^2+s+1)}$$

in its region of convergence?

Solution

We may find different signals with the given Laplace transform by choosing different region of convergence. Consider the given Laplace transform

$$\begin{aligned} X(s) &= \frac{s-1}{(s+2)(s+3)(s^2+s+1)} \\ &= \frac{s-1}{(s+2)(s+3)\left(s+\frac{1}{2}-j\sqrt{3}/2\right)\left(s+\frac{1}{2}+j\sqrt{3}/2\right)} \end{aligned}$$

The poles of the given Laplace transform are

$$\begin{aligned} s_1 &= -2 \quad s_2 = -3 \\ s_3 &= -\frac{1}{2} + j\frac{\sqrt{3}}{2} \quad s_4 = -\frac{1}{2} - j\frac{\sqrt{3}}{2} \end{aligned}$$

The pole-zero diagram is shown in Fig. 9.33. Based on the location of these poles, we may choose from the following ROCs:

- $\Re\{s\} > -1/2$
- $-2 < \Re\{s\} < -1/2$
- $-3 < \Re\{s\} < -2$
- $\Re\{s\} < -3$

Therefore, we may find four different signals with the given Laplace transform.

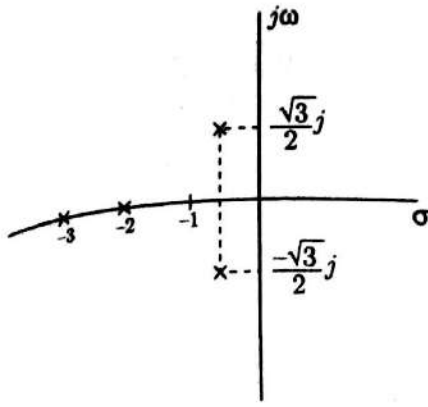


Fig. 9.33 Pole-zero plot for Example 9.62

Example 9.63 Let $x(t)$ be a signal that has a rational Laplace transform with exactly two poles located at $s = -1$ and $s = -3$. If $g(t) = e^{2t}x(t)$ and $G(\omega) = \mathcal{F}[g(t)]$ converges, determine whether $x(t)$ is left sided, right sided, or two sided.

Solution

Using shifting in the s -domain property, we know that if

$$x(t) \longleftrightarrow X(s) \quad \text{with ROC} = R$$

then

$$\underbrace{e^{2t}x(t)}_{g(t)} \longleftrightarrow \underbrace{X(s-2)}_{G(s)} \quad \text{with ROC} = R+2$$

That is, the ROC of $G(s)$ is the ROC of $X(s)$ shifted to the right by 2.

We are also given that $X(s)$ has exactly two poles, located at

$$s_1 = -1 \quad \text{and} \quad s_2 = -3$$

Since $G(s) = x(s-2)$, $G(s)$ also has exactly two poles, located at

$$s_1 = -1 + 2 = 1 \quad \text{and} \quad s_2 = -3 + 2 = -1$$

Since, we are given that $G(\omega) = \mathcal{F}[g(t)]$ converges, we may infer that the $j\omega$ -axis lies in the ROC of $G(s)$. That is, the ROC of $G(s)$ is the region $-1 < \Re\{s\} < 1$ in the s -plane. Therefore, we may conclude that $g(t)$ is a two-sided signal and hence $x(t) = e^{-2t}g(t)$ will also be a two-sided signal.

Example 9.64 Suppose we are given the following three facts about the signal $x(t)$:

1. $x(t) = 0$, for $t < 0$.
2. $x\left(\frac{k}{80}\right) = 0$, for $k = 1, 2, 3, \dots$

$$3. \quad x\left(\frac{1}{160}\right) = e^{-120}.$$

Let $X(s)$ denote the Laplace transform of $x(t)$, and determine which of the following statements is consistent with the given information about $x(t)$:

- (a) $X(s)$ has only one pole in the finite s -plane.
- (b) $X(s)$ has only two poles in the finite s -plane.
- (c) $X(s)$ has more than two poles in the finite s -plane.

Solution

(a) If $X(s)$ has only one pole in the finite s -plane, then $x(t)$ would be of the form $A e^{-at}u(t)$. Clearly, such a signal violates condition 2. Therefore, this statement is inconsistent with the given information.

(b) If $X(s)$ has only two poles in the finite s -plane, then $x(t)$ would be of the form $A e^{-at} \sin(\omega_0 t)u(t)$. Clearly, such a signal could be made to satisfy all three conditions:

(1) $x(t) = 0$, for $t < 0$ because of $u(t)$.

(2) $x\left(\frac{k}{80}\right) = 0$, for $k = 1, 2, 3, \dots$, if we choose $\omega_0 = 80\pi$.

(3) $x(t) = A e^{-at} \sin(\omega_0 t)u(t)$

$$x\left(\frac{1}{160}\right) = A e^{-a/160} \sin\left(80\pi \frac{1}{160}\right) u\left(\frac{1}{160}\right)$$

$$e^{-120} = A e^{-a/160} \sin\left(\frac{\pi}{2}\right)$$

$$e^{-120} = A e^{-a/160}$$

Thus,

$$a \frac{1}{160} = 120$$

$$a = 19200$$

Therefore, this statement is consistent with the given information.

(c) If $X(s)$ has more than two poles in the finite s -plane, then $x(t)$ would be of the form $A e^{-at} \sin(\omega_0 t)u(t) + B e^{-bt} \sin(\omega_0 t)u(t)$. Clearly, such a signal could be made to satisfy all the three conditions. Therefore, this statement is consistent with the given information.

Example 9.65 Suppose the following facts are given about the signal $x(t)$ with Laplace transform $X(s)$:

1. $x(t)$ is real and even

2. $X(s)$ has four poles and no zeros in the finite s -plane
3. $X(s)$ has a pole at $s = \frac{1}{2} e^{j\pi/4}$
4. $\int_{-\infty}^{\infty} x(t) dt = 4$

Determine $X(s)$ and its ROC.

Solution

Since $X(s)$ has four poles and no zeros in the finite s -plane, we may assume that $X(s)$ is of the form

$$X(s) = \frac{A}{(s-a)(s-b)(s-c)(s-d)}$$

Since $x(t)$ is real, the poles of $X(s)$ must occur in conjugate reciprocal pairs. Therefore, we assume that $b = a^*$ and $d = c^*$. This results in

$$X(s) = \frac{A}{(s-a)(s-a^*)(s-c)(s-c^*)}$$

Since the signal $x(t)$ is also even, the Laplace transform $X(s)$ must also be even [Eq. (9.34)]. This implies that the poles have to be symmetric about the $j\omega$ -axis. Therefore, we may assume that $c = -a^*$. This results in

$$X(s) = \frac{A}{(s-a)(s-a^*)(s+a^*)(s+a)}$$

The location of one of the poles is $a = 1/2 e^{j\pi/4}$. Therefore, we have

$$\begin{aligned} X(s) &= \frac{A}{(s - (1/2) e^{j\pi/4})(s - (1/2) e^{-j\pi/4})} \\ &\quad \times \frac{1}{(s + (1/2) e^{-j\pi/4})(s + (1/2) e^{j\pi/4})} \\ X(s) &= \frac{A}{(s^2 - (s/\sqrt{2}) + \frac{1}{4})(s^2 + (s/\sqrt{2}) + \frac{1}{4})} \end{aligned}$$

We know that

$$\begin{aligned} X(s) &= \int_{-\infty}^{\infty} x(t) e^{-st} dt \\ X(s) \Big|_{s=0} &= X(0) = \int_{-\infty}^{\infty} x(t) dt \end{aligned}$$

$$\begin{aligned} \frac{A}{(0 - 0 + (1/4))(0 + 0 + (1/4))} &= 4 \\ \frac{A}{1/16} &= 4 \\ A &= \frac{1}{4} \end{aligned}$$

Therefore,

$$X(s) = \frac{1/4}{(s^2 - (s/\sqrt{2}) + \frac{1}{4})(s^2 + (s/\sqrt{2}) + \frac{1}{4})}$$

Example 9.66 Consider two right-sided signals $x(t)$ and $y(t)$ related through differential equations

$$\frac{dx(t)}{dt} = -2y(t) + \delta(t)$$

and

$$\frac{dy(t)}{dt} = 2x(t)$$

Determine $Y(s)$ and $X(s)$, along with their ROC.

Solution

Consider the given differential equations

$$\frac{dx(t)}{dt} = -2y(t) + \delta(t) \quad \text{and} \quad \frac{dy(t)}{dt} = 2x(t)$$

Taking the Laplace transform of both sides of the two differential equations, we have

$$sX(s) = -2Y(s) + 1 \quad \text{and} \quad sY(s) = 2X(s)$$

solving for $X(s)$ and $Y(s)$, we obtain

$$Y(s) = \frac{2}{s^2 + 4} \quad \text{and} \quad X(s) = \frac{s}{s^2 + 4}$$

The ROC for both $X(s)$ and $Y(s)$ is $\Re\{s\} > 0$ because both are right-sided signals.

Example 9.67 A causal LTI system S with impulse response $h(t)$ has its input $x(t)$ and output $y(t)$ related through a linear constant-coefficient differential equation of the form

$$\begin{aligned} \frac{d^3 y(t)}{dt^3} + (1 + \alpha) \frac{d^2 y(t)}{dt^2} \\ + \alpha(1 + \alpha) \frac{dy(t)}{dt} + \alpha^2 y(t) = x(t) \end{aligned}$$

(a) If

$$g(t) = \frac{dh(t)}{dt} + h(t)$$

how many poles does $G(s)$ have?(b) For what real values of the parameter α is S guaranteed to be stable.**Solution**

Consider the given differential equation

$$\frac{d^3 y(t)}{dt^3} + (1 + \alpha) \frac{d^2 y(t)}{dt^2} + \alpha(1 + \alpha) \frac{dy(t)}{dt} + \alpha^2 y(t) = x(t)$$

Taking the Laplace transform of both sides of the given differential equation, we obtain

$$s^3 Y(s) + (1 + \alpha)s^2 Y(s) + \alpha(1 + \alpha)s Y(s) + \alpha^2 Y(s) = X(s)$$

$$Y(s)[s^3 + (1 + \alpha)s^2 + \alpha(1 + \alpha)s + \alpha^2] = X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^3 + (1 + \alpha)s^2 + \alpha(1 + \alpha)s + \alpha^2}$$

$$H(s) = \frac{1}{(s + 1)(s^2 + \alpha s + \alpha^2)}$$

(a) Consider the given differential equation

$$g(t) = \frac{dh(t)}{dt} + h(t)$$

Taking the Laplace transform of both sides of the given differential equation, we obtain

$$G(s) = sH(s) + H(s) = H(s)(s + 1)$$

$$G(s) = \frac{1}{s^2 + \alpha s + \alpha^2}$$

Clearly, $G(s)$ has two poles.

(b) We know that

$$\begin{aligned} H(s) &= \frac{1}{(s + 1)(s^2 + \alpha s + \alpha^2)} \\ &= \frac{1}{(s + 1)(s + (\alpha/2) - j(\sqrt{3}\alpha/2))} \\ &\quad \times \frac{1}{(s + (\alpha/2) + j(\sqrt{3}\alpha/2))} \end{aligned}$$

Therefore, $H(s)$ has poles at -1 , $-(\alpha/2) + j(\sqrt{3}\alpha/2)$, and $-(\alpha/2) - j(\sqrt{3}\alpha/2)$. We know that for a causal system to be stable, all the poles of the system must lie in the left half of the s -plane. For this to be true, $-(\alpha/2) < 0$, i.e., $\alpha > 0$.

Example 9.68 We are given the following five facts about a real signal $x(t)$ with Laplace transform $X(s)$:

1. $X(s)$ has exactly two poles
2. $X(s)$ has no zeros in the finite s -plane
3. $X(s)$ has a pole at $s = -1 + j$
4. $e^{2t}x(t)$ is not absolutely integrable
5. $X(0) = 8$

Determine $X(s)$ and specify its ROC.**Solution**Since $X(s)$ has exactly two poles in the finite s -plane, we may assume that $X(s)$ is of the form

$$X(s) = \frac{A}{(s - a)(s - b)}$$

Since $x(t)$ is real, the poles of $X(s)$ must occur in conjugate reciprocal pairs. Therefore, we assume that $b = a^*$. This results in

$$X(s) = \frac{A}{(s - a)(s - a^*)}$$

The location of one of the poles is $a = -1 + j$. Therefore, we have

$$X(s) = \frac{A}{(s + 1 - j)(s + 1 + j)}$$

We know that

$$X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

$$X(s) \Big|_{s=0} = X(0) = \int_{-\infty}^{\infty} x(t) dt$$

From fact 5, we have

$$X(0) = \frac{A}{(0 + 1 - j)(0 + 1 + j)} = \frac{A}{2} = 8$$

$$A = 16$$

Therefore,

$$X(s) = \frac{16}{(s+1-j)(s+1+j)}$$

$$= \frac{16}{s^2 + 2s + 2}$$

From the pole locations we know that there are two possible choices of ROC, either $\Re\{s\} < -1$ or $\Re\{s\} > -1$. From fact 4, we have

$$y(t) = e^{2t}x(t) \longleftrightarrow Y(s) = X(s-2)$$

The ROC of $Y(s)$ is the ROC of $X(s)$ shifted by 2 to the right. The poles of $Y(s)$ are located at $-1+j+2=1+j$ and $-1-j+2=1-j$. Since it is given that $y(t)$ is not absolutely integrable, the ROC of $Y(s)$ should not include the $j\omega$ -axis. This is possible only if ROC of $X(s)$ is $\Re\{s\} > -1$.

Example 9.69 Show that for an LTI system, when the input is $x(t) = e^{s_0 t}$, the output is of the form $y(t) = H(s_0) e^{s_0 t}$. How is $H(s_0)$ related to the impulse response of the system?

Solution

We know that the input and output of an LTI system are related by

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau) e^{s_0(t-\tau)} d\tau$$

$$= e^{s_0 t} \int_{-\infty}^{\infty} h(\tau) e^{-s_0 \tau} d\tau$$

$$y(t) = H(s_0) e^{s_0 t}$$

where

$$H(s_0) = \int_{-\infty}^{\infty} h(\tau) e^{-s_0 \tau} d\tau = \mathcal{L}[h(t)] \Big|_{s=s_0}$$

Example 9.70 Determine the impulse response $h(t)$ of a system having a double-order pole at $s = -a$ and a zero at $s = -b$, where $a, b > 0$ and $b - a = B$. It is also given that $h(0) = 2$.

Solution

Since $h(t)$ of the system has a double-order pole at $s = -a$ and a zero at $s = -b$, we may assume that $H(s)$ is of the form

$$X(s) = \frac{K(s+b)}{(s+a)^2}$$

$$X(s) = \frac{Ks}{(s+a)^2} + \frac{Kb}{(s+a)^2}$$

$$X(s) = KsG(s) + KbG(s)$$

where

$$G(s) = \frac{1}{(s+a)^2}$$

Taking its inverse Laplace transform, we get

$$g(t) = t e^{-at} u(t)$$

Also, because

$$X(s) = KsG(s) + KbG(s)$$

Taking its inverse Laplace transform, we get

$$x(t) = K \frac{dg(t)}{dt} + Kbg(t)$$

$$= K \frac{d}{dt} t e^{-at} u(t) + Kbt e^{-at} u(t)$$

$$= K \left[t \left(e^{-at} \delta(t) - a e^{-at} u(t) \right) + e^{-at} u(t) \right]$$

$$+ Kbt e^{-at} u(t)$$

$$= K \left[0 - Kat e^{-at} u(t) + K e^{-at} u(t) \right]$$

$$+ Kbt e^{-at} u(t)$$

$$= K(b-a)t e^{-at} u(t) + K e^{-at} u(t)$$

$$h(t) = KBt e^{-at} u(t) + K e^{-at} u(t)$$

$$h(t) = K(Bt+1) e^{-at} u(t)$$

Also, given that

$$h(0) = 2 \quad K = 2$$

Therefore,

$$h(t) = 2(Bt+1) e^{-at} u(t)$$

Example 9.71 Determine the impulse response $h(t)$ of the system $H(s)$ of an LTI system from the following facts:

1. When the input to the system is $x(t) = e^{2t}$ and the output is $y(t) = 1/6 e^{2t}$
2. When $h(t)$ satisfies the differential equation

$$\frac{dh(t)}{dt} + 2h(t) = e^{-4t}u(t) - bu(t)$$

where b is an unknown constant. Your answer must not contain any unknown constant.

Solution

We know that if we apply a complex exponential input $x(t) = e^{s_0 t}$ to an LTI system with impulse response $h(t)$, the system output will be $y(t) = H(s_0) e^{s_0 t}$, i.e.,

$$x(t) \longrightarrow y(t)$$

$$e^{s_0 t} \longrightarrow H(s_0) e^{s_0 t}$$

Given that

$$e^{2t} \longrightarrow \frac{1}{6} e^{2t} = H(2) e^{2t}$$

Therefore, $H(2) = 1/6$. Now, consider the given differential equation

$$\frac{dh(t)}{dt} + 2h(t) = e^{-4t}u(t) - bu(t)$$

Taking the Laplace transform of the above equation, we obtain

$$sH(s) + 2H(s) = \frac{1}{s+4} - \frac{b}{s}$$

$$(s+2)H(s) = \frac{s-b(s+4)}{s(s+4)}$$

$$H(s) = \frac{s-b(s+4)}{s(s+2)(s+4)}$$

Also, given that

$$H(s) \Big|_{s=2} = H(2) = \frac{2-6b}{2 \times 4 \times 6} = \frac{1}{6}$$

$$b = -1$$

Therefore,

$$H(s) = \frac{s+(s+4)}{s(s+2)(s+4)}$$

$$= \frac{2(s+2)}{s(s+2)(s+4)}$$

$$H(s) = \frac{2}{s(s+4)}$$

Using partial fraction expansion, we obtain

$$H(s) = \frac{1/2}{s} - \frac{1/2}{s+4}$$

The inverse Laplace transform of the above equation yields

$$h(t) = \frac{1}{2}[1 - e^{-4t}]u(t)$$

Example 9.72 Solve, by using Laplace transform, the following set of simultaneous differential equations for $x(t)$:

$$2\frac{dx(t)}{dt} + 4x(t) + \frac{dy(t)}{dt} + 7y(t) = 5u(t)$$

$$\frac{dx(t)}{dt} + x(t) + \frac{dy(t)}{dt} + 3y(t) = 5\delta(t)$$

The initial conditions are $x(0^-) = y(0^-) = 0$.

Solution

Consider the given differential equation

$$2\frac{dx(t)}{dt} + 4x(t) + \frac{dy(t)}{dt} + 7y(t) = 5u(t)$$

Taking the Laplace transform of the above equation, we obtain

$$2sX(s) + 4X(s) + sY(s) + 7Y(s) = \frac{5}{s}$$

$$(2s+4)X(s) + (s+7)Y(s) = \frac{5}{s}$$

$$X(s) = \frac{5}{s(2s+4)} - \frac{s+7}{2s+4}Y(s)$$

Now, consider the second differential equation

$$\frac{dx(t)}{dt} + x(t) + \frac{dy(t)}{dt} + 3y(t) = 5\delta(t)$$

Taking the Laplace transform of the above equation, we obtain

$$sX(s) + X(s) + sY(s) + 3Y(s) = 5$$

$$(s+1)X(s) + (s+3)Y(s) = 5$$

$$Y(s) = \frac{5}{s+3} - \frac{s+1}{s+3}X(s)$$

Substituting this expression of $Y(s)$ into the expression of $X(s)$, we obtain

$$\begin{aligned} X(s) &= \frac{5}{s(2s+4)} - \frac{s+7}{2s+4} Y(s) \\ &= \frac{5}{s(2s+4)} - \frac{s+7}{2s+4} \left[\frac{5}{s+3} - \frac{s+1}{s+3} X(s) \right] \\ X(s) &= \frac{5}{s(2s+4)} - \frac{5(s+7)}{(s+3)(2s+4)} \\ &\quad + \frac{(s+1)(s+7)}{(s+3)(2s+4)} X(s) \\ X(s) \left[1 - \frac{(s+1)(s+7)}{(s+3)(2s+4)} \right] &= \frac{5}{s(2s+4)} \\ &\quad - \frac{5(s+7)}{(s+3)(2s+4)} \end{aligned}$$

$$\begin{aligned} X(s) [(s+3)(2s+4) - (s+1)(s+7)] \\ = \frac{5(s+3)}{s} - 5(s+7) \end{aligned}$$

$$X(s)[s^2 + 2s + 5] = -30 - 5s + \frac{15}{s}$$

$$X(s)[(s+1)^2 + 4] = -25 - 5(s+1) + \frac{15}{s}$$

$$\begin{aligned} X(s) &= -\frac{25}{(s+1)^2 + 4} - \frac{5(s+1)}{(s+1)^2 + 4} \\ &\quad + \frac{15}{s[(s+1)^2 + 4]} \end{aligned}$$

Applying partial fraction expansion on the last term of the RHS of the above equation, we get

$$\begin{aligned} X(s) &= -\frac{25}{(s+1)^2 + 4} - \frac{5(s+1)}{(s+1)^2 + 4} \\ &\quad + \frac{3}{s} + \frac{-3s-6}{(s+1)^2 + 4} \\ &= -\frac{25}{(s+1)^2 + 4} - \frac{5(s+1)}{(s+1)^2 + 4} \\ &\quad + \frac{3}{s} - \frac{3(s+1)}{(s+1)^2 + 4} - \frac{3}{(s+1)^2 + 4} \end{aligned}$$

$$X(s) = \frac{3}{s} - \frac{28}{(s+1)^2 + 4} - \frac{8(s+1)}{(s+1)^2 + 4}$$

$$X(s) = \frac{3}{s} - 14 \frac{2}{(s+1)^2 + 4} - 8 \frac{(s+1)}{(s+1)^2 + 4}$$

The inverse Laplace transform of the above equation yields

$$x(t) = 3u(t) - 14e^{-t} \sin(2t)u(t) - 8e^{-t} \cos(2t)u(t)$$

Example 9.73 An LTI system has an impulse response $h(t) = e^{-at}u(t)$, and when it is excited by an input signal $x(t)$, its output is $y(t) = (e^{-bt} - e^{-ct})u(t)$. Determine its input $x(t)$.

Solution

Given that the impulse response

$$h(t) = e^{-at}u(t)$$

Its Laplace transform is given by

$$H(s) = \frac{1}{s+a}$$

Also, given that the system output

$$y(t) = (e^{-bt} - e^{-ct})u(t)$$

The Laplace transform of the above equation yields

$$Y(s) = \frac{1}{s+b} - \frac{1}{s+c} = \frac{c-b}{(s+b)(s+c)}$$

We know that

$$X(s) = \frac{Y(s)}{H(s)}$$

$$X(s) = \frac{c-b(s+a)}{(s+b)(s+c)}$$

Using partial fraction expansion, we obtain

$$X(s) = \frac{a-b}{s+b} + \frac{c-a}{s+c}$$

The inverse Laplace transform of the above equation yields

$$x(t) = (a-b)e^{-bt}u(t) + (c-a)e^{-ct}u(t)$$

Example 9.74 Consider the LTI system shown in Fig. 9.34(a) for which we are given the following information:

$$X(s) = \frac{s+2}{s-2}$$

$$x(t) = 0, \quad t > 0$$

and

$$y(t) = -\frac{2}{3} e^{2t} u(t) + \frac{1}{3} e^{-t} u(t)$$

$y(t)$ is shown in Fig. 9.34(b).

- Determine $H(s)$ and its region of convergence.
- Determine $h(t)$.
- Using the system function $H(s)$ found in part (a), determine the output $y(t)$ if the input is $x(t) = e^{3t}$, $-\infty < t < \infty$.

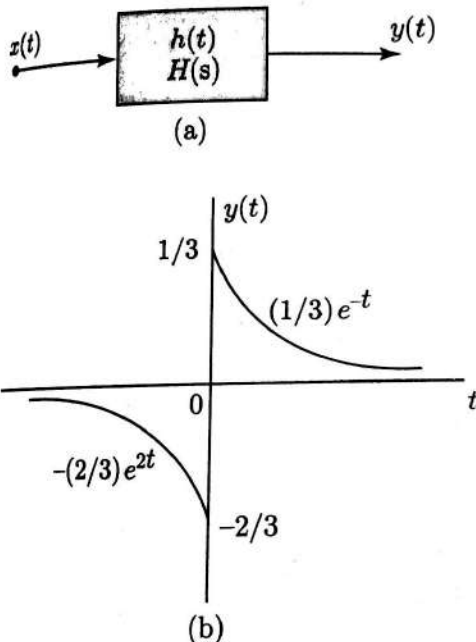


Fig. 9.34

Solution

(a) Given that

$$y(t) = -\frac{2}{3} e^{2t} u(t) + \frac{1}{3} e^{-t} u(t)$$

Taking the Laplace transform of the above equation, we get

$$Y(s) = \frac{2/3}{s-2} + \frac{1/3}{s+1}, \quad -1 < \Re\{s\} < 2$$

$$Y(s) = \frac{s}{(s-2)(s+1)}, \quad -1 < \Re\{s\} < 2$$

Given that $x(t) = 0$, for $t > 0$, i.e., $x(t)$ is left sided. Therefore, the ROC for $X(s)$ is $\Re\{s\} < 2$.

We know that

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s}{(s+1)(s+2)}$$

We know that the ROC of $Y(s)$ has to be the intersection of the ROCs of $X(s)$ and $H(s)$. This leads to conclude that the ROC of $H(s)$ is $\Re\{s\} > -1$.

(b) The partial fraction expansion of $H(s)$ is

$$H(s) = \frac{-1}{s+1} + \frac{2}{s+2}$$

Therefore,

$$h(t) = -e^{-t} u(t) + 2e^{-2t} u(t)$$

(c) We know that

$$e^{s_0 t} \longrightarrow H(s_0) e^{s_0 t}$$

Therefore,

$$x(t) = e^{3t} \longrightarrow H(3) e^{3t} = y(t)$$

Hence,

$$y(t) = H(3) e^{3t} = \frac{3}{(3+1)(3+2)} e^{3t} = \frac{3}{20} e^{3t}$$

Example 9.75 A causal LTI system S has the block diagram representation shown in Fig. 9.35. Determine a differential equation relating the input $x(t)$ to the output $y(t)$ of this system.

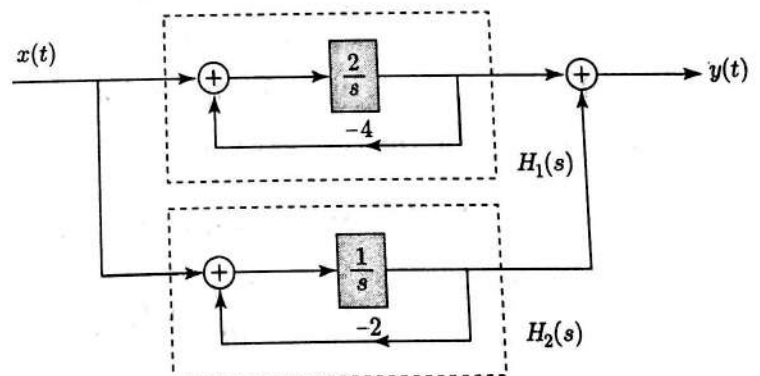


Fig. 9.35

Solution

The overall system shown in Fig. 9.35 may be treated as two feedback systems $H_1(s)$ and $H_2(s)$ connected in parallel as shown in Fig. 9.36. Using

Eq. (9.82) [and Fig. 9.25(d)], the system function of the upper feedback system is

$$H_1(s) = \frac{2/s}{1 + 4(2/s)} = \frac{2}{s + 8}$$

Similarly, the system function of lower feedback system is

$$H_2(s) = \frac{1/s}{1 + 2(1/s)} = \frac{1}{s + 2}$$

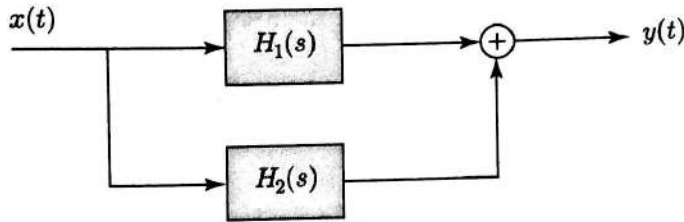


Fig. 9.36

Since $H_1(s)$ and $H_2(s)$ are connected in parallel, the system function of the overall system is

$$\begin{aligned} H(s) &= H_1(s) + H_2(s) \\ &= \frac{2}{s + 8} + \frac{1}{s + 2} \\ H(s) &= \frac{3s + 12}{s^2 + 10s + 16} \end{aligned}$$

Also because $H(s) = Y(s)/X(s)$, we may write,

$$\begin{aligned} \frac{Y(s)}{X(s)} &= \frac{3s + 12}{s^2 + 10s + 16} \\ Y(s)[s^2 + 10s + 16] &= X(s)[3s + 12] \\ s^2 Y(s) + 10s Y(s) + 16Y(s) &= 3sX(s) + 12X(s) \end{aligned}$$

Taking the inverse Laplace transform, we obtain

$$\frac{d^2 y(t)}{dt^2} + 10 \frac{dy(t)}{dt} + 16y(t) = 3 \frac{dx(t)}{dt} + 12x(t)$$

Example 9.76 The input $x(t)$ and output $y(t)$ of a causal LTI system are related through the block diagram representation shown in Fig. 9.37.

(a) Determine a differential equation relating $y(t)$ and $x(t)$.

(b) Is this system stable?

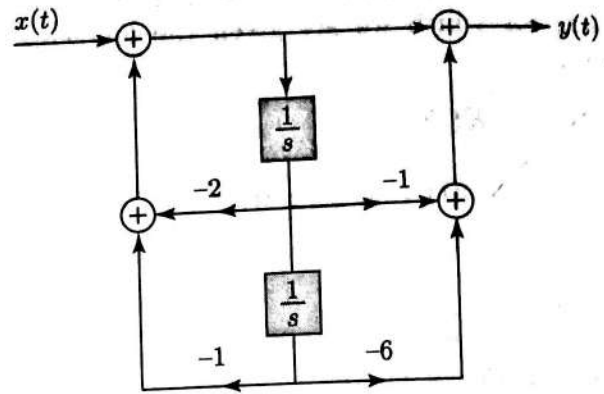


Fig. 9.37

Solution

(a) We may redraw the block diagram as shown in Fig. 9.38. From the left-half section of the Fig. 9.38, it is clear that

$$\begin{aligned} V(s) &= X(s) - 2 \frac{V(s)}{s} - \frac{V(s)}{s^2} \\ V(s) \left[1 + \frac{2}{s} + \frac{1}{s^2} \right] &= X(s) \\ \frac{V(s)}{X(s)} &= \frac{1}{1 + (2/s) + (1/s^2)} \end{aligned}$$

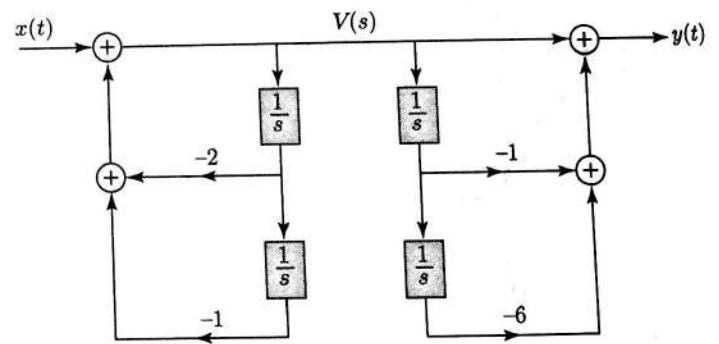


Fig. 9.38

Now, from the right-half section of the Fig. 9.38, it is clear that

$$\begin{aligned} Y(s) &= V(s) - \frac{V(s)}{s} - 6 \frac{V(s)}{s^2} \\ Y(s) &= V(s) \left[1 - \frac{1}{s} - \frac{6}{s^2} \right] \\ \frac{Y(s)}{V(s)} &= 1 - \frac{1}{s} - \frac{6}{s^2} \end{aligned}$$

Therefore,

$$\begin{aligned} H(s) &= \frac{Y(s)}{X(s)} = \left(\frac{V(s)}{X(s)} \right) \left(\frac{Y(s)}{V(s)} \right) \\ &= \left(\frac{1}{1 + (2/s) + (1/s^2)} \right) \\ &\quad \left(1 - \frac{1}{s} - \frac{6}{s^2} \right) \\ &= \frac{1 - (1/s) - (6/s^2)}{1 + (2/s) + 1/s^2} \end{aligned}$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{s^2 - s - 6}{s^2 + 2s + 1}$$

$$Y(s)[s^2 + 2s + 1] = X(s)[s^2 - s - 6]$$

$$\begin{aligned} s^2 Y(s) + 2s Y(s) + Y(s) \\ = s^2 X(s) - s X(s) - 6 X(s) \end{aligned}$$

The inverse Laplace transform of the above equation yields

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = \frac{d^2 x(t)}{dt^2} - \frac{dx(t)}{dt} - 6x(t)$$

(b) Consider the overall system function

$$\begin{aligned} H(s) &= \frac{s^2 - s - 6}{s^2 + 2s + 1} \\ &= \frac{(s - 3)(s + 2)}{(s + 1)^2} \end{aligned}$$

We know that for a causal system to be stable, all the poles must lie in the left half of the s -plane. The two poles of the system are at -1 . We may conclude that this system is stable.

Example 9.77 Find the transfer function and frequency response (magnitude and phase response) for the following:

(a) An ideal delay of T seconds.

(b) An ideal differentiator.

(c) An ideal integrator.

Also plot the magnitude response and phase response.

Solution

(a) *Ideal delay* For an ideal delay of T seconds, the input $x(t)$ and output $y(t)$ are related by

$$y(t) = x(t - T)$$

The Laplace transform of this equation yields

$$Y(s) = X(s) e^{-sT}$$

$$H(s) = \frac{Y(s)}{X(s)} = e^{-sT}$$

The frequency response is given by

$$H(s) \Big|_{s=j\omega} = H(j\omega) = e^{-j\omega T} = |H(j\omega)| e^{j\angle H(j\omega)}$$

Consequently,

$$|H(j\omega)| = 1 \quad \text{and} \quad \angle H(j\omega) = -\omega T$$

These magnitude and phase responses are shown in Fig. 9.39(a). The magnitude response is constant (unity) for all frequencies. The phase response increases linearly with frequency with a slope of $-T$.

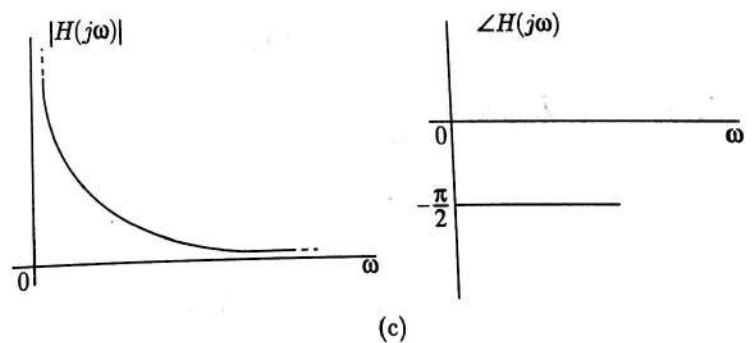
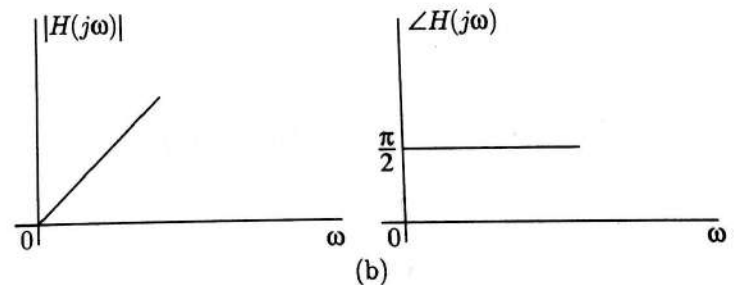
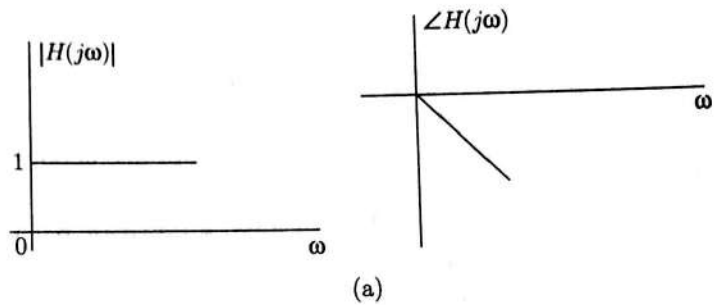


Fig. 9.39 Magnitude and phase response of (a) an ideal delay, (b) an ideal differentiator and (c) an ideal integrator

(b) *An ideal differentiator.* For an ideal differentiator, the input $x(t)$ and the output $y(t)$ are related by

$$y(t) = \frac{dx(t)}{dt}$$

The Laplace transform of this equation yields

$$Y(s) = sX(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = s$$

The frequency response is given by

$$H(s) \Big|_{s=j\omega} = H(j\omega) = j\omega$$

$$= \omega e^{j(\pi/2)} = |H(j\omega)| e^{j\angle H(j\omega)}$$

Consequently,

$$|H(j\omega)| = \omega, \quad \text{and} \quad \angle H(j\omega) = \frac{\pi}{2}$$

These magnitude and phase responses are shown in Fig. 9.39(b). The magnitude response increases linearly with frequency, and the phase response is constant ($\pi/2$) for all frequencies.

(c) *An ideal integrator.* For an ideal integrator with zero initial state, i.e., $y(0^-) = 0$, the input $x(t)$ and the output $y(t)$ are related by

$$y(t) = \int_0^t x(\tau) d\tau$$

The Laplace transform of this equation yields

$$Y(s) = \frac{1}{s} X(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s}$$

The frequency response is given by

$$H(s) \Big|_{s=j\omega} = H(j\omega) = \frac{1}{j\omega}$$

$$= \frac{-j}{\omega}$$

$$= \frac{1}{\omega} e^{-j(\pi/2)} = |H(j\omega)| e^{j\angle H(j\omega)}$$

Consequently,

$$|H(j\omega)| = \frac{1}{\omega} \quad \text{and} \quad \angle H(j\omega) = -\frac{\pi}{2}$$

These magnitude and phase responses are shown in Fig. 9.39(c). The magnitude response is inversely proportional to frequency, and the phase response is constant ($-\pi/2$) for all frequencies.

Example 9.78 Find the inverse unilateral Laplace transform of

$$(a) X(s) = \frac{2s^3 - 9s^2 + 4s + 10}{s^2 - 3s - 4}$$

$$(b) X(s) = \frac{s^2 + s - 3}{s^2 + 3s + 2}$$

Solution

(a) Since the order of the numerator polynomial is greater than the order of the denominator polynomial, the given rational Laplace transform is improper. We use long division to express $X(s)$ as the sum of a proper rational function and a polynomial in s :

$$\begin{array}{r} 2s - 3 \\ s^2 - 3s - 4 \overline{) 2s^3 - 9s^2 + 4s + 10} \\ \underline{2s^3 - 6s^2 - 8s} \\ -3s^2 + 12s + 10 \\ \underline{-3s^2 + 9s + 12} \\ 3s - 2 \end{array}$$

Thus, we may write

$$X(s) = 2s - 3 + \frac{3s - 2}{s^2 - 3s - 4}$$

$$X(s) = 2s - 3 + \frac{3s - 2}{(s + 1)(s - 4)}$$

Using a partial fraction expansion to expand the rational function, we obtain

$$X(s) = 2s - 3 + \frac{1}{s + 1} + \frac{2}{s - 4}$$

The inverse Laplace transform of this equation yields

$$x(t) = 2 \frac{d\delta(t)}{dt} - 3\delta(t) + e^{-t}u(t) + 2e^{4t}u(t)$$

(b) Since the order of the numerator denominator polynomial is equal, the given rational Laplace transform is improper. We use long division to

express $X(s)$ as the sum of a proper rational function and a polynomial in s :

$$\begin{array}{r} 1 \\ s^2 + 3s + 2 \overline{) s^2 + s - 3} \\ \underline{s^2 + 3s + 2} \\ -2s - 5 \end{array}$$

Thus, we may write

$$X(s) = 1 + \frac{-2s - 5}{s^2 + 3s + 2}$$

$$X(s) = 1 - \frac{2s + 5}{(s + 1)(s + 2)}$$

Using a partial fraction expansion to expand the rational function, we obtain

$$X(s) = 1 - \frac{3}{s + 1} + \frac{1}{s + 2}$$

The inverse Laplace transform of this equation yields

$$x(t) = \delta(t) - 3e^{-t}u(t) + e^{-2t}u(t)$$

Example 9.79 Find the inverse unilateral Laplace transform of

$$X(s) = \frac{4s^2 + 6}{s^3 + s^2 - 2}$$

Solution

Consider the given Laplace transform

$$X(s) = \frac{4s^2 + 6}{s^3 + s^2 - 2}$$

$$X(s) = \frac{4s^2 + 6}{(s - 1)(s^2 + 2s + 2)}$$

The partial fraction expansion for $X(s)$ takes the form

$$X(s) = \frac{A}{s - 1} + \frac{Bs + C}{s^2 + 2s + 2} \quad (9.91)$$

Multiply both sides of the above equation by $s - 1$ and evaluate at $s = 1$ to obtain

$$A = (s - 1)X(s) \Big|_{s=1}$$

$$A = (s - 1) \frac{4s^2 + 6}{(s - 1)(s^2 + 2s + 2)} \Big|_{s=1}$$

$$A = 2$$

Now consider Eq. (9.91),

$$X(s) = \frac{A}{s - 1} + \frac{Bs + C}{s^2 + 2s + 2}$$

$$\frac{4s^2 + 6}{(s - 1)(s^2 + 2s + 2)} = \frac{2}{s - 1} + \frac{Bs + C}{s^2 + 2s + 2}$$

$$\begin{aligned} 4s^2 + 6 &= 2(s^2 + 2s + 2) + (Bs + C)(s - 1) \\ &= (2 + B)s^2 + (4 - B + C)s + (4 - C) \end{aligned}$$

Equating coefficients of s^2 gives $B = 2$, and equating coefficients of s^0 gives $C = -2$. Hence,

$$\begin{aligned} X(s) &= \frac{2}{s - 1} + \frac{2s - 2}{s^2 + 2s + 2} \\ &= \frac{2}{s - 1} + \frac{2s - 2 + 2 - 2}{(s + 1)^2 + 1} \\ &= \frac{2}{s - 1} + 2 \frac{s + 1}{(s + 1)^2 + 1} - 4 \frac{1}{(s + 1)^2 + 1} \end{aligned}$$

The inverse Laplace transform of this equation yields

$$x(t) = 2e^t u(t) + 2e^{-t} \cos(t) u(t) - 4e^{-t} \sin(t) u(t)$$

Example 9.80 Find the inverse unilateral Laplace transform of

$$X(s) = \frac{1}{e^{s+3}} \frac{s^2}{(s + 1)(s + 2)}$$

Solution

Given that

$$\begin{aligned} X(s) &= \frac{1}{e^{s+3}} \frac{s^2}{(s + 1)(s + 2)} \\ &= \frac{1}{e^{s+3}} G(s) \end{aligned}$$

where

$$\begin{aligned} G(s) &= \frac{s^2}{(s + 1)(s + 2)} \\ &= \frac{s^2}{s^2 + 3s + 2} \end{aligned}$$

Since the order of the numerator denominator polynomial is equal, the rational Laplace transform $G(s)$ is improper. We use long division to express $G(s)$ as the sum of a proper rational function and a polynomial in s :

$$\begin{array}{r} 1 \\ s^2 + 3s + 2 \overline{) s^2} \\ \underline{s^2 + 3s + 2} \\ -3s - 2 \end{array}$$

Thus, we may write

$$G(s) = 1 + \frac{-3s - 2}{s^2 + 3s + 2}$$

$$G(s) = 1 - \frac{3s + 2}{(s + 1)(s + 2)}$$

Using a partial fraction expansion to expand the rational function, we obtain

$$G(s) = 1 + \frac{1}{s + 1} - \frac{4}{s + 2}$$

The inverse Laplace transform of this equation yields

$$g(t) = \delta(t) + e^{-t}u(t) - 4e^{-2t}u(t)$$

Also, because

$$\begin{aligned} X(s) &= \frac{1}{e^{s+3}} G(s) \\ &= \frac{1}{e^3} e^{-s} G(s) \end{aligned}$$

Therefore,

$$\begin{aligned} x(t) &= \frac{1}{e^3} g(t - 1) \\ &= \frac{1}{e^3} [\delta(t - 1) + e^{-(t-1)}u(t - 1) \\ &\quad - 4e^{-2(t-1)}u(t - 1)] \end{aligned}$$

Example 9.81 Find the Laplace transform of the half-wave rectification of $\sin(\omega t)$ shown in Fig. 9.40(a).

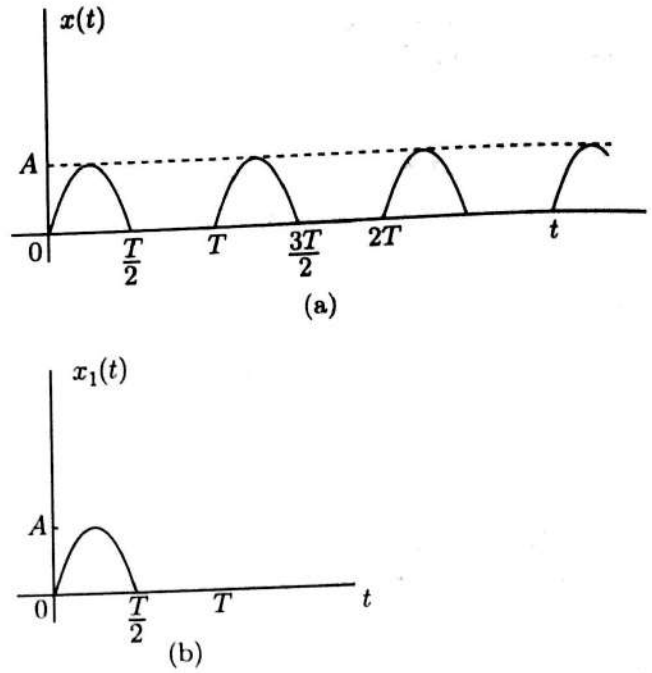


Fig. 9.40 (a) Half-wave rectified sine wave and (b) its first cycle

Solution

We know that the Laplace transform of a causal periodic signal is given by [Eq. (9.49)]

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

where $X_1(s)$ is the Laplace transform of the first cycle $x_1(t)$ of the causal periodic signal $x(t)$. The given half-wave rectified sinusoidal wave is a causal periodic signal with period $T_0 = T$. The first cycle $x_1(t)$ of $x(t)$ is as shown in Fig. 9.40(b). The first cycle $x_1(t)$ can be written as

$$x_1(t) = \begin{cases} A \sin(\omega t), & 0 \leq t \leq \frac{T}{2} \\ 0, & \frac{T}{2} \leq t \leq T \end{cases}$$

$$x_1(t) = A \sin(\omega t)u(t)$$

$$+ A \sin\left(\omega \left(t - \frac{T}{2}\right)\right)u\left(t - \frac{T}{2}\right)$$

Taking the Laplace transform of the above equation, we obtain

$$\mathcal{L}[x_1(t)] = A\mathcal{L}[\sin(\omega t)u(t)]$$

$$+ A\mathcal{L}\left[\sin\left(\omega \left(t - \frac{T}{2}\right)\right)u\left(t - \frac{T}{2}\right)\right]$$

$$X_1(s) = A \frac{\omega}{s^2 + \omega^2} + A \frac{\omega}{s^2 + \omega^2} e^{-sT/2}$$

$$= \frac{A\omega}{s^2 + \omega^2} [1 + e^{-sT/2}]$$

Substituting $X_1(s)$ into the expression of $X(s)$, we get

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

$$= \frac{A\omega}{s^2 + \omega^2} \frac{1 + e^{-sT/2}}{1 - e^{-sT}}$$

$$= \frac{A\omega}{s^2 + \omega^2} \frac{1 + e^{-sT/2}}{(1 + e^{-sT/2})(1 - e^{-sT/2})}$$

$$X(s) = \frac{A\omega}{s^2 + \omega^2} \left(\frac{1}{1 - e^{-sT/2}} \right)$$

Example 9.82 Find the Laplace transform of the full-wave rectification of $\sin(\omega_0 t)$ shown in Fig. 9.41(a).

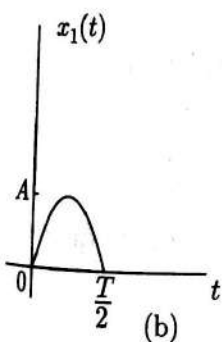
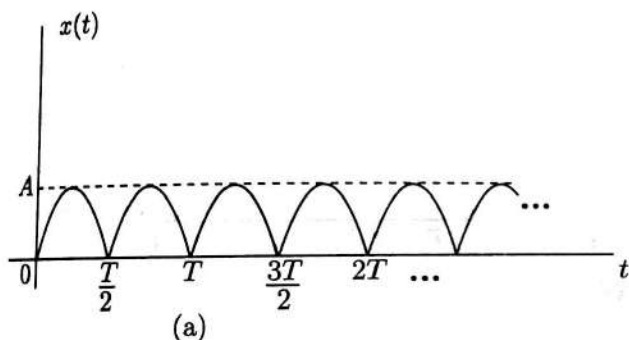


Fig. 9.41 (a) Full-wave rectified sine wave and (b) its first cycle.

Solution

We know that the Laplace transform of a causal periodic signals is given by [Eq. (9.49)]

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

where $X_1(s)$ is the Laplace transform of the first cycle $x_1(t)$ of the causal periodic signal $x(t)$. The given half-wave rectified sinusoidal wave is a causal periodic signal with period $T_0 = T/2$. The first cycle $x_1(t)$ of $x(t)$ is as shown in Fig. 9.41(b). The first cycle $x_1(t)$ can be written as

$$x_1(t) = A \sin(\omega t) \quad 0 \leq t \leq \frac{T}{2} \quad \omega = \frac{2\pi}{T}$$

$$x_1(t) = A \sin(\omega t) u(t)$$

$$+ A \sin\left(\omega\left(t - \frac{T}{2}\right)\right) u\left(t - \frac{T}{2}\right)$$

Taking the Laplace transform of the above equation, we obtain

$$\mathcal{L}[x_1(t)] = A\mathcal{L}[\sin(\omega t)u(t)]$$

$$+ A\mathcal{L}\left[\sin\left(\omega\left(t - \frac{T}{2}\right)\right)u\left(t - \frac{T}{2}\right)\right]$$

$$X_1(s) = A \frac{\omega}{s^2 + \omega^2} + A \frac{\omega}{s^2 + \omega^2} e^{-sT/2}$$

$$= \frac{A\omega}{s^2 + \omega^2} [1 + e^{-sT/2}]$$

Substituting $X_1(s)$ into the expression of $X(s)$, we get

$$X(s) = \frac{X_1(s)}{1 - e^{-sT_0}}$$

$$= \frac{A\omega}{s^2 + \omega^2} \frac{1 + e^{-sT/2}}{1 - e^{-sT/2}}$$

$$= \frac{A\omega}{s^2 + \omega^2} \frac{e^{-sT/4} (e^{sT/4} + e^{-sT/4})}{e^{-sT/4} (e^{sT/4} - e^{-sT/4})}$$

$$X(s) = \frac{A\omega}{s^2 + \omega^2} \frac{(e^{sT/4} + e^{-sT/4})}{(e^{sT/4} - e^{-sT/4})}$$

$$X(s) = \frac{A\omega}{s^2 + \omega^2} \coth\left(\frac{sT}{4}\right)$$

Example 9.83 Using convolution, find the inverse Laplace transform $h(t)$ of

$$(a) H(s) = \frac{1}{(s^2 + 1)^2}$$

$$(b) H(s) = \frac{1}{s^2(s - a)}$$

Solution

(a) Given that

$$H(s) = \frac{1}{(s^2 + 1)^2}$$

$$= \left(\frac{1}{s^2 + 1} \right) \left(\frac{1}{s^2 + 1} \right)$$

Taking the inverse Laplace transform of this equation yields

$$\mathcal{L}^{-1}[H(s)] = \mathcal{L}^{-1} \left[\frac{1}{s^2 + 1} \right] * \mathcal{L}^{-1} \left[\frac{1}{s^2 + 1} \right]$$

$$h(t) = \sin(t)u(t) * \sin(t)u(t)$$

$$= \int_{-\infty}^{\infty} \sin(\tau)u(\tau) \sin(t - \tau)u(t - \tau) d\tau$$

$$= \int_0^t \sin(\tau) \sin(t - \tau) d\tau$$

$$= \int_0^t \frac{1}{2} [\cos(2\tau - t) - \cos(t)] d\tau$$

$$= \frac{1}{2} \int_0^t \cos(2\tau - t) d\tau - \frac{1}{2} \int_0^t \cos(\tau) d\tau$$

$$h(t) = \frac{1}{2} \sin(t) - \frac{1}{2} t \cos(t) \quad t \geq 0$$

(b) Given that

$$H(s) = \frac{1}{s^2(s - a)}$$

$$= \left(\frac{1}{s^2} \right) \left(\frac{1}{s - a} \right)$$

Taking the inverse Laplace transform of this equation yields

$$\mathcal{L}^{-1}[H(s)] = \mathcal{L}^{-1} \left[\frac{1}{s^2} \right] * \mathcal{L}^{-1} \left[\frac{1}{s - a} \right]$$

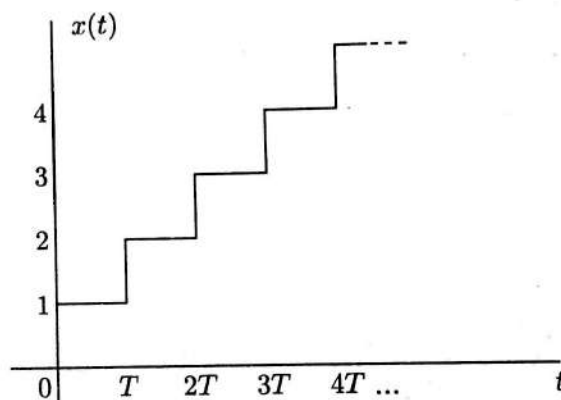
$$h(t) = tu(t) * e^{at}u(t)$$

$$h(t) = \int_{-\infty}^{\infty} \tau u(\tau) e^{a(t-\tau)} u(t - \tau) d\tau$$

$$= \int_0^t \tau e^{a(t-\tau)} d\tau$$

$$= e^{at} \int_0^t \tau e^{-a\tau} d\tau$$

$$h(t) = \frac{1}{a^2} [e^{at} - at - 1] \quad t \geq 0$$

Example 9.84 Find the Laplace transform of the staircase function shown in Fig. 9.42.**Fig. 9.42****Solution**Using step function $u(t)$, the staircase function (shown in Fig. 9.42) can be written as

$$x(t) = u(t) + u(t - T) + u(t - 2T) + \dots$$

Taking the Laplace transform of the above equation, we obtain

$$X(s) = \frac{1}{s} + \frac{1}{s} e^{-sT} + \frac{1}{s} e^{-2sT} + \dots$$

$$= \frac{1}{s} [1 + e^{-sT} + e^{-2sT} + \dots]$$

$$X(s) = \frac{1}{s(1 - e^{-sT})}$$

Example 9.85 Find the Laplace transform of the signals shown in Fig. 9.43.

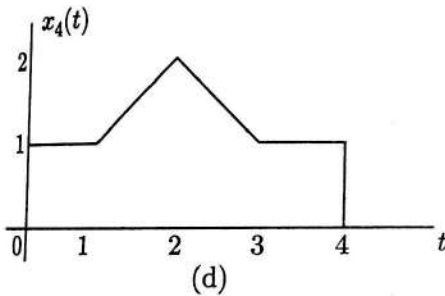
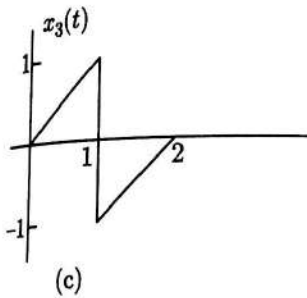
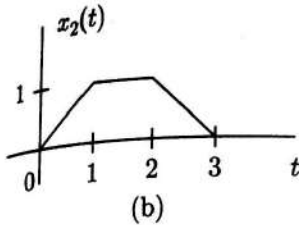
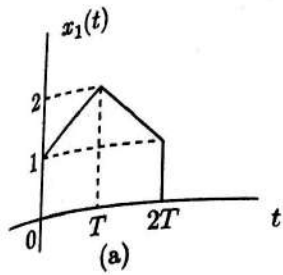


Fig. 9.43

Solution

(a) Using step function $u(t)$ and ramp function $r(t)$, the given signal $x_1(t)$ can be written as

$$\begin{aligned} x_1(t) &= u(t) + r(t) - r(t-1) - r(t-1) \\ &\quad + r(t-2) - u(t-2) \\ &= u(t) + r(t) - 2r(t-1) + r(t-2) \\ &\quad - u(t-2) \end{aligned}$$

Taking the Laplace transform of the above equation, we obtain

$$\begin{aligned} X_1(s) &= \frac{1}{s} + \frac{1}{s^2} - \frac{2}{s^2} e^{-s} + \frac{1}{s^2} e^{-2s} - \frac{1}{s} e^{-2s} \\ &= \frac{1}{s} [1 - e^{-2s}] + \frac{1}{s^2} [1 - 2e^{-s} + e^{-2s}] \end{aligned}$$

(b) Using ramp function $r(t)$, the given signal $x_2(t)$ can be written as

$$x_2(t) = r(t) - r(t-1) - r(t-2) + r(t-3)$$

Taking the Laplace transform of the above equation, we obtain

$$\begin{aligned} X_2(s) &= \frac{1}{s^2} - \frac{1}{s^2} e^{-s} - \frac{1}{s^2} e^{-2s} + \frac{1}{s^2} e^{-3s} \\ &= \frac{1}{s^2} [1 - e^{-s} - e^{-2s} + e^{-3s}] \end{aligned}$$

(c) Using step function $u(t)$ and ramp function $r(t)$, the given signal $x_3(t)$ can be written as

$$x_3(t) = r(t) - 2u(t-1) - r(t-2)$$

Taking the Laplace transform of the above equation, we obtain

$$\begin{aligned} X_3(s) &= \frac{1}{s^2} - \frac{2}{s} e^{-s} - \frac{1}{s^2} e^{-2s} \\ &= \frac{1}{s^2} [1 - 2s e^{-s} - e^{-2s}] \end{aligned}$$

(d) Using step function $u(t)$ and ramp function $r(t)$, the given signal $x_4(t)$ can be written as

$$\begin{aligned} x_4(t) &= u(t) + r(t-1) - r(t-2) - r(t-2) \\ &\quad + r(t-3) - u(t-4) \\ &= u(t) + r(t-1) - 2r(t-2) + r(t-3) \\ &\quad - u(t-4) \end{aligned}$$

Taking the Laplace transform of the above equation, we obtain

$$\begin{aligned} X_4(s) &= \frac{1}{s} + \frac{1}{s^2} e^{-s} - \frac{2}{s^2} e^{-2s} + \frac{1}{s^2} e^{-3s} - \frac{1}{s} e^{-4s} \\ &= \frac{1}{s} [1 - e^{-4s}] + \frac{1}{s^2} [e^{-s} - 2e^{-2s} + e^{-3s}] \end{aligned}$$

Example 9.86 Realize

$$H(s) = \frac{s(s+2)}{(s+1)(s+3)(s+4)}$$

by canonic direct (direct form II), series, parallel, and the transposed form of canonic direct.

Solution(a) *Canonic realization:* Given that

$$\begin{aligned}
 H(s) &= \frac{s(s+2)}{(s+1)(s+3)(s+4)} \\
 &= \frac{s^2 + 2s}{s^3 + 8s^2 + 19s + 12} \\
 &= \frac{(1/s) + (2/s^2)}{1 + (8/s) + (19/s^2) + (12/s^3)} \\
 &= \underbrace{\left(\frac{1}{1 + (8/s) + (19/s^2) + (12/s^3)} \right)}_{H_2(s)} \underbrace{(1/s + 2/s^2)}_{H_1(s)}
 \end{aligned}$$

where

$$\begin{aligned}
 H_2(s) &= \frac{V(s)}{X(s)} \\
 &= \frac{1}{1 + (8/s) + (19/s^2) + (12/s^3)} \\
 V(s) &= X(s) - 8\frac{V(s)}{s} - 19\frac{V(s)}{s^2} - 12\frac{V(s)}{s^3}
 \end{aligned}$$

and

$$\begin{aligned}
 H_1(s) &= \frac{Y(s)}{V(s)} = \frac{1}{s} + \frac{2}{s^2} \\
 Y(s) &= \frac{V(s)}{s} + 2\frac{V(s)}{s^2}
 \end{aligned}$$

The given transfer function $H(s) = (s(s+2))/((s+1)(s+3)(s+4))$ is of the third order; therefore, we need only three integrator for its realization as shown in Fig. 9.44(a).

(b) *Cascade realization* Given that

$$\begin{aligned}
 H(s) &= \frac{s(s+2)}{(s+1)(s+3)(s+4)} \\
 H(s) &= \left(\frac{s}{s+1} \right) \left(\frac{s+2}{s+3} \right) \left(\frac{1}{s+4} \right) = H_1(s)H_2(s)
 \end{aligned}$$

where

$$H_1(s) = \frac{s}{s+1} = \frac{1}{1 + (1/s)}$$

$$H_2(s) = \frac{s+2}{s+3} = \frac{1 + (2/s)}{1 + (3/s)}$$

and

$$H_3(s) = \frac{1}{s+4} = \frac{1/s}{1 + 4/s}$$

We can realize $H(s)$ as a cascade of $H_1(s)$, $H_2(s)$, and $H_3(s)$. $H_1(s)$, $H_2(s)$, and $H_3(s)$ are first-order transfer functions and can be implemented by using canonic realization as shown in Fig. 9.44(b).

(c) *Parallel realization* We can express $H(s)$ as a sum of partial fractions as

$$\begin{aligned}
 H(s) &= \frac{s(s+2)}{(s+1)(s+3)(s+4)} \\
 &= \frac{-1/6}{s+1} + \frac{-3/2}{s+3} + \frac{8/3}{s+4} \\
 H(s) &= H_1(s) + H_2(s) + H_3(s)
 \end{aligned}$$

where

$$\begin{aligned}
 H_1(s) &= \frac{-1/6}{s+1} = \frac{-(1/6)(1/s)}{1 + (1/s)} \\
 H_2(s) &= \frac{-3/2}{s+3} = \frac{-(3/2)(1/s)}{1 + 3/s}
 \end{aligned}$$

and

$$H_3(s) = \frac{8/3}{s+4} = \frac{(8/3)(1/s)}{1 + 4/s}$$

We can realize $H(s)$ as a sum of $H_1(s)$, $H_2(s)$, and $H_3(s)$. $H_1(s)$, $H_2(s)$, and $H_3(s)$ are first-order transfer functions and can be implemented by using canonic realization as shown in Fig. 9.44(c).

(d) *Transposed realization of canonic direct form* Fig. 9.44(d) shows the transposed version of the canonic direct form realization in Fig. 9.44(a) found by

- (i) reversing all paths.
- (ii) replacing pick-off nodes by adders and vice versa.
- (iii) interchanging the input and output nodes.

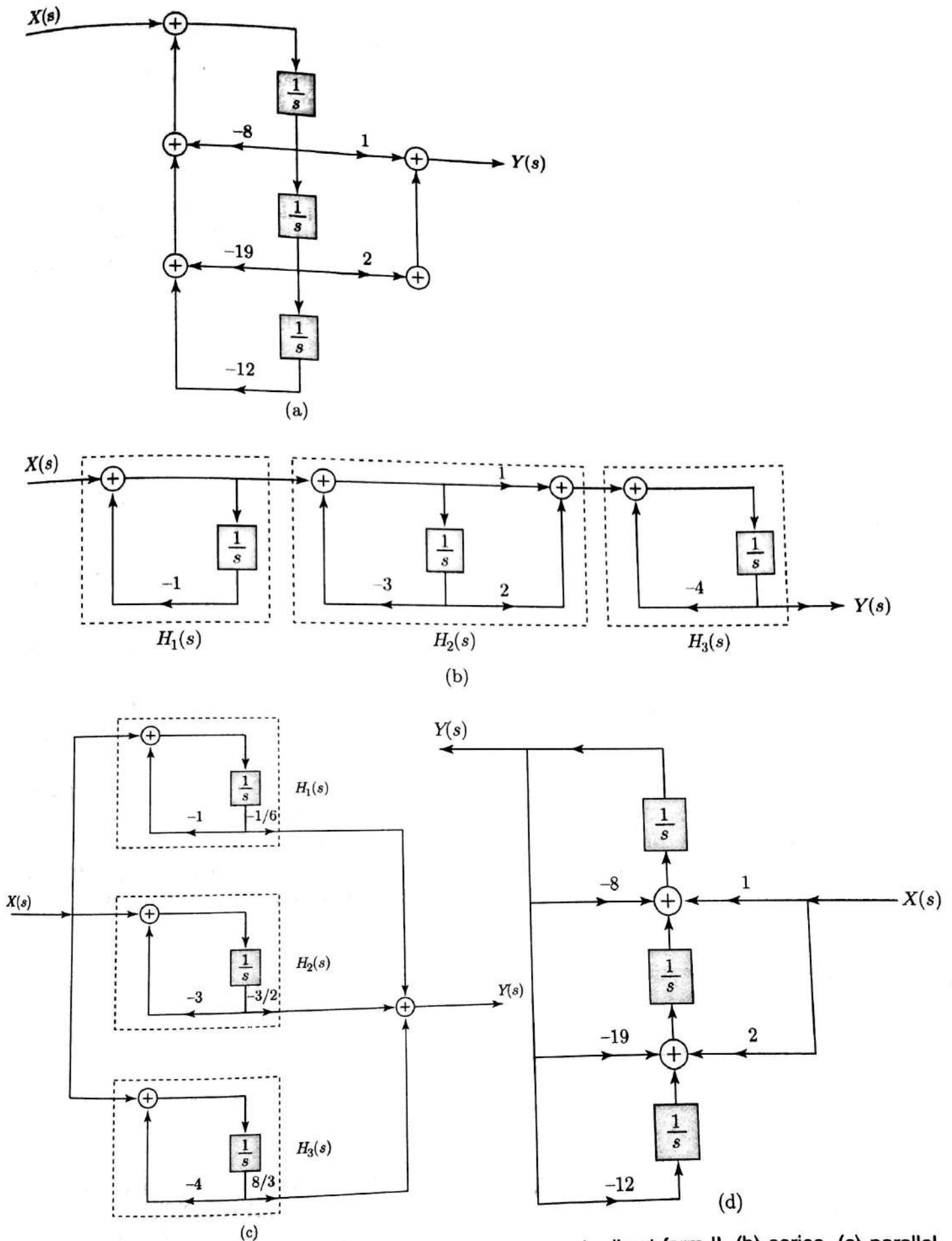


Fig. 9.44 Realization of $H(s)$ for Example 9.86 using (a) canonic direct form-II, (b) series, (c) parallel and (d) transposed form of canonic direct form-II

SUMMARY

1. The Laplace transform is a function of s and is represented in a complex plane termed the s -plane.
2. The Fourier transform is obtained by evaluating the Laplace transform on the $j\omega$ -axis.
3. The bilateral Laplace transform of $x(t)$ is defined by

$$\mathcal{L}[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

and the inverse Laplace transform of $X(s)$ is defined by

$$x(t) = \frac{1}{2\pi j} \int_{\sigma-j\infty}^{\sigma+j\infty} X(s) e^{st} ds$$

4. The values of s for which $X(s)$ converges (i.e., $X(s)$ exists) constitute the region of convergence (ROC).
5. The transformation $x(t) \longleftrightarrow X(s)$ is not one to one unless ROC is specified.
6. The transfer function $H(s)$ of an LTI system is the Laplace transform of its impulse response $h(t)$. It may also be defined as a ratio of the Laplace transform of the output to the Laplace transform of the input when all initial conditions are zero (system in zero state).
7. If $X(s)$ is the Laplace transform of the input $x(t)$ and $Y(s)$ is the Laplace transform of the corresponding output $y(t)$ (when

all initial conditions are zero), then $Y(s) = X(s)H(s)$.

8. The LTI system response to an everlasting exponential e^{st} is also an everlasting exponential $H(s)e^{st}$.
9. A causal system with rational system function $H(s)$ is stable if and only if all the poles of $H(s)$ lie in the left-half of the s -plane.
10. The unilateral, or one-sided Laplace transform applies to causal signals and provides a convenient tool for solving system problems involving differential equations with initial conditions.
11. The unilateral Laplace transform is defined as

$$\mathcal{L}_u = X(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt$$

12. The bilateral and unilateral Laplace transforms are related by

$$X(s) = \mathcal{L}_u[x(t)u(t)] + \mathcal{L}_u[x_-(-t)u(t)] \Big|_{s \rightarrow -s}$$

where $\mathcal{L}_u[x(t)u(t)]$ is the unilateral Laplace transform of the causal part of $x(t)$ and $x_-(t)$ is the noncausal part of $x(t)$.

13. The natural response (zero-input response) represents the system output when the input is zero. The forced response (zero-state response) represents the system output when the initial conditions are zero.

MULTIPLE-CHOICE QUESTIONS

1. The impulse response of a system described by the differential equation $(d^2y(t))/dt^2 + y(t) = x(t)$ will be
 - (a) a constant
 - (b) an impulse function
 - (c) a sinusoid
 - (d) an exponentially decaying function
2. Laplace transform converts convolution of time-signals to
 - (a) addition
 - (b) subtraction
 - (c) multiplication
 - (d) division
3. Region of convergence of a causal LTI system
 - (a) is the entire s -plane
 - (b) is the right-half of s -plane
 - (c) left-half of s -plane
 - (d) does not exist

4. Laplace transform of t^n is always equal to
 (a) n/s^n
 (b) $n!/s^n$
 (c) $n!/s^{n+1}$
 (d) all
5. The final value of $x(t) \xrightarrow{\mathcal{L}_u} X(s) = (5s+6)/(s^3+6s^2+3s)$ is
 (a) 2
 (b) 1
 (c) 5/6
 (d) 5
6. The Laplace transform of $u(t)$ is
 (a) $1/s$
 (b) s
 (c) $1/s^2$
 (d) 1
7. The Laplace transform of $u(t)$ is $A(s)$ and the Fourier transform of $u(t)$ is $B(j\omega)$. Then
 (a) $B(j\omega) = A(s)|_{s=j\omega}$
 (b) $A(s) = 1/s$, but $B(j\omega) \neq \frac{1}{j\omega}$
 (c) $A(s) \neq 1/s$, but $B(j\omega) = \frac{1}{j\omega}$
 (d) $A(s) \neq 1/s$, but $B(j\omega) \neq \frac{1}{j\omega}$
8. The Laplace transform of $x(t-t_0)$ is
 (a) $e^{st_0}X(s)$
 (b) $e^{-st_0}X(s)$
 (c) $X(s-s_0)$
 (d) none of the above
9. If $x(t) \xrightarrow{\mathcal{L}_u} X(s)$, then $\mathcal{L}_u \left[\frac{dx(t)}{dt} \right]$ is given by
 (a) $\frac{dX(s)}{ds}$
 (b) $\frac{X(s)}{s} - \frac{x(0^-)}{s}$
 (c) $sX(s) - x(0^-)$
 (d) $sX(s) - sx(0^-)$
10. If the Laplace transform of $x(t)$ is $\omega/(s^2+\omega^2)$, then the value of $\lim_{t \rightarrow \infty} x(t)$ is
 (a) cannot be determined
 (b) is zero
 (c) is unity
 (d) is infinity
11. The unit impulse response of an LTI system is the unit step function $u(t)$. For $t > 0$, the response of the system to an excitation $e^{-at}u(t)$, $a > 0$, will be
 (a) ae^{-at}
 (b) $(1-e^{-at})/a$
 (c) $a(1-e^{-at})$
 (d) $1-e^{-at}$
12. If R_x is the ROC of $x(t)$ and R_y is the ROC of $y(t)$, then the ROC of $x(t) * y(t)$ is
 (a) $R_x + R_y$
 (b) $R_x - R_y$
 (c) $R_x \cap R_y$
 (d) $R_x \cup R_y$
13. If the notation $*$ is used to denote the convolution and $x(t) \xleftrightarrow{\mathcal{L}} X(s)$ and $y(t) \xleftrightarrow{\mathcal{L}} Y(s)$ make the Laplace transform pair, then $x(t)y(t) \xleftrightarrow{\mathcal{L}} F(s)$ given by
 (a) $X(s) * Y(s)$
 (b) $X(s)Y(s)$
 (c) $\frac{1}{2\pi} [X(s) * Y(s)]$
 (d) $\frac{1}{2\pi j} [X(s) * Y(s)]$
14. Region of convergence of $X(s)$ contain
 (a) zeros
 (b) poles
 (c) no zero
 (d) no pole
15. If the system is causal and stable, the system poles must lie
 (a) on the $j\omega$ -axis
 (b) on the left-half of the s -plane
 (c) on the right-half of the s -plane
 (d) both (a) and (c)

PROBLEMS

9.1 Find the Laplace transform of the following signals:

- (a) $x_1(t) = 3u(t) - 3u(t-3)$
 (b) $x_2(t) = r(t) - r(t-1) - u(t-2)$
 (c) $x_3(t) = 10u(t) - 10e^{-t}u(t)$

9.2 Find the Laplace transform of the following signals:

- (a) $x_1(t) = (t-1)u(t)$
 (b) $x_2(t) = e^{-(t+4)}u(t)$
 (c) $x_3(t) = e^{(t-3)}u(t-4)$

9.3 Find the Laplace transform of the following signals:

(a) $x_1(t) = e^{-5t}[u(t) - u(t - 5)]$

(b) $x_2(t) = [e^{-3t} - e^{-10t} + 2 \cos(3t)]u(t)$

9.4 Use the Laplace transform to determine the output $y(t)$ of the system represented by the differential equation:

$$\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 3y(t) = e^{-t}$$

$$\dot{y}(0^-) = y(0^-) = 1$$

9.5 State the initial-value and final-value theorems of Laplace transforms. Compute the initial and final values for

$$X(s) = \frac{3s + 4}{s(s + 1)(s + 2)^2}$$

9.6 Consider the signal $x(t) = e^{-t}u(t) + e^{-2t}u(t)$. Express its Laplace transform in the form $X(s) = K(N(s)/D(s))$, where K = system constant. Identify the ROC. Indicate the poles and zeros in the s -plane.

9.7 Determine the forced and natural responses of the system described by the following differential equations with the specified input and initial conditions:

$$\frac{dy(t)}{dt} + 3y(t) = 4x(t) \quad x(t) = \cos(2t)u(t)$$

$$y(0^-) = -2$$

$$\frac{d^2 y(t)}{dt^2} + 4y(t) = 8x(t) \quad x(t) = u(t),$$

$$y(0^-) = 1 \quad \dot{y}(0^-) = 2$$

9.8 The differential equation characterizing a continuous LTI system is given by

$$2 \frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + y(t) = u(t)$$

With initial conditions $y(0^-) = -1$ and $\dot{y}(0^-) = 1$. Determine $Y(s)$ as the sum of zero-input response and zero-state response. Also determine $y(t)$.

9.9 Using the Laplace transform, solve the following differential equations and in each case determine the zero-input and zero-state components of the solution:

(a)

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt}$$

if $y(0^-) = \dot{y}(0^-) = 0$ and $x(t) = u(t)$.

(b)

$$\frac{d^2 y(t)}{dt^2} + 4 \frac{dy(t)}{dt} + 4y(t) = \frac{dx(t)}{dt} + x(t)$$

if $y(0^-) = 2$, $\dot{y}(0^-) = 1$ and $x(t) = e^{-t}u(t)$.

9.10 For a system with transfer function

$$H(s) = \frac{2s + 3}{s^2 + 2s + 5}$$

(a) Find the zero-state response for input $x(t)$ of (i) $10u(t)$ and (ii) $u(t - 5)$.

(b) For this system, write the differential equation relating the output $y(t)$ to the input $x(t)$.

9.11 An LTI system has a step response given by $s(t) = e^{-t}u(t) - e^{-2t}u(t)$. Determine the output of this system $y(t)$ given an input $x(t) = \delta(t - \pi) - \cos(\sqrt{3}t)u(t)$.

9.12 Find the inverse Laplace transform of the following functions:

(a) $X(s) = \frac{2s + 5}{(s + 2)(s + 3)} \quad -3 < \Re\{s\} < -2$

(b) $X(s) = \frac{2s - 5}{(s - 2)(s - 3)} \quad 2 < \Re\{s\} < 3$

(c) $X(s) = \frac{2s + 3}{(s + 1)(s + 2)} \quad \Re\{s\} > -1$

(d) $X(s) = \frac{2s + 3}{(s + 1)(s + 2)} \quad \Re\{s\} < -2$

9.13 An absolutely integrable signal $x(t)$ has a pole at $s = \pi$. It is possible that other poles may be present.

(a) Can $x(t)$ be left-sided? Explain.

(b) Can $x(t)$ be right-sided? Explain.

(c) Can $x(t)$ be two-sided? Explain.

(d) Can $x(t)$ be of finite duration? Explain.

9.14 The Laplace transform of a signal $x(t)$ that is zero for $t < 0$ is

$$X(s) = \frac{s^3 + 2s^2 + 3s + 2}{s^4 + 2s^3 + 2s^2 + 2s + 2}$$

Determine the Laplace transform of the following signals:

(a) $y(t) = 3x\left(\frac{t}{3}\right)$

(b) $y(t) = tx(t)$

(c) $y(t) = tx(t - 1)$

(d) $y(t) = \frac{dx(t)}{dt}$

(e) $y(t) = (t - 1)x(t - 1) + \frac{dx(t)}{dt}$

$$(f) y(t) = \int_0^t x(\tau) d\tau$$

9.15 Determine the initial and final values of each of the signals whose Laplace transform are as follows without computing the inverse Laplace transform. If there is no final value, state why not.

$$(a) X(s) = \frac{1}{s+a}$$

$$(b) X(s) = \frac{1}{(s+a)^n}$$

$$(c) X(s) = \frac{6}{s(s^2+25)}$$

$$(d) X(s) = \frac{s+2}{s+3}$$

$$(e) X(s) = \frac{s^2+s+3}{s^3+4s^2+2s+2}$$

9.16 Find the following convolutions using Laplace transform:

$$(a) e^{at}u(t) * e^{bt}u(t), a \neq b.$$

$$(b) e^{at}u(t) * e^{at}u(t)$$

$$(c) tu(t) * e^{at}u(t)$$

$$(d) \text{rect}\left(\frac{t}{T}\right) * \text{rect}\left(\frac{t}{T}\right)$$

9.17 Realize

(a)

$$H(s) = \frac{s(s+2)}{(s+1)(s+3)(s+4)}$$

(b)

$$H(s) = \frac{s^3}{(s+1)^2(s+2)(s+3)}$$

by canonic, cascade, and parallel forms.

9.18 Realize the transfer function in Problem 9.17

by using the transposed form of the realizations found in Problem 9.17.

9.19 Let

$$g(t) = x(t) + \alpha x(-t)$$

where

$$x(t) = \beta e^{-t}u(t)$$

and the Laplace transform of $g(t)$ is

$$G(s) = \frac{s}{s^2-1} \quad -1 < \Re\{s\} < 1$$

Determine the values of the constant α and β .

9.20 Obtain the time function $f(t)$ whose Laplace transform is

$$F(s) = \frac{s^2+3s+1}{(s+1)^3(s+2)^2}$$

9.21 Find the inverse Laplace transform of the following:

$$(a) X(s) = \frac{s^2}{(s+a)^2+b^2}$$

$$(b) X(s) = \ln\left(\frac{s+1}{s+2}\right)$$

9.22 The autocorrelation function $r_{xx}(t)$ of a signal is given by

$$r_{xx}(t) = \int_{-\infty}^{\infty} x(\tau)x(\tau+t) d\tau$$

Derive an expression for $R_{ss}(s) = \mathcal{L}[r_{xx}(\tau)]$ in terms of $X(s)$, where $X(s) = \mathcal{L}[x(t)]$.

ANSWERS TO MULTIPLE-CHOICE QUESTIONS

- | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|--------|--------|--------|
| 1. (c) | 2. (c) | 3. (b) | 4. (b) | 5. (a) | 6. (a) | 7. (b) | 8. (b) | 9. (c) |
| 10. (a) | 11. (b) | 12. (c) | 13. (d) | 14. (d) | 15. (b) | | | |